

“Sparling identity” with differential forms

Initialization

```
In[*]:= xAct`xTerior`Diff;  
  
In[*]:= Block[{Print}, << xAct`xTerior`  
|bloquea |escribe
```

Global settings

```
In[*]:= << xAct`ShowTime1`  
  
In[*]:= $ShowTimeThreshold = 1;  
  
In[*]:= $PrePrint = ScreenDollarIndices;  
|pre-escribe
```

We define a suite of functions used to apply rules:

```
In[*]:= ApplyRule[expr_] :=  
|evalúa |lista MakeRule[Evaluate[List @@ expr], MetricOn → All, ContractMetrics → True] |todo |verdade  
  
In[*]:= ApplyRuleN[expr_] := MakeRule[Evaluate[List @@ expr] |evalúa |lista  
  
In[*]:= ApplyRuleNS[expr_] := MakeRule[Evaluate[List @@ expr], UseSymmetries → False] |evalúa |lista |falso  
  
In[*]:= ApplyRuleAll[expr_Equal] :=  
|igual  
|evalúa MakeRule[{Evaluate@expr[[1]], Evaluate@expr[[2]]}, MetricOn → All] |evalúa |todo  
  
In[*]:= ApplyRuleNoSym[expr_] := MakeRule[Evaluate[List @@ expr], |evalúa |lista  
|falso UseSymmetries → False, MetricOn → All, ContractMetrics → True] |todo |verdade
```

We introduce a list and an association to store the equations that we obtain

```
In[*]:= $Equations = {};  
  
In[*]:= SetAttributes[AddEquation, HoldAll] |asigna atributos |impide evaluación de todo argumento  
AddEquation[expr_Symbol] := AppendTo[$Equations, Hold[expr]] |símbolo |añade al final |impide evaluació  
  
In[*]:= PrintEquations := Print[(# → ScreenDollarIndices /@ ReleaseHold@# &) /@ $Equations] |escribe |desbloquea evaluación  
  
In[*]:= Equations := Association[(# → ScreenDollarIndices /@ ReleaseHold@# &) /@ $Equations] |asociación |desbloquea evaluación
```

This is to avoid problems canonicalising expressions involving covariant derivatives:

```
In[*]:= $MixedDers = False;  
|falso
```

Define 4-dimensional manifold

```
In[*]:= DefConstantSymbol[Dim]
      ** DefConstantSymbol: Defining constant symbol Dim.
```

```
In[*]:= Dim = 4;
```

```
In[*]:= DefManifold[M4, Dim, IndexRange[a, m]]
      ** DefManifold: Defining manifold M4.
      ** DefVBundle: Defining vbundle TangentM4.
```

We introduce a metric

```
In[*]:= DefMetric[{1, 3, 0}, metricg[-a, -b], CD, PrintAs -> "g"]
      ** DefTensor: Defining symmetric metric tensor metricg[-a, -b].
      ** DefTensor: Defining antisymmetric tensor epsilon metricg[-a, -b, -c, -d].
      ** DefTensor: Defining tetrametric Tetrametricg[-a, -b, -c, -d].
      ** DefTensor: Defining tetrametric Tetrametricg†[-a, -b, -c, -d].
      ** DefCovD: Defining covariant derivative CD[-a].
      ** DefTensor: Defining vanishing torsion tensor TorsionCD[a, -b, -c].
      ** DefTensor: Defining symmetric Christoffel tensor ChristoffelCD[a, -b, -c].
      ** DefTensor: Defining Riemann tensor RiemannCD[-a, -b, -c, -d].
      ** DefTensor: Defining symmetric Ricci tensor RicciCD[-a, -b].
      ** DefCovD: Contractions of Riemann automatically replaced by Ricci.
      ** DefTensor: Defining Ricci scalar RicciScalarCD[].
      ** DefCovD: Contractions of Ricci automatically replaced by RicciScalar.
      ** DefTensor: Defining symmetric Einstein tensor EinsteinCD[-a, -b].
      ** DefTensor: Defining Weyl tensor WeylCD[-a, -b, -c, -d].
      ** DefTensor: Defining symmetric TFRicci tensor TFRicciCD[-a, -b].
      ** DefTensor: Defining Kretschmann scalar KretschmannCD[].
      ** DefCovD: Computing RiemannToWeylRules for dim 4
      ** DefCovD: Computing RicciToTFRicci for dim 4
      ** DefCovD: Computing RicciToEinsteinRules for dim 4
      ** DefInertHead: Defining inert head CovDCD.
      ** DefTensor: Defining weight +2 density Detmetricg[[]].      Determinant.
```

Pretty print:

```
In[*]:= PrintAs@epsilon metricg ^= "η";
In[*]:= PrintAs@RiemannCD ^= "R";
```

Nester-Witten 2-form

We define the “super-potential” 2-form (sometimes known as Nester-Witten 2-form):

```
In[*]:= DefDiffForm[L[-a], M4, 2]
** DefTensor: Defining tensor L[-a].
```

The super-potential 2-form is given by

```
In[*]:= SuperPotential2Form =
  L[-a] == 1/2 epsilonmetricg[-a, -b, -c, -d] x Coframe[M4][b] ^ ChristoffelForm[CD][c, d]
Out[*]=
  L_a == 1/2 eta_abcd theta^b ^ Gamma[nabla]^cd
```

```
In[*]:= AddEquation[SuperPotential2Form];
```

Exterior derivative of the Nester-Witten 2-form (Sparling identity)

The exterior covariant derivative of the volume element vanish:

```
In[*]:= Diff[epsilonmetricg[-a, -b, -c, -d], CD] == 0
[diff]
```

```
Out[*]=
  d[nabla][eta_abcd] == 0
```

We use this relation to isolate the exterior derivative of the volume element as follows:

```
In[*]:= ChangeExtD[%, CD, PD]
Out[*]=
  d[nabla][eta_abcd] - eta_ebcd Gamma[nabla]^e_a - eta_aecd Gamma[nabla]^e_b - eta_abed Gamma[nabla]^e_c - eta_abce Gamma[nabla]^e_d == 0
```

Cartan eqns. enable us to compute the exterior derivative of the volume element:

```
In[*]:= UseCartan[#, CD] & /@ %
Out[*]=
  - eta_ebcd Gamma[nabla]^e_a - eta_aecd Gamma[nabla]^e_b - eta_abed Gamma[nabla]^e_c - eta_abce Gamma[nabla]^e_d + eta_abcd Gamma[nabla]^e_e == 0

In[*]:= % /. expr_ChristoffelForm[CD][a_, -a_] -> Hold[expr ChristoffelForm[CD][a, -a]]
[impide evaluación]
Out[*]=
  Hold[eta_abcd Gamma[nabla]^e_e] - eta_ebcd Gamma[nabla]^e_a - eta_aecd Gamma[nabla]^e_b - eta_abed Gamma[nabla]^e_c - eta_abce Gamma[nabla]^e_d == 0
```

```
In[*]:= Solve[%, Cases[%, _Hold, Infinity]] // Flatten // ReleaseHold
[resuelve] [casos] [impide...] [infinito] [aplana] [desbloquea evaluación]
Out[*]=
  { eta_abcd Gamma[nabla]^i_i -> eta_ebcd Gamma[nabla]^e_a + eta_afcd Gamma[nabla]^f_b + eta_abgd Gamma[nabla]^g_c + eta_abch Gamma[nabla]^h_d }
```

```
In[*]:= Equal@@First@%
[igual] [primero]
Out[*]=
  eta_abcd Gamma[nabla]^e_e == eta_ebcd Gamma[nabla]^e_a + eta_aecd Gamma[nabla]^e_b + eta_abed Gamma[nabla]^e_c + eta_abce Gamma[nabla]^e_d
```

```
In[*]:= etaGamma = %;
```

```
In[*]:= AddEquation[etaGamma];
```

We compute the exterior derivative of the super-potential 2-form:

```
In[*]:= Diff /@ SuperPotential2Form
|diff
```

```
Out[*]=
```

$$d[L_a] = \frac{1}{2} \left(\eta_{abcd} \left(d[\theta^b] \wedge \Gamma[\nabla]^{cd} - \theta^b \wedge d[\Gamma[\nabla]^{cd}] \right) + d[\eta_{abcd}] \wedge \theta^b \wedge \Gamma[\nabla]^{cd} \right)$$

We use now Cartan structure equations to replace the different exterior derivatives in the right hand side:

```
In[*]:= % /. MakeRule[{ChristoffelForm[CD][a, -b], ChristoffelForm[CD][a, -b]},
MetricOn -> All, ContractMetrics -> False]
|todo |false
```

```
Out[*]=
```

$$d[L_a] = \frac{1}{2} \left(g^{dg} d[\eta_{abcd}] \wedge \theta^b \wedge \Gamma[\nabla]_g^c + \eta_{abcd} \left(g^{de} d[\theta^b] \wedge \Gamma[\nabla]_e^c - g^{df} \theta^b \wedge d[\Gamma[\nabla]_f^c] - \theta^b \wedge d[g^{df}] \wedge \Gamma[\nabla]_f^c \right) \right)$$

```
In[*]:= UseCartan[#, CD] & /@%
```

```
Out[*]=
```

$$d[L_a] = \frac{1}{2} \left(\eta_{abcd} g^{dg} \Gamma[\nabla]_h^c \wedge \theta^b \wedge \Gamma[\nabla]_g^c + \eta_{abcd} \left(-g^{de} \Gamma[\nabla]_i^b \wedge \theta^i \wedge \Gamma[\nabla]_e^c - g^{df} \left(\theta^b \wedge R[\nabla]_f^c - \theta^b \wedge \Gamma[\nabla]_j^c \wedge \Gamma[\nabla]_f^j \right) + \theta^b \wedge \Gamma[\nabla]^{df} \wedge \Gamma[\nabla]_f^c + \theta^b \wedge \Gamma[\nabla]^{fd} \wedge \Gamma[\nabla]_f^c \right) \right)$$

```
In[*]:= ToCanonical@ContractMetric@%
```

```
Out[*]=
```

$$d[L_a] = -\frac{1}{2} \eta_a^{bcd} \theta_b \wedge R[\nabla]_{cd} - \frac{1}{2} \eta_a^{bcd} \Gamma[\nabla]_{bc} \wedge \Gamma[\nabla]_d^e \wedge \theta_e + \frac{1}{2} \eta_a^{bcd} \Gamma[\nabla]_{bc} \wedge \Gamma[\nabla]_e^e \wedge \theta_d - \frac{1}{2} \eta_a^{bcd} \Gamma[\nabla]_b^e \wedge \Gamma[\nabla]_{ce} \wedge \theta_d$$

We do some manipulations to simplify the above:

```
In[*]:= % /. epsilonmetricg[inds__] ×
ChristoffelForm[CD][inds1__] ^ ChristoffelForm[CD][a_, -a_] ^ Coframe[M4][inds2_] :->
ChristoffelForm[CD][inds1] ^ ReplaceAll[epsilonmetricg[inds] ×
|sustituye todos
ChristoffelForm[CD][a, -a], ApplyRule@etaGamma] ^ Coframe[M4][inds2]
```

```
Out[*]=
```

$$d[L_a] = -\frac{1}{2} \eta_a^{bcd} \theta_b \wedge R[\nabla]_{cd} - \frac{1}{2} \eta_a^{bcd} \Gamma[\nabla]_{bc} \wedge \Gamma[\nabla]_d^e \wedge \theta_e + \frac{1}{2} \left(\eta_e^{bcd} \Gamma[\nabla]_{bc} \wedge \Gamma[\nabla]_a^e \wedge \theta_d + \eta_{af}^{cd} \Gamma[\nabla]_{bc} \wedge \Gamma[\nabla]^{fb} \wedge \theta_d + \eta_a^{b d} \Gamma[\nabla]_{bc} \wedge \Gamma[\nabla]^{gc} \wedge \theta_d + \eta_a^{bc h} \Gamma[\nabla]_{bc} \wedge \Gamma[\nabla]^{hd} \wedge \theta_d \right) - \frac{1}{2} \eta_a^{bcd} \Gamma[\nabla]_b^e \wedge \Gamma[\nabla]_{ce} \wedge \theta_d$$

```
In[*]:= ToCanonical@ContractMetric@%
```

```
Out[*]=
```

$$d[L_a] = -\frac{1}{2} \eta_a^{bcd} \theta_b \wedge R[\nabla]_{cd} - \frac{1}{2} \eta^{bcde} \Gamma[\nabla]_{ba} \wedge \Gamma[\nabla]_{cd} \wedge \theta_e - \frac{1}{2} \eta_a^{bcd} \Gamma[\nabla]_b^e \wedge \Gamma[\nabla]_{ec} \wedge \theta_d$$

This is the so-called “Sparling identity”:

```
In[*]:= ScreenDollarIndices /@%
```

```
Out[*]=
```

$$d[L_a] = -\frac{1}{2} \eta_a^{bcd} \Theta_b \wedge R[\nabla]_{cd} - \frac{1}{2} \eta^{bcde} \Gamma[\nabla]_{ba} \wedge \Gamma[\nabla]_{cd} \wedge \Theta_e - \frac{1}{2} \eta_a^{bcd} \Gamma[\nabla]_b^e \wedge \Gamma[\nabla]_{ec} \wedge \Theta_d$$

```
In[*]:= SparlingIdentity = %;
```

```
In[*]:= AddEquation[SparlingIdentity];
```

We introduce now the Einstein 3-form and the Sparling 3-form

```
In[*]:= DefDiffForm[E[-b], M4, 3]
```

```
** DefTensor: Defining tensor E[-b].
```

```
In[*]:= DefDiffForm[S[-b], M4, 3]
```

```
** DefTensor: Defining tensor S[-b].
```

By definition

```
In[*]:= EinsteinForm = E[-a] == SparlingIdentity[[2, 1]]
```

```
Out[*]=
```

$$E_a = -\frac{1}{2} \eta_a^{bcd} \Theta_b \wedge R[\nabla]_{cd}$$

```
In[*]:= AddEquation[EinsteinForm];
```

```
In[*]:= SparlingForm = S[-a] == SparlingIdentity[[2, 2]] + SparlingIdentity[[2, 3]]
```

```
Out[*]=
```

$$S_a = -\frac{1}{2} \eta^{bcde} \Gamma[\nabla]_{ba} \wedge \Gamma[\nabla]_{cd} \wedge \Theta_e - \frac{1}{2} \eta_a^{bcd} \Gamma[\nabla]_b^e \wedge \Gamma[\nabla]_{ec} \wedge \Theta_d$$

```
In[*]:= AddEquation[SparlingForm];
```

The Sparling identity is rendered in the form

```
In[*]:= First@SparlingIdentity == First@EinsteinForm + First@SparlingForm
|primero |primero |primero
```

```
Out[*]=
```

$$d[L_a] = E_a + S_a$$

```
In[*]:= SparlingIdentityShort = %;
```

```
In[*]:= AddEquation[SparlingIdentityShort];
```

We compute the Hodge dual of the Einstein 3-form

```
In[*]:= EinsteinForm
```

```
Out[*]=
```

$$E_a = -\frac{1}{2} \eta_a^{bcd} \Theta_b \wedge R[\nabla]_{cd}$$

```
In[*]:= CurvatureFormToTensor[#, Coframe] & /@%
```

```
Out[*]=
```

$$E_a = \frac{1}{4} \eta_a^{bcd} R_{efdc} \Theta_b \wedge \Theta^e \wedge \Theta^f$$

```
In[*]:= Hodge[metricg] /@%
```

```
Out[*]=
```

$$\star_g[\varepsilon_a] = \frac{1}{4} \eta_a{}^{bcd} R_{efdc} \star_g[\theta_b \wedge \theta^e \wedge \theta^f]$$

```
In[*]:= ExpandHodgeDual[#, Coframe[M4], metricg] & /@%
```

```
Out[*]=
```

$$\star_g[\varepsilon_a] = \frac{1}{4} \left(-\delta_a^e \delta_f^d g^{cb} + \delta_a^b \delta_f^d g^{ce} + \delta_a^e \delta_f^c g^{db} - g_{af} g^{ce} g^{db} - \delta_a^b \delta_f^c g^{de} + g_{af} g^{cb} g^{de} \right) R_{bedc} \theta^f$$

```
In[*]:= ToCanonical /@ (ContractMetric /@%)
```

```
Out[*]=
```

$$\star_g[\varepsilon_a] = -\frac{1}{2} R[\nabla] \theta_a + R[\nabla]_{ab} \theta^b$$

```
In[*]:= % // RicciToEinstein
```

```
Out[*]=
```

$$\star_g[\varepsilon_a] = -\frac{1}{2} R[\nabla] \theta_a + \left(G[\nabla]_{ab} + \frac{1}{2} g_{ba} R[\nabla] \right) \theta^b$$

```
In[*]:= ToCanonical /@ (ContractMetric /@%)
```

```
Out[*]=
```

$$\star_g[\varepsilon_a] = G[\nabla]_{ab} \theta^b$$

The 1-form $\star[\varepsilon_a]$ is called the “matter current”.

```
In[*]:= MatterCurrent = %;
```

```
In[*]:= AddEquation[MatterCurrent];
```

The quantity $\star[\mathcal{S}_a]$ shall be called “sparling current”. The sum of the “matter current” and the “Sparling current” is always conserved:

```
In[*]:= SparlingIdentityShort
```

```
Out[*]=
```

$$d[L_a] = \varepsilon_a + \mathcal{S}_a$$

```
In[*]:= Hodge[metricg] /@%
```

```
Out[*]=
```

$$\star_g[d[L_a]] = \star_g[\varepsilon_a] + \star_g[\mathcal{S}_a]$$

```
In[*]:= Codiff[metricg] /@%
```

```
Out[*]=
```

$$\delta_g[\star_g[d[L_a]]] = \delta_g[\star_g[\varepsilon_a]] + \delta_g[\star_g[\mathcal{S}_a]]$$

```
In[*]:= CodiffToDiff@First@% == Last@%
```

|primero |último

```
Out[*]=
```

$$0 = \delta_g[\star_g[\varepsilon_a]] + \delta_g[\star_g[\mathcal{S}_a]]$$

This equation says that the sum of the matter and the Einstein currents is conserved. While the current $\star[\varepsilon_a]$ is tensorial, the “sparling current” $\star[\mathcal{S}_a]$ is not. Therefore one cannot define the energy momentum of the gravitational field as a tensor in the tangent bundle.

List of equations used

```
In[*]:= Equations[]
Out[*]=
```

$$\left\langle \begin{aligned} &\text{Hold}[\text{SuperPotential2Form}] \rightarrow L_a = \frac{1}{2} \eta_{abcd} \theta^b \wedge \Gamma[\nabla]^{cd}, \text{Hold}[\text{etaGamma}] \rightarrow \\ &\eta_{abcd} \Gamma[\nabla]^e_e = \eta_{ebcd} \Gamma[\nabla]^e_a + \eta_{aecd} \Gamma[\nabla]^e_b + \eta_{abed} \Gamma[\nabla]^e_c + \eta_{abce} \Gamma[\nabla]^e_d, \\ &\text{Hold}[\text{SparlingIdentity}] \rightarrow d[L_a] = \\ &\quad -\frac{1}{2} \eta_a^{bcd} \theta_b \wedge R[\nabla]_{cd} - \frac{1}{2} \eta^{bcde} \Gamma[\nabla]_{ba} \wedge \Gamma[\nabla]_{cd} \wedge \theta_e - \frac{1}{2} \eta_a^{bcd} \Gamma[\nabla]_b^e \wedge \Gamma[\nabla]_{ec} \wedge \theta_d, \\ &\text{Hold}[\text{EinsteinForm}] \rightarrow \varepsilon_a = -\frac{1}{2} \eta_a^{bcd} \theta_b \wedge R[\nabla]_{cd}, \\ &\text{Hold}[\text{SparlingForm}] \rightarrow S_a = -\frac{1}{2} \eta^{bcde} \Gamma[\nabla]_{ba} \wedge \Gamma[\nabla]_{cd} \wedge \theta_e - \frac{1}{2} \eta_a^{bcd} \Gamma[\nabla]_b^e \wedge \Gamma[\nabla]_{ec} \wedge \theta_d, \\ &\text{Hold}[\text{SparlingIdentityShort}] \rightarrow d[L_a] = \varepsilon_a + S_a, \\ &\text{Hold}[\text{MatterCurrent}] \rightarrow *_g[\varepsilon_a] = G[\nabla]_{ab} \theta^b \end{aligned} \right\rangle$$

List of used types

```
In[*]:= types = {$Manifolds, $AbstractIndices, $Mappings,
                 $VBundles, $Tensors, $Metrics, $CovDs, $InertHeads, $ScalarFunctions,
                 $ConstantSymbols, $Parameters, $Products, $Charts, $Bases};
In[*]:= DeleteCases[types, {}]
Out[*]=
```

elimina casos

```
{ {M4}, {a, b, c, d, e, f, g, h, i, j, k, l, m}, {TM4},
  {metricg, epsilonmetricg, Tetrametricg, Tetrametricg†, TorsionCD,
    ChristoffelCD, RiemannCD, RicciCD, RicciScalarCD, EinsteinCD,
    WeylCD, TFRicciCD, KretschmannCD, Detmetricg, L, ε, S}, {metricg},
  {PD, CD}, {ERROR, Keep, Dagger, At, xAct`xTensor`Private`TMPChristoffel,
    xAct`xTensor`Private`TMPCovD, Diff, Hodge, Codiff, Int, CartanD, KoszulCD, CovDCD},
  {1, 1, 1, 1, 1, 1, 4}, {Wedge, CircleTimes} }
```

Schwarzschild in Klotsch-Strobl coordinates

```
In[*]:= SparlingForm
Out[*]=
```

$$S_a = -\frac{1}{2} \eta^{bcde} \Gamma[\nabla]_{ba} \wedge \Gamma[\nabla]_{cd} \wedge \theta_e - \frac{1}{2} \eta_a^{bcd} \Gamma[\nabla]_b^e \wedge \Gamma[\nabla]_{ec} \wedge \theta_d$$

```
In[*]:= % /. MakeRule[{Coframe[M4][a], Coframe[M4][a]}, MetricOn → All, ContractMetrics → False]
```

|todo

|false

```
Out[*]=
```

$$S_a = -\frac{1}{2} \eta^{bcde} g_{fe} \Gamma[\nabla]_{ba} \wedge \Gamma[\nabla]_{cd} \wedge \theta^f - \frac{1}{2} \eta_a^{bcd} g_{fd} \Gamma[\nabla]_b^e \wedge \Gamma[\nabla]_{ec} \wedge \theta^f$$

```
In[*]:= SparlingFormUp = %
```

```
Out[*]=
```

$$S_a = -\frac{1}{2} \eta^{bcde} g_{fe} \Gamma[\nabla]_{ba} \wedge \Gamma[\nabla]_{cd} \wedge \theta^f - \frac{1}{2} \eta_a^{bcd} g_{fd} \Gamma[\nabla]_b^e \wedge \Gamma[\nabla]_{ec} \wedge \theta^f$$

```
In[*]:= << xAct`xIdeal`
```

```
-----
```

```
Package xAct`xIdeal` version 0.1.0, {2025, 10, 29}
```

```
Copyright (C) 2023, Alfonso Garcia-Parrado Gomez-Lobo  
and Salvador Mengual Sendra, under the General Public License.
```

```
-----
```

```
These packages come with ABSOLUTELY NO WARRANTY; for details type  
Disclaimer[]. This is free software, and you are welcome to redistribute  
it under certain conditions. See the General Public License for details.
```

```
-----
```

```
In[*]:= xActSolutionData[{"Schwarzschild", "KlotschStroblCoordinates"}, "Properties"]
```

|propiedades

```
Out[*]=
```

```
{CoordinateAssumptions, CoordinateNames, Metric, ParameterNames, ParameterAssumptions,  
ParameterValues, ScalarFunctionNames, ScalarFunctionValues, SignatureOfMetric}
```

```
In[*]:= xActSolutionData[{"Schwarzschild", "KlotschStroblCoordinates"}, "CoordinateNames"]
```

```
Out[*]=
```

```
{x, y,  $\theta$ ,  $\phi$ }
```

```
In[*]:= $DefInfoQ = False;
```

|false

```
In[*]:= $CVVerbose = False;
```

|false

```
In[*]:= DefChart[KS, M4, {1, 2, 3, 4}, {x[], y[],  $\theta$ [],  $\phi$ []}]
```

```
In[*]:= xActSolutionData[{"Schwarzschild", "KlotschStroblCoordinates"}, "ParameterNames"]
```

```
Out[*]=
```

```
{m}
```

```
In[*]:= DefConstantSymbol[mass, PrintAs → "m"]
```

```
In[*]:= metricTensor = xActSolutionData[{"Schwarzschild", "KlotschStroblCoordinates" → KS,  
"ScalarFunctionValues" → {}, "ParameterValues" → {mass}}, "Metric"]
```

```
Out[*]=
```

$$\text{CTensor}\left[\left\{\left\{\frac{8 m y^2}{2 m + x y}, 4 m, 0, 0\right\}, \{4 m, 0, 0, 0\},\right.\right. \\ \left.\left.\{0, 0, (2 m + x y)^2, 0\}, \{0, 0, 0, \sin[\theta]^2 (2 m + x y)^2\}\right\}, \{-KS, -KS\}, 0\right]$$


```
In[*]:= SetCMetric[metricTensor, KS, SignatureOfMetric → {3, 1, 0}]
```

SetCMetric: There are already metrics {metricg} in vbundle TM4.

```
In[*]:= cd = CovDOfMetric[metricTensor]
```

```
Out[*]=
```

$$\begin{aligned} & \text{CCovD} \left[\text{PDKS}, \text{CTensor} \left[\left\{ \left\{ \left\{ -\frac{y (4m + xy)}{(2m + xy)^2}, 0, 0, 0 \right\}, \{0, 0, 0, 0\}, \left\{ 0, 0, -\frac{x (2m + xy)}{4m}, 0 \right\}, \right. \right. \right. \\ & \quad \left. \left\{ 0, 0, 0, -\frac{\sin[\theta]^2 x (2m + xy)}{4m} \right\} \right\}, \left\{ \left\{ \frac{y^3 (6m + xy)}{(2m + xy)^3}, \frac{y (4m + xy)}{(2m + xy)^2}, 0, 0 \right\}, \right. \right. \\ & \quad \left. \left\{ \frac{y (4m + xy)}{(2m + xy)^2}, 0, 0, 0 \right\}, \left\{ 0, 0, \frac{y (-2m + xy)}{4m}, 0 \right\}, \left\{ 0, 0, 0, \frac{\sin[\theta]^2 y (-2m + xy)}{4m} \right\} \right\}, \\ & \quad \left\{ \left\{ 0, 0, \frac{y}{2m + xy}, 0 \right\}, \left\{ 0, 0, \frac{x}{2m + xy}, 0 \right\}, \left\{ \frac{y}{2m + xy}, \frac{x}{2m + xy}, 0, 0 \right\}, \right. \\ & \quad \left. \left\{ 0, 0, 0, -\cos[\theta] \sin[\theta] \right\} \right\}, \left\{ \left\{ 0, 0, 0, \frac{y}{2m + xy} \right\}, \left\{ 0, 0, 0, \frac{x}{2m + xy} \right\}, \right. \\ & \quad \left. \left\{ 0, 0, 0, \cot[\theta] \right\} \right\}, \left\{ \frac{y}{2m + xy}, \frac{x}{2m + xy}, \cot[\theta], 0 \right\} \right\}, \{KS, -KS, -KS, 0\}, \\ & \quad \text{CTensor} \left[\left\{ \left\{ \frac{8m y^2}{2m + xy}, 4m, 0, 0 \right\}, \{4m, 0, 0, 0\}, \{0, 0, (2m + xy)^2, 0\}, \right. \right. \\ & \quad \left. \left. \{0, 0, 0, \sin[\theta]^2 (2m + xy)^2\} \right\}, \{-KS, -KS, 0\} \right] \end{aligned}$$

```
In[*]:= $Metrics = Reverse[$Metrics];
```

[\[invierte orden\]](#)

```
In[*]:= TangentM4 /: MetricsOfVBundle[TangentM4] = Reverse[MetricsOfVBundle[TangentM4]];
```

[\[invierte orden\]](#)

```
In[*]:= MetricsOfVBundle[TangentM4]
```

```
Out[*]=
```

$$\begin{aligned} & \left\{ \text{CTensor} \left[\left\{ \left\{ \frac{8m y^2}{2m + xy}, 4m, 0, 0 \right\}, \{4m, 0, 0, 0\}, \right. \right. \right. \\ & \quad \left. \left\{ 0, 0, (2m + xy)^2, 0 \right\}, \left\{ 0, 0, 0, \sin[\theta]^2 (2m + xy)^2 \right\} \right\}, \{-KS, -KS, 0\}, \text{metricg} \right\} \end{aligned}$$

```
In[*]:= Ricci[cd]
```

```
Out[*]=
```

Zero

```
In[*]:= SparlingFormUp
```

```
Out[*]=
```

$$S_a = -\frac{1}{2} \eta^{bcde} g_{fe} \Gamma[\nabla]_{ba} \wedge \Gamma[\nabla]_{cd} \wedge \theta^f - \frac{1}{2} \eta_a^{bcd} g_{fd} \Gamma[\nabla]_b^e \wedge \Gamma[\nabla]_{ec} \wedge \theta^f$$

```
In[*]:= ToBasis[KS] /@ %
```

```
Out[*]=
```

$$S_a = -\frac{1}{2} \eta^{bcde} g_{fe} \Gamma[\nabla]_{ba} \wedge \Gamma[\nabla]_{cd} \wedge \theta^f - \frac{1}{2} \eta_a^{bcd} g_{fd} \Gamma[\nabla]_b^e \wedge \Gamma[\nabla]_{ec} \wedge \theta^f$$

```
In[*]:= TraceBasisDummy /@ %;
```

```
In[*]:= % /. epsilonmetricg → epsilon[metricTensor];
```

2.21663

In[]:= % /. CD → cd

1.4125

Out[]:=

$$S_a = \frac{g_{11} \left(-e_a^3 \cos[\theta] \sin[\theta] \theta^3 \wedge \theta^4 \wedge \theta^1 x (2m+xy)^3 - e_a^4 \cos[\theta] \sin[\theta]^3 \theta^4 \wedge \theta^4 \wedge \theta^1 x (2m+xy)^3 - 4m e_a^1 \cos[\theta] \sin[\theta] \theta^1 \wedge \theta^4 \wedge \theta^1 y (4m+xy) \right)}{8 \sqrt{m^2 \sin[\theta]^2 (2m+xy)^4}} -$$

$$\frac{1}{8 \sqrt{m^2 \sin[\theta]^2 (2m+xy)^4}} g_{11} \left(e_a^3 \cos[\theta] \sin[\theta] \theta^3 \wedge \theta^4 \wedge \theta^1 x (2m+xy)^3 + e_{\dots 49 \dots}^4 \cos[\theta] \sin[\theta]^3 \right.$$

$$\left. \theta^{\dots 49 \dots} \wedge \theta^{\dots 49 \dots} \wedge \theta^{\dots 49 \dots} x (2m+xy)^3 + 4m e_{\dots 49 \dots}^1 \cos[\theta] \sin[\theta] \theta^1 \wedge \theta^4 \wedge \theta^1 y (4m+xy) \right) +$$

$$m e_{\dots 270 \dots}^1 \dots \left(\cos[\theta] \sin[\theta] \theta^3 \wedge \theta^4 \wedge \theta^4 y (2m+xy) - \frac{\dots 1 \dots}{\dots 1 \dots} - \frac{\dots 1 \dots}{\dots 1 \dots} + \frac{\dots 1 \dots - \dots 1 \dots}{(\dots 1 \dots)^3} \right) - \frac{1}{8 \sqrt{m^2 \sin[\theta]^2 (2m+xy)^4}}$$

$$e_a^3 g_{42} (2m+xy)^2 \left(\cos[\theta] \sin[\theta] \theta^3 \wedge \theta^4 \wedge \theta^4 y (2m+xy) - \frac{\sin[\theta]^2 \theta^1 \wedge \theta^4 \wedge \theta^4 y^2 (4m+xy)}{2m+xy} - \right.$$

$$\left. \sin[\theta]^2 y \left(\theta^4 \wedge \theta^2 \wedge \theta^4 x + \theta^4 \wedge \theta^1 \wedge \theta^4 y + \cot[\theta] \theta^4 \wedge \theta^3 \wedge \theta^4 (2m+xy) \right) + \right.$$

$$\left. \frac{\sin[\theta]^2 \theta^1 \wedge \theta^4 \wedge \theta^4 x y^3 (2m+xy) (10m+3xy) - \sin[\theta]^2 \theta^2 \wedge \theta^4 \wedge \theta^4 xy (2m+xy) (8m^2+6mxy+x^2y^2)}{(2m+xy)^3} \right)$$

La expresión completa no está disponible (tamaño de memoria original: 1.6 MB)

In[]:= ToValues /@%

Out[]:=

$$S_a = \frac{g_{11} \left(-e_a^3 \cos[\theta] \sin[\theta] \theta^3 \wedge \theta^4 \wedge \theta^1 x (2m+xy)^3 - e_a^4 \cos[\theta] \sin[\theta]^3 \theta^4 \wedge \theta^4 \wedge \theta^1 x (2m+xy)^3 - 4m e_a^1 \cos[\theta] \sin[\theta] \theta^1 \wedge \theta^4 \wedge \theta^1 y (4m+xy) \right)}{8 \sqrt{m^2 \sin[\theta]^2 (2m+xy)^4}} -$$

$$\frac{1}{8 \sqrt{m^2 \sin[\theta]^2 (2m+xy)^4}} g_{11} \left(e_a^3 \cos[\theta] \sin[\theta] \theta^3 \wedge \theta^4 \wedge \theta^1 x (2m+xy)^3 + e_{\dots 49 \dots}^4 \cos[\theta] \sin[\theta]^3 \right.$$

$$\left. \theta^{\dots 49 \dots} \wedge \theta^{\dots 49 \dots} \wedge \theta^{\dots 49 \dots} x (2m+xy)^3 + 4m e_{\dots 49 \dots}^1 \cos[\theta] \sin[\theta] \theta^1 \wedge \theta^4 \wedge \theta^1 y (4m+xy) \right) +$$

$$m e_{\dots 270 \dots}^1 \dots \left(\cos[\theta] \sin[\theta] \theta^3 \wedge \theta^4 \wedge \theta^4 y (2m+xy) - \frac{\dots 1 \dots}{\dots 1 \dots} - \frac{\dots 1 \dots}{\dots 1 \dots} + \frac{\dots 1 \dots - \dots 1 \dots}{(\dots 1 \dots)^3} \right) - \frac{1}{8 \sqrt{m^2 \sin[\theta]^2 (2m+xy)^4}}$$

$$e_a^3 g_{42} (2m+xy)^2 \left(\cos[\theta] \sin[\theta] \theta^3 \wedge \theta^4 \wedge \theta^4 y (2m+xy) - \frac{\sin[\theta]^2 \theta^1 \wedge \theta^4 \wedge \theta^4 y^2 (4m+xy)}{2m+xy} - \right.$$

$$\left. \sin[\theta]^2 y \left(\theta^4 \wedge \theta^2 \wedge \theta^4 x + \theta^4 \wedge \theta^1 \wedge \theta^4 y + \cot[\theta] \theta^4 \wedge \theta^3 \wedge \theta^4 (2m+xy) \right) + \right.$$

$$\left. \frac{\sin[\theta]^2 \theta^1 \wedge \theta^4 \wedge \theta^4 x y^3 (2m+xy) (10m+3xy) - \sin[\theta]^2 \theta^2 \wedge \theta^4 \wedge \theta^4 xy (2m+xy) (8m^2+6mxy+x^2y^2)}{(2m+xy)^3} \right)$$

La expresión completa no está disponible (tamaño de memoria original: 1.6 MB)

In[]:= PowerExpand //@%;

|expande potencias

1.6255

In[*]:= ComponentArray@%

Out[*]=

$$\{S_1 = \frac{g_{11} \sin[\theta] \theta^3 \wedge \theta^4 \wedge \theta^1 x y}{4 m} - \frac{g_{21} \sin[\theta] \theta^3 \wedge \theta^4 \wedge \theta^2 x y}{4 m} - \frac{g_{31} \sin[\theta] \theta^3 \wedge \theta^4 \wedge \theta^3 x y}{4 m} - \frac{g_{\dots 49 \dots} \dots 49 \dots \dots 1 \dots \dots 1 \dots x y}{4 m} + \dots 152 \dots + \frac{\dots 1 \dots}{2 \dots 1 \dots^2} + \frac{\text{Csc}[\theta] g_{\dots 1 \dots} \left(\cos[\theta] \sin[\theta] \theta^{\dots 49 \dots} \wedge \theta^{\dots 49 \dots} \wedge \theta^{\dots 49 \dots} y (2 m + x y) - \dots 1 \dots - \dots 1 \dots + \frac{\dots 1 \dots}{\dots 1 \dots^3} \right)}{2 (2 m + x y)^2} + \frac{1}{2 (2 m + x y)^2} \text{Csc}[\theta] g_{\dots 49 \dots \dots 49 \dots} \left(\cos[\theta] \sin[\theta] \theta^3 \wedge \theta^4 \wedge \theta^3 y (2 m + x y) - \frac{\dots 1 \dots^2 \dots 1 \dots \dots 1 \dots \dots 1 \dots (\dots 1 \dots)}{2 m + x y} - \frac{\sin[\theta]^2 \theta^{\dots 49 \dots} \wedge \theta^{\dots 49 \dots} \wedge \theta^{\dots 49 \dots} x y^3 (2 m + x y) (10 m + 3 x y) - \dots 1 \dots^2 \dots 4 \dots (\dots 1 \dots)}{(2 m + x y)^3} \right) + \frac{1}{2 (2 m + x y)^2} \text{Csc}[\theta] g_{43} \left(\cos[\theta] \sin[\theta] \theta^3 \wedge \theta^4 \wedge \theta^4 y (2 m + x y) - \frac{\sin[\theta]^2 \theta^1 \wedge \theta^4 \wedge \theta^4 y^2 (4 m + x y)}{2 m + x y} - \sin[\theta]^2 y \left(\theta^4 \wedge \theta^2 \wedge \theta^4 x + \theta^4 \wedge \theta^1 \wedge \theta^4 y + \text{Cot}[\theta] \theta^4 \wedge \theta^3 \wedge \theta^4 (2 m + x y) \right) + \frac{\sin[\theta]^2 \theta^1 \wedge \theta^4 \wedge \theta^4 x y^3 (2 m + x y) (10 m + 3 x y) - \sin[\theta]^2 \theta^2 \wedge \theta^4 \wedge \theta^4 x y (2 m + x y) (8 m^2 + 6 m x y + x^2 y^2)}{(2 m + x y)^3} \right), \dots 2 \dots, S_{\dots 49 \dots} = \dots 1 \dots \}$$

La expresión completa no está disponible (tamaño de memoria original: 1.3 MB)

In[*]:= % /. metricg → metricTensor

Out[*]=

$$\{S_1 = -\sin[\theta] \theta^3 \wedge \theta^4 \wedge \theta^2 x y + \sin[\theta] \theta^4 \wedge \theta^3 \wedge \theta^2 x y + \dots 39 \dots + \frac{1}{2} \text{Csc}[\theta] \left(\cos[\theta] \sin[\theta] \theta^3 \wedge \theta^4 \wedge \theta^3 y (2 m + x y) - \frac{\sin[\theta]^2 \theta^1 \wedge \theta^4 \wedge \theta^3 y^2 (4 m + x y)}{2 m + x y} - \sin[\theta]^2 y \left(\theta^4 \wedge \theta^2 \wedge \theta^3 x + \theta^4 \wedge \theta^1 \wedge \theta^3 y + \text{Cot}[\theta] \theta^4 \wedge \theta^3 \wedge \theta^3 (2 m + x y) \right) + \frac{\sin[\theta]^2 \theta^1 \wedge \theta^4 \wedge \theta^3 x y^3 (2 m + x y) (10 m + 3 x y) - \sin[\theta]^2 \theta^2 \wedge \theta^4 \wedge \theta^3 x y (2 m + x y) (8 m^2 + 6 m x y + x^2 y^2)}{(2 m + x y)^3} \right), S_2 = -\sin[\theta] \theta^3 \wedge \theta^4 \wedge \theta^2 x^2 + \sin[\theta] \theta^4 \wedge \theta^3 \wedge \theta^2 x^2 + \dots 19 \dots + \frac{1}{2} \text{Csc}[\theta] \left(\dots 1 \dots \right) + \frac{2 \dots 1 \dots \dots 1 \dots \left(-\frac{\dots 1 \dots}{4 m} - \frac{\dots 1 \dots}{4 \dots 1 \dots} - \dots 1 \dots + \dots 1 \dots \right)}{(2 m + x y)^2} - \frac{2 m \text{Csc}[\theta] (\dots 1 \dots)}{(2 m + x y)^2}, S_3 = \dots 1 \dots, S_4 = -\cos[\theta] \sin[\theta]^2 \theta^4 \wedge \theta^1 \wedge \theta^4 y (4 m + x y) - \frac{\sin[\theta] y^2 \left(-\left(\theta^2 \wedge \theta^3 \wedge \theta^1 \right) x^2 (2 m + x y) - \theta^1 \wedge \theta^3 \wedge \theta^1 x y (2 m + x y) - \text{Cot}[\theta] \theta^3 \wedge \theta^3 \wedge \theta^1 x (2 m + x y)^2 \right)}{(2 m + x y)^2} + \dots 18 \dots + \frac{1}{2} \sin[\theta] \left(-y \left(\theta^3 \wedge \theta^2 \wedge \theta^1 x + \theta^3 \wedge \theta^1 \wedge \theta^1 y \right) - \cos[\theta] \sin[\theta] \theta^4 \wedge \theta^4 \wedge \theta^1 y (2 m + x y) - \frac{\theta^1 \wedge \theta^3 \wedge \theta^1 y^2 (4 m + x y)}{2 m + x y} + \frac{\theta^1 \wedge \theta^3 \wedge \theta^1 x y^3 (2 m + x y) (10 m + 3 x y) - \theta^2 \wedge \theta^3 \wedge \theta^1 x y (2 m + x y) (8 m^2 + 6 m x y + x^2 y^2)}{(2 m + x y)^3} \right) \}$$

La expresión completa no está disponible (tamaño de memoria original: 311.2 kB)

In[*]:= % /. Coframe → dx;

In[]:= ToValues /@%

Out[]:=

$$\left\{ \begin{aligned} S_1 &= -\sin[\theta] d[\theta] \wedge d[\phi] \wedge d[y] x y + \sin[\theta] d[\phi] \wedge d[\theta] \wedge d[y] x y + \dots 39 \dots + \\ &\quad \frac{1}{2} \operatorname{Csc}[\theta] \left(\cos[\theta] \sin[\theta] d[\theta] \wedge d[\phi] \wedge d[\theta] y (2m+xy) - \frac{\sin[\theta]^2 d[x] \wedge d[\phi] \wedge d[\theta] y^2 (4m+xy)}{2m+xy} - \right. \\ &\quad \left. \frac{\sin[\theta]^2 y (d[\phi] \wedge d[y] \wedge d[\theta] x + d[\phi] \wedge d[x] \wedge d[\theta] y + \cot[\theta] d[\phi] \wedge d[\theta] \wedge d[\theta] (2m+xy)) + \sin[\theta]^2 d[x] \wedge d[\phi] \wedge d[\theta] x y^3 (2m+xy) (10m+3xy) - \sin[\theta]^2 d[y] \wedge d[\phi] \wedge d[\theta] x y (2m+xy) (8m^2+6mxy+x^2y^2)}{(2m+xy)^3} \right), \\ S_2 &= -\sin[\theta] d[\theta] \wedge d[\phi] \wedge d[y] x^2 + \dots 1 \dots + \dots 20 \dots + \dots 1 \dots - \frac{2 \dots 2 \dots (\dots 1 \dots)}{(\dots 1 \dots)^2}, \\ &\quad \dots 1 \dots, \\ S_4 &= -\cos[\theta] \sin[\theta]^2 d[\phi] \wedge d[x] \wedge d[\phi] y (4m+xy) - \\ &\quad \frac{\sin[\theta] y^2 (-d[y] \wedge d[\theta] \wedge d[x] x^2 (2m+xy) - \dots 1 \dots - \cot[\theta] d[\theta] \wedge d[\theta] \wedge d[x] x (\dots 1 \dots)^2)}{(2m+xy)^2} + \dots 18 \dots + \\ &\quad \frac{1}{2} \sin[\theta] \left(-y (d[\theta] \wedge d[y] \wedge d[x] x + d[\theta] \wedge d[x] \wedge d[x] y) - \cos[\theta] \sin[\theta] d[\phi] \wedge d[\phi] \wedge d[x] y (2m+xy) - \right. \\ &\quad \left. \frac{d[x] \wedge d[\theta] \wedge d[x] y^2 (4m+xy)}{2m+xy} + \frac{d[x] \wedge d[\theta] \wedge d[x] x y^3 (2m+xy) (10m+3xy) - d[y] \wedge d[\theta] \wedge d[x] x y (2m+xy) (8m^2+6mxy+x^2y^2)}{(2m+xy)^3} \right) \end{aligned} \right\}$$

La expresión completa no está disponible (tamaño de memoria original: 260.5 kB)



In[]:= Map[ToCanonical, %, {2}]

[aplica a todos](#)

Out[]:=

$$\left\{ \begin{aligned} S_1 &= -3 \sin[\theta] d[y] \wedge d[\theta] \wedge d[\phi] x y + \\ &\quad \frac{\sin[\theta] d[x] \wedge d[\theta] \wedge d[\phi] y^2}{(2m+xy)^4} - \frac{64 m^4 \cos[\theta] d[x] \wedge d[y] \wedge d[\phi] y}{(2m+xy)^4} - \\ &\quad \frac{80 m^3 \cos[\theta] d[x] \wedge d[y] \wedge d[\phi] x y^2}{(2m+xy)^4} - \frac{32 m^2 \cos[\theta] d[x] \wedge d[y] \wedge d[\phi] x^2 y^3}{(2m+xy)^4} - \\ &\quad \frac{4 m \cos[\theta] d[x] \wedge d[y] \wedge d[\phi] x^3 y^4}{(2m+xy)^4} + \frac{16 m^4 \cos[\theta] \cot[\theta] d[y] \wedge d[\theta] \wedge d[\phi]}{(2m+xy)^3} - \\ &\quad \frac{16 m^3 \cos[\theta] d[x] \wedge d[y] \wedge d[\phi] y}{(2m+xy)^3} + \frac{24 m^3 \cos[\theta] \cot[\theta] d[y] \wedge d[\theta] \wedge d[\phi] x y}{(2m+xy)^3} + \\ &\quad \frac{16 m^3 \sin[\theta] d[y] \wedge d[\theta] \wedge d[\phi] x y}{(2m+xy)^3} - \frac{16 m^2 \cos[\theta] d[x] \wedge d[y] \wedge d[\phi] x y^2}{(2m+xy)^3} + \\ &\quad \frac{12 m^2 \cos[\theta] \cot[\theta] d[y] \wedge d[\theta] \wedge d[\phi] x^2 y^2}{(2m+xy)^3} + \frac{20 m^2 \sin[\theta] d[y] \wedge d[\theta] \wedge d[\phi] x^2 y^2}{(2m+xy)^3} - \\ &\quad \frac{20 m^2 \sin[\theta] d[x] \wedge d[\theta] \wedge d[\phi] x y^3}{(2m+xy)^3} - \frac{4 m \cos[\theta] d[x] \wedge d[y] \wedge d[\phi] x^2 y^3}{(2m+xy)^3} + \\ &\quad \frac{2 m \cos[\theta] \cot[\theta] d[y] \wedge d[\theta] \wedge d[\phi] x^3 y^3}{(2m+xy)^3} + \frac{8 m \sin[\theta] d[y] \wedge d[\theta] \wedge d[\phi] x^3 y^3}{(2m+xy)^3} - \\ &\quad \frac{16 m \sin[\theta] d[x] \wedge d[\theta] \wedge d[\phi] x^2 y^4}{(2m+xy)^3} + \frac{\sin[\theta] d[y] \wedge d[\theta] \wedge d[\phi] x^4 y^4}{(2m+xy)^3} - \\ &\quad \frac{3 \sin[\theta] d[x] \wedge d[\theta] \wedge d[\phi] x^3 y^5}{(2m+xy)^3} - \frac{8 m^3 \cos[\theta] \cot[\theta] d[y] \wedge d[\theta] \wedge d[\phi]}{(2m+xy)^2} + \\ &\quad \frac{24 m^2 \cos[\theta] d[x] \wedge d[y] \wedge d[\phi] y}{(2m+xy)^2} - \frac{8 m^2 \cos[\theta] \cot[\theta] d[y] \wedge d[\theta] \wedge d[\phi] x y}{(2m+xy)^2} - \end{aligned} \right\}$$

$$\begin{aligned}
& \frac{8 m^2 \sin[\theta] d[y] \wedge d[\theta] \wedge d[\phi] x y}{(2 m+x y)^2} + \frac{8 m^2 \sin[\theta] d[x] \wedge d[\theta] \wedge d[\phi] y^2}{(2 m+x y)^2} + \\
& \frac{8 m \cos[\theta] d[x] \wedge d[y] \wedge d[\phi] x y^2}{(2 m+x y)^2} - \frac{2 m \cos[\theta] \cot[\theta] d[y] \wedge d[\theta] \wedge d[\phi] x^2 y^2}{(2 m+x y)^2} - \\
& \frac{12 m \sin[\theta] d[y] \wedge d[\theta] \wedge d[\phi] x^2 y^2}{(2 m+x y)^2} + \frac{14 m \sin[\theta] d[x] \wedge d[\theta] \wedge d[\phi] x y^3}{(2 m+x y)^2} - \\
& \frac{4 \sin[\theta] d[y] \wedge d[\theta] \wedge d[\phi] x^3 y^3}{(2 m+x y)^2} + \frac{3 \sin[\theta] d[x] \wedge d[\theta] \wedge d[\phi] x^2 y^4}{(2 m+x y)^2} - \\
& \frac{2 m \sin[\theta] d[y] \wedge d[\theta] \wedge d[\phi] x y}{2 m+x y} - \frac{10 m \sin[\theta] d[x] \wedge d[\theta] \wedge d[\phi] y^2}{2 m+x y} + \\
& \frac{2 \sin[\theta] d[y] \wedge d[\theta] \wedge d[\phi] x^2 y^2}{2 m+x y} - \frac{3 \sin[\theta] d[x] \wedge d[\theta] \wedge d[\phi] x y^3}{2 m+x y}, \\
S_2 = & -\sin[\theta] d[y] \wedge d[\theta] \wedge d[\phi] x^2 + 3 \sin[\theta] d[x] \wedge d[\theta] \wedge d[\phi] x y - \\
& \frac{16 m^4 \cos[\theta] \cot[\theta] d[x] \wedge d[\theta] \wedge d[\phi]}{(2 m+x y)^3} - \frac{16 m^3 \cos[\theta] d[x] \wedge d[y] \wedge d[\phi] x}{(2 m+x y)^3} - \\
& \frac{24 m^3 \cos[\theta] \cot[\theta] d[x] \wedge d[\theta] \wedge d[\phi] x y}{(2 m+x y)^3} - \frac{16 m^2 \cos[\theta] d[x] \wedge d[y] \wedge d[\phi] x^2 y}{(2 m+x y)^3} - \\
& \frac{12 m^2 \cos[\theta] \cot[\theta] d[x] \wedge d[\theta] \wedge d[\phi] x^2 y^2}{(2 m+x y)^3} - \frac{4 m \cos[\theta] d[x] \wedge d[y] \wedge d[\phi] x^3 y^2}{(2 m+x y)^3} - \\
& \frac{2 m \cos[\theta] \cot[\theta] d[x] \wedge d[\theta] \wedge d[\phi] x^3 y^3}{(2 m+x y)^3} + \frac{8 m^3 \cos[\theta] \cot[\theta] d[x] \wedge d[\theta] \wedge d[\phi]}{(2 m+x y)^2} + \\
& \frac{8 m^2 \cos[\theta] d[x] \wedge d[y] \wedge d[\phi] x}{(2 m+x y)^2} + \frac{8 m^2 \cos[\theta] \cot[\theta] d[x] \wedge d[\theta] \wedge d[\phi] x y}{(2 m+x y)^2} - \\
& \frac{8 m^2 \sin[\theta] d[x] \wedge d[\theta] \wedge d[\phi] x y}{(2 m+x y)^2} + \frac{4 m \cos[\theta] d[x] \wedge d[y] \wedge d[\phi] x^2 y}{(2 m+x y)^2} + \\
& \frac{2 m \cos[\theta] \cot[\theta] d[x] \wedge d[\theta] \wedge d[\phi] x^2 y^2}{(2 m+x y)^2} - \frac{4 m \sin[\theta] d[x] \wedge d[\theta] \wedge d[\phi] x^2 y^2}{(2 m+x y)^2} - \\
& \frac{2 m \sin[\theta] d[y] \wedge d[\theta] \wedge d[\phi] x^2}{2 m+x y} - \frac{10 m \sin[\theta] d[x] \wedge d[\theta] \wedge d[\phi] x y}{2 m+x y} - \\
& \frac{\sin[\theta] d[y] \wedge d[\theta] \wedge d[\phi] x^3 y}{2 m+x y} - \frac{7 \sin[\theta] d[x] \wedge d[\theta] \wedge d[\phi] x^2 y^2}{2 m+x y}, \\
S_3 = & -m \cos[\theta] d[y] \wedge d[\theta] \wedge d[\phi] x - 3 m \cos[\theta] d[x] \wedge d[\theta] \wedge d[\phi] y - \\
& \sin[\theta] d[x] \wedge d[y] \wedge d[\phi] x y - \frac{1}{2} \cos[\theta] d[y] \wedge d[\theta] \wedge d[\phi] x^2 y - \\
& \frac{9}{2} \cos[\theta] d[x] \wedge d[\theta] \wedge d[\phi] x y^2 + \frac{8 m^3 \sin[\theta] d[x] \wedge d[y] \wedge d[\phi] x y}{(2 m+x y)^3} + \\
& \frac{10 m^2 \sin[\theta] d[x] \wedge d[y] \wedge d[\phi] x^2 y^2}{(2 m+x y)^3} + \frac{4 m \sin[\theta] d[x] \wedge d[y] \wedge d[\phi] x^3 y^3}{(2 m+x y)^3} + \\
& \frac{\sin[\theta] d[x] \wedge d[y] \wedge d[\phi] x^4 y^4}{2 (2 m+x y)^3} - \frac{8 m^2 \sin[\theta] d[x] \wedge d[y] \wedge d[\phi] x y}{(2 m+x y)^2} -
\end{aligned}$$

$$\begin{aligned}
& \frac{4 m^2 \cos[\theta] d[x] \wedge d[\theta] \wedge d[\phi] x y^2}{(2 m + x y)^2} - \frac{12 m \sin[\theta] d[x] \wedge d[y] \wedge d[\phi] x^2 y^2}{(2 m + x y)^2} - \\
& \frac{4 m \cos[\theta] d[x] \wedge d[\theta] \wedge d[\phi] x^2 y^3}{(2 m + x y)^2} - \frac{4 \sin[\theta] d[x] \wedge d[y] \wedge d[\phi] x^3 y^3}{(2 m + x y)^2} - \\
& \frac{\cos[\theta] d[x] \wedge d[\theta] \wedge d[\phi] x^3 y^4}{(2 m + x y)^2} - \frac{2 m^2 \cos[\theta] d[y] \wedge d[\theta] \wedge d[\phi] x}{2 m + x y} + \\
& \frac{2 m^2 \cos[\theta] d[x] \wedge d[\theta] \wedge d[\phi] y}{2 m + x y} - \frac{2 m \cos[\theta] d[y] \wedge d[\theta] \wedge d[\phi] x^2 y}{2 m + x y} + \\
& \frac{8 m \cos[\theta] d[x] \wedge d[\theta] \wedge d[\phi] x y^2}{2 m + x y} + \frac{5 \sin[\theta] d[x] \wedge d[y] \wedge d[\phi] x^2 y^2}{2 (2 m + x y)} - \\
& \frac{\cos[\theta] d[y] \wedge d[\theta] \wedge d[\phi] x^3 y^2}{2 (2 m + x y)} + \frac{7 \cos[\theta] d[x] \wedge d[\theta] \wedge d[\phi] x^2 y^3}{2 (2 m + x y)}, \\
S_4 = & \sin[\theta] d[x] \wedge d[y] \wedge d[\theta] x y - \frac{8 m^3 \sin[\theta] d[x] \wedge d[y] \wedge d[\theta] x y}{(2 m + x y)^3} - \\
& \frac{10 m^2 \sin[\theta] d[x] \wedge d[y] \wedge d[\theta] x^2 y^2}{(2 m + x y)^3} - \\
& \frac{4 m \sin[\theta] d[x] \wedge d[y] \wedge d[\theta] x^3 y^3}{(2 m + x y)^3} - \frac{\sin[\theta] d[x] \wedge d[y] \wedge d[\theta] x^4 y^4}{2 (2 m + x y)^3} + \\
& \frac{8 m^2 \sin[\theta] d[x] \wedge d[y] \wedge d[\theta] x y}{(2 m + x y)^2} + \frac{12 m \sin[\theta] d[x] \wedge d[y] \wedge d[\theta] x^2 y^2}{(2 m + x y)^2} + \\
& \frac{4 \sin[\theta] d[x] \wedge d[y] \wedge d[\theta] x^3 y^3}{(2 m + x y)^2} - \frac{5 \sin[\theta] d[x] \wedge d[y] \wedge d[\theta] x^2 y^2}{2 (2 m + x y)} \}
\end{aligned}$$

```
In[*]:= Map[Simplify, %, {2}]
|apl... |simplifica
```

```
Out[*]=
```

$$\begin{aligned}
\{ S_1 = & -2 \sin[\theta] y (2 d[y] \wedge d[\theta] \wedge d[\phi] x + d[x] \wedge d[\theta] \wedge d[\phi] y), \\
S_2 = & -2 \sin[\theta] x (d[y] \wedge d[\theta] \wedge d[\phi] x + 2 d[x] \wedge d[\theta] \wedge d[\phi] y), \\
S_3 = & -\cos[\theta] d[y] \wedge d[\theta] \wedge d[\phi] x (2 m + x y) - \\
& 2 y (\sin[\theta] d[x] \wedge d[y] \wedge d[\phi] x + \cos[\theta] d[x] \wedge d[\theta] \wedge d[\phi] (m + x y)), \\
S_4 = & 2 \sin[\theta] d[x] \wedge d[y] \wedge d[\theta] x y \}
\end{aligned}$$

```
In[*]:= % /. {{x[] -> 0}, {y[] -> 0}}
```

```
Out[*]=
```

$$\{ \{ S_1 = 0, S_2 = 0, S_3 = 0, S_4 = 0 \}, \{ S_1 = 0, S_2 = 0, S_3 = 0, S_4 = 0 \} \}$$

Schwarzschild in Eddington-Finkelstein coordinates

```
In[*]:= SparlingForm
```

```
Out[*]=
```

$$S_a = -\frac{1}{2} \eta^{bcde} \Gamma[\nabla]_{ba} \wedge \Gamma[\nabla]_{cd} \wedge \theta_e - \frac{1}{2} \eta_a^{bcd} \Gamma[\nabla]_b{}^e \wedge \Gamma[\nabla]_{ec} \wedge \theta_d$$

```

In[*]:= % /. MakeRule[{Coframe[M4][a], Coframe[M4][a]}, MetricOn → All, ContractMetrics → False]
                                         |todo                                     |false

Out[*]=

$$S_a = -\frac{1}{2} \eta^{bcde} g_{fe} \Gamma[\nabla]_{ba} \wedge \Gamma[\nabla]_{cd} \wedge \theta^f - \frac{1}{2} \eta_a^{bcd} g_{fd} \Gamma[\nabla]_b^e \wedge \Gamma[\nabla]_{ec} \wedge \theta^f$$


In[*]:= SparlingFormUp = %
Out[*]=

$$S_a = -\frac{1}{2} \eta^{bcde} g_{fe} \Gamma[\nabla]_{ba} \wedge \Gamma[\nabla]_{cd} \wedge \theta^f - \frac{1}{2} \eta_a^{bcd} g_{fd} \Gamma[\nabla]_b^e \wedge \Gamma[\nabla]_{ec} \wedge \theta^f$$


In[*]:= << xAct`xIdeal`

-----

Package xAct`xIdeal` version 0.1.0, {2025, 10, 29}

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it under certain conditions. See the General Public License for details.

-----

In[*]:= xActSolutionData[{"Schwarzschild", "EddingtonFinkelsteinCoordinates"}, "Properties"]
                                         |propiedades

Out[*]=
{CoordinateAssumptions, CoordinateNames, Metric, ParameterNames, ParameterAssumptions,
ParameterValues, ScalarFunctionNames, ScalarFunctionValues, SignatureOfMetric}

In[*]:= xActSolutionData[
{"Schwarzschild", "EddingtonFinkelsteinCoordinates"}, "CoordinateNames"]
Out[*]=
{v, r,  $\theta$ ,  $\phi$ }

In[*]:= xActSolutionData[
{"Schwarzschild", "EddingtonFinkelsteinCoordinates"}, "SignatureOfMetric"]
Out[*]=
{3, 1, 0}

In[*]:= $DefInfoQ = False;
                                         |false

In[*]:= $CVVerbose = False;
                                         |false

In[*]:= DefChart[EF, M4, {0, 1, 2, 3}, {v[], r[],  $\theta$ [],  $\phi$ []}]

In[*]:= xActSolutionData[
{"Schwarzschild", "EddingtonFinkelsteinCoordinates"}, "ParameterNames"]
Out[*]=
{m,  $\epsilon$ }

```

```

In[*]:= xActSolutionData[
  {"Schwarzschild", "EddingtonFinkelsteinCoordinates"}, "ParameterAssumptions"]

Out[*]=
Function[
  {xAct`xIdeal`Private`coords, xAct`xIdeal`Private`params, xAct`xIdeal`Private`scfuncs},
  With[{xAct`xIdeal`Private`m = xAct`xIdeal`Private`params[[1]],
    epsilon = xAct`xIdeal`Private`params[[2]]},
    xAct`xIdeal`Private`m > 0 && (epsilon === 1 || epsilon === -1)] ]

In[*]:= DefConstantSymbol[mass, PrintAs -> "m"]; DefConstantSymbol[ε]

In[*]:= metricTensor =
  xActSolutionData[{"Schwarzschild", "EddingtonFinkelsteinCoordinates" -> EF,
    "ScalarFunctionValues" -> {}, "ParameterValues" -> {mass, -1}}, "Metric"]

Out[*]=
CTensor[
  {
    {-1 +  $\frac{2m}{r}$ , -1, 0, 0}, {-1, 0, 0, 0}, {0, 0, r2, 0}, {0, 0, 0, r2 Sin[θ]2}, {-EF, -EF}, 0
  }

In[*]:= SetCMetric[metricTensor, EF, SignatureOfMetric -> {3, 1, 0}]

... SetCMetric: There are already metrics {metricg} in vbundle TM4.

In[*]:= cd = CovDOfMetric[metricTensor]

Out[*]=
CCovD[PDEF, CTensor[
  {
    {- $\frac{m}{r^2}$ , 0, 0, 0}, {0, 0, 0, 0}, {0, 0, r, 0}, {0, 0, 0, r Sin[θ]2},
    { $\frac{m(-2m+r)}{r^3}$ ,  $\frac{m}{r^2}$ , 0, 0}, { $\frac{m}{r^2}$ , 0, 0, 0}, {0, 0, 2m-r, 0}, {0, 0, 0, (2m-r) Sin[θ]2},
    {0, 0, 0, 0}, {0, 0,  $\frac{1}{r}$ , 0}, {0,  $\frac{1}{r}$ , 0, 0}, {0, 0, 0, -Cos[θ] Sin[θ]},
    {0, 0, 0, 0}, {0, 0, 0,  $\frac{1}{r}$ }, {0, 0, 0, Cot[θ]}, {0,  $\frac{1}{r}$ , Cot[θ], 0}
  }], {EF, -EF, -EF}, 0],
  CTensor[
    {
      {-1 +  $\frac{2m}{r}$ , -1, 0, 0}, {-1, 0, 0, 0}, {0, 0, r2, 0}, {0, 0, 0, r2 Sin[θ]2},
      {-EF, -EF}, 0
    }

This should work automatically:

In[*]:= $Metrics = Reverse[$Metrics];
           |invierte orden

In[*]:= TangentM4 /: MetricsOfVBundLe[TangentM4] = Reverse[MetricsOfVBundLe[TangentM4]];
           |invierte orden

In[*]:= MetricsOfVBundLe[TangentM4]

Out[*]=
{
  CTensor[
    {
      {-1 +  $\frac{2m}{r}$ , -1, 0, 0}, {-1, 0, 0, 0}, {0, 0, r2, 0}, {0, 0, 0, r2 Sin[θ]2},
      {-EF, -EF}, 0
    }, metricg
  }

In[*]:= Ricci[cd]

Out[*]=
Zero

```


In[*]:= **KerrSolutionQ**[metricTensor]

In[*]:= **SparlingFormUp**

Out[*]=

$$S_a = -\frac{1}{2} \eta^{bcde} g_{fe} \Gamma[\nabla]_{ba} \wedge \Gamma[\nabla]_{cd} \wedge \theta^f - \frac{1}{2} \eta_a^{bcd} g_{fd} \Gamma[\nabla]_b^e \wedge \Gamma[\nabla]_{ec} \wedge \theta^f$$

In[*]:= **ToBasis**[EF] /@%

Out[*]=

$$S_a = -\frac{1}{2} \eta^{bcde} g_{fe} \Gamma[\nabla]_{ba} \wedge \Gamma[\nabla]_{cd} \wedge \theta^f - \frac{1}{2} \eta_a^{bcd} g_{fd} \Gamma[\nabla]_b^e \wedge \Gamma[\nabla]_{ec} \wedge \theta^f$$

In[*]:= **TraceBasisDummy** /@%;

In[*]:= % /. **epsilonMetricg** → **epsilon**[metricTensor];

2.06151

In[*]:= % /. **CD** → **cd**

Out[*]=

$$S_a = -\frac{g_{03} \left(\frac{m e^{\frac{1}{2} \theta} r^2 \sin^2 \theta}{r} - \frac{m e^{\frac{1}{2} \theta} r^2 \sin^2 \theta}{r} \right)}{2 \sqrt{r^4 \sin^2 \theta}} + \frac{g_{03} \left(\frac{m e^{\frac{1}{2} \theta} r^2 \sin^2 \theta}{r} + \frac{m e^{\frac{1}{2} \theta} r^2 \sin^2 \theta}{r} \right)}{2 \sqrt{r^4 \sin^2 \theta}} + \frac{m e^{\frac{1}{2} \theta} g_{03} \left(-\left((2m-r) r \theta^{\frac{1}{2}} \sin^2 \theta \right) - r^2 \theta^{\frac{1}{2}} \sin^2 \theta \right)}{2 r^3 \sqrt{r^4 \sin^2 \theta}} +$$

$$\dots 223 \dots + \frac{g_{30} \left(-e^{\frac{1}{2} \theta} r^2 \sin^2 \theta \theta^{\frac{1}{2}} \sin^2 \theta - e^{\frac{1}{2} \theta} r^2 \sin^2 \theta \theta^{\frac{1}{2}} \sin^2 \theta + e^{\frac{1}{2} \theta} \cos[\theta] r^3 \sin^2 \theta \theta^{\frac{1}{2}} \sin^2 \theta \right)}{2 \sqrt{r^4 \sin^2 \theta}}$$

La expresión completa no está disponible (tamaño de memoria original: 0.6 MB)

In[*]:= **ToValues** /@%

Out[*]=

$$S_a = -\frac{g_{03} \left(\frac{m e^{\frac{1}{2} \theta} r^2 \sin^2 \theta}{r} - \frac{m e^{\frac{1}{2} \theta} r^2 \sin^2 \theta}{r} \right)}{2 \sqrt{r^4 \sin^2 \theta}} + \frac{g_{03} \left(\frac{m e^{\frac{1}{2} \theta} r^2 \sin^2 \theta}{r} + \frac{m e^{\frac{1}{2} \theta} r^2 \sin^2 \theta}{r} \right)}{2 \sqrt{r^4 \sin^2 \theta}} + \frac{m e^{\frac{1}{2} \theta} g_{03} \left(-\left((2m-r) r \theta^{\frac{1}{2}} \sin^2 \theta \right) - r^2 \theta^{\frac{1}{2}} \sin^2 \theta \right)}{2 r^3 \sqrt{r^4 \sin^2 \theta}} +$$

$$\dots 223 \dots + \frac{g_{30} \left(-e^{\frac{1}{2} \theta} r^2 \sin^2 \theta \theta^{\frac{1}{2}} \sin^2 \theta - e^{\frac{1}{2} \theta} r^2 \sin^2 \theta \theta^{\frac{1}{2}} \sin^2 \theta + e^{\frac{1}{2} \theta} \cos[\theta] r^3 \sin^2 \theta \theta^{\frac{1}{2}} \sin^2 \theta \right)}{2 \sqrt{r^4 \sin^2 \theta}}$$

La expresión completa no está disponible (tamaño de memoria original: 0.6 MB)

In[*]:= **PowerExpand** /@%;

[expande potencias](#)

1.49661

In[*]:= ComponentArray@%

Out[*]=

$$\left\{ \begin{aligned} S_0 &= \frac{m \cos[\theta] g_{00} \theta^0 \wedge \theta^3 \wedge \theta^0}{r^2} + \frac{m \cos[\theta] g_{10} \theta^0 \wedge \theta^3 \wedge \theta^1}{r^2} + \frac{m \cos[\theta] g_{20} \theta^0 \wedge \theta^3 \wedge \theta^2}{r^2} + \\ &\frac{m \cos[\theta] g_{30} \theta^0 \wedge \theta^3 \wedge \theta^3}{r^2} + \frac{m \csc[\theta] g_{00} \dots 49 \dots \theta^1 \wedge \theta^2 \wedge \theta^0}{r^3} + \frac{m \csc[\theta] g_{\dots 49 \dots \dots 49 \dots} \left(\dots 1 \dots \right)}{2 r^5} + \dots 98 \dots, \\ S_1 &= \frac{m \csc[\theta] g_{\dots 49 \dots \dots 49 \dots} \theta^{\dots 49 \dots} \wedge \theta^{\dots 49 \dots} \wedge \theta^{\dots 49 \dots}}{r^3} + \frac{\dots 1 \dots}{r^3} + \frac{\dots 71 \dots}{2} + \frac{\dots 1 \dots}{2} - \frac{\dots 1 \dots}{2 r^2}, \\ S_{\dots 1 \dots} &= \dots 1 \dots, S_3 = \frac{m g_{01} \sin[\theta] \left(- \left((2m-r) r \theta^0 \wedge \theta^2 \wedge \theta^0 \right) - r^2 \theta^1 \wedge \theta^2 \wedge \theta^0 \right)}{2 r^3} + \frac{m g_{11} \sin[\theta] \left(- \left((2m-r) r \theta^0 \wedge \theta^2 \wedge \theta^1 \right) - r^2 \theta^1 \wedge \theta^2 \wedge \theta^1 \right)}{2 r^3} + \\ &\dots 47 \dots + \frac{1}{2} g_{30} \sin[\theta] \left(\theta^1 \wedge \theta^2 \wedge \theta^3 + \frac{m \theta^2 \wedge \theta^0 \wedge \theta^3}{r} - \cos[\theta] r \sin[\theta] \theta^3 \wedge \theta^3 \wedge \theta^3 \right) - \\ &\frac{1}{2} g_{30} \sin[\theta] \left(\frac{m \theta^0 \wedge \theta^2 \wedge \theta^3}{r} - \theta^2 \wedge \theta^1 \wedge \theta^3 - \cos[\theta] r \sin[\theta] \theta^3 \wedge \theta^3 \wedge \theta^3 \right) \end{aligned} \right\}$$

La expresión completa no está disponible (tamaño de memoria original: 481.4 kB)

In[*]:= % /. metricg -> metricTensor

Out[*]=

$$\left\{ \begin{aligned} S_0 &= \frac{m \cos[\theta] \left(-1 + \frac{2m}{r} \right) \theta^0 \wedge \theta^3 \wedge \theta^0}{r^2} - \frac{m \cos[\theta] \theta^0 \wedge \theta^3 \wedge \theta^1}{r^2} + \\ &\frac{m \sin[\theta] \theta^1 \wedge \theta^2 \wedge \theta^3}{r} + \frac{m \sin[\theta] \left(- \left((2m-r) r \theta^0 \wedge \theta^2 \wedge \theta^3 \right) - r^2 \theta^1 \wedge \theta^2 \wedge \theta^3 \right)}{2 r^3} - \\ &\frac{m \cos[\theta] \theta^1 \wedge \theta^3 \wedge \theta^0}{r^2} - \frac{m \sin[\theta] \theta^1 \wedge \theta^3 \wedge \theta^2}{r} - \\ &\frac{m \csc[\theta] \left(- \left((2m-r) r \sin[\theta]^2 \theta^0 \wedge \theta^3 \wedge \theta^2 \right) - r^2 \sin[\theta]^2 \theta^1 \wedge \theta^3 \wedge \theta^2 \right)}{2 r^3} - \\ &\frac{1}{2} \sin[\theta] \left(\frac{m \left(-2m+r \right) \theta^2 \wedge \theta^0 \wedge \theta^3}{r^2} + \frac{m \theta^2 \wedge \theta^1 \wedge \theta^3}{r} \right) + \\ &\frac{1}{2} \csc[\theta] \left(- \frac{m \left(2m-r \right) \sin[\theta]^2 \theta^3 \wedge \theta^0 \wedge \theta^2}{r^2} + \frac{m \sin[\theta]^2 \theta^3 \wedge \theta^1 \wedge \theta^2}{r} \right) + \frac{1}{2 r^3} \\ &\csc[\theta] \left(2m-r \right) \left(\cos[\theta] r \sin[\theta] \theta^1 \wedge \theta^3 \wedge \theta^0 + \cos[\theta]^2 r^2 \theta^2 \wedge \theta^3 \wedge \theta^0 + \right. \\ &\quad \left. \cos[\theta] r \sin[\theta] \theta^3 \wedge \theta^1 \wedge \theta^0 + \left(2m-r \right) r \sin[\theta]^2 \theta^3 \wedge \theta^2 \wedge \theta^0 \right) - \frac{1}{2 r^2} \\ &\csc[\theta] \left(-1 + \frac{2m}{r} \right) \left(\cos[\theta] r \sin[\theta] \theta^1 \wedge \theta^3 \wedge \theta^0 + \cos[\theta]^2 r^2 \theta^2 \wedge \theta^3 \wedge \theta^0 + \right. \\ &\quad \left. \cos[\theta] r \sin[\theta] \theta^3 \wedge \theta^1 \wedge \theta^0 + \left(2m-r \right) r \sin[\theta]^2 \theta^3 \wedge \theta^2 \wedge \theta^0 \right) + \frac{1}{2 r^2} \\ &\csc[\theta] \left(-1 + \frac{2m}{r} \right) \left(-\cos[\theta] r \sin[\theta] \theta^1 \wedge \theta^3 \wedge \theta^0 - r \left(-2m+r \right) \sin[\theta]^2 \theta^2 \wedge \theta^3 \wedge \theta^0 - \right. \\ &\quad \left. \cos[\theta] r \sin[\theta] \left(\theta^3 \wedge \theta^1 \wedge \theta^0 + \cot[\theta] r \theta^3 \wedge \theta^2 \wedge \theta^0 \right) \right) + \frac{1}{2 r^3} \\ &\csc[\theta] \left(-2m+r \right) \left(-\cos[\theta] r \sin[\theta] \theta^1 \wedge \theta^3 \wedge \theta^0 - r \left(-2m+r \right) \sin[\theta]^2 \theta^2 \wedge \theta^3 \wedge \theta^0 - \right. \\ &\quad \left. \cos[\theta] r \sin[\theta] \left(\theta^3 \wedge \theta^1 \wedge \theta^0 + \cot[\theta] r \theta^3 \wedge \theta^2 \wedge \theta^0 \right) \right) + \end{aligned} \right\}$$

$$\begin{aligned}
& \frac{1}{2r^2} \text{Csc}[\theta] \left(\text{Cos}[\theta] r \text{Sin}[\theta] \theta^1 \wedge \theta^3 \wedge \theta^1 + \text{Cos}[\theta]^2 r^2 \theta^2 \wedge \theta^3 \wedge \theta^1 + \right. \\
& \quad \left. \text{Cos}[\theta] r \text{Sin}[\theta] \theta^3 \wedge \theta^1 \wedge \theta^1 + (2m-r) r \text{Sin}[\theta]^2 \theta^3 \wedge \theta^2 \wedge \theta^1 \right) - \frac{1}{2r^2} \\
& \text{Csc}[\theta] \left(-\text{Cos}[\theta] r \text{Sin}[\theta] \theta^1 \wedge \theta^3 \wedge \theta^1 - r(-2m+r) \text{Sin}[\theta]^2 \theta^2 \wedge \theta^3 \wedge \theta^1 - \right. \\
& \quad \left. \text{Cos}[\theta] r \text{Sin}[\theta] \left(\theta^3 \wedge \theta^1 \wedge \theta^1 + \text{Cot}[\theta] r \theta^3 \wedge \theta^2 \wedge \theta^1 \right) \right) - \frac{1}{2r} \\
& \text{Csc}[\theta] (-2m+r) \left(\text{Sin}[\theta]^2 \theta^1 \wedge \theta^3 \wedge \theta^2 + \text{Cos}[\theta] r \text{Sin}[\theta] \theta^2 \wedge \theta^3 \wedge \theta^2 + \right. \\
& \quad \left. \frac{m \text{Sin}[\theta]^2 \theta^3 \wedge \theta^0 \wedge \theta^2}{r} + \text{Cos}[\theta] r \text{Sin}[\theta] \theta^3 \wedge \theta^2 \wedge \theta^2 \right) - \frac{1}{2r} \\
& \text{Csc}[\theta] (2m-r) \left(\frac{m \text{Sin}[\theta]^2 \theta^0 \wedge \theta^3 \wedge \theta^2}{r} + \text{Cos}[\theta] r \text{Sin}[\theta] \theta^2 \wedge \theta^3 \wedge \theta^2 - \right. \\
& \quad \left. \text{Sin}[\theta]^2 \left(\theta^3 \wedge \theta^1 \wedge \theta^2 + \text{Cot}[\theta] r \theta^3 \wedge \theta^2 \wedge \theta^2 \right) \right) - \\
& \frac{(2m-r) \text{Sin}[\theta] \left(\theta^1 \wedge \theta^2 \wedge \theta^3 + \frac{m \theta^2 \wedge \theta^0 \wedge \theta^3}{r} - \text{Cos}[\theta] r \text{Sin}[\theta] \theta^3 \wedge \theta^3 \wedge \theta^3 \right)}{2r} - \\
& \frac{(-2m+r) \text{Sin}[\theta] \left(\frac{m \theta^0 \wedge \theta^2 \wedge \theta^3}{r} - \theta^2 \wedge \theta^1 \wedge \theta^3 - \text{Cos}[\theta] r \text{Sin}[\theta] \theta^3 \wedge \theta^3 \wedge \theta^3 \right)}{2r}, \\
S_1 = & \frac{m \text{Sin}[\theta] \theta^0 \wedge \theta^2 \wedge \theta^3}{r} - \frac{m \text{Cos}[\theta] \theta^0 \wedge \theta^3 \wedge \theta^0}{r^2} - \frac{m \text{Sin}[\theta] \theta^0 \wedge \theta^3 \wedge \theta^2}{r} - \\
& \frac{m \text{Sin}[\theta] \theta^2 \wedge \theta^0 \wedge \theta^3}{r} - \left(-1 + \frac{2m}{r} \right) \text{Sin}[\theta] \theta^2 \wedge \theta^3 \wedge \theta^0 + \text{Sin}[\theta] \theta^2 \wedge \theta^3 \wedge \theta^1 + \\
& \frac{m \text{Sin}[\theta] \theta^3 \wedge \theta^0 \wedge \theta^2}{r} + \left(-1 + \frac{2m}{r} \right) \text{Sin}[\theta] \theta^3 \wedge \theta^2 \wedge \theta^0 - \frac{1}{2r^2} \\
& \text{Csc}[\theta] \left(\text{Cos}[\theta] r \text{Sin}[\theta] \theta^1 \wedge \theta^3 \wedge \theta^0 + \text{Cos}[\theta]^2 r^2 \theta^2 \wedge \theta^3 \wedge \theta^0 + \right. \\
& \quad \left. \text{Cos}[\theta] r \text{Sin}[\theta] \theta^3 \wedge \theta^1 \wedge \theta^0 + (2m-r) r \text{Sin}[\theta]^2 \theta^3 \wedge \theta^2 \wedge \theta^0 \right) + \frac{1}{2r^2} \\
& \text{Csc}[\theta] \left(-\text{Cos}[\theta] r \text{Sin}[\theta] \theta^1 \wedge \theta^3 \wedge \theta^0 - r(-2m+r) \text{Sin}[\theta]^2 \theta^2 \wedge \theta^3 \wedge \theta^0 - \right. \\
& \quad \left. \text{Cos}[\theta] r \text{Sin}[\theta] \left(\theta^3 \wedge \theta^1 \wedge \theta^0 + \text{Cot}[\theta] r \theta^3 \wedge \theta^2 \wedge \theta^0 \right) \right) - \\
& \text{Sin}[\theta] \theta^3 \wedge \theta^2 \wedge \theta^1 - \frac{1}{2} \text{Csc}[\theta] \left(\text{Sin}[\theta]^2 \theta^1 \wedge \theta^3 \wedge \theta^2 + \text{Cos}[\theta] r \text{Sin}[\theta] \theta^2 \wedge \theta^3 \wedge \theta^2 + \right. \\
& \quad \left. \frac{m \text{Sin}[\theta]^2 \theta^3 \wedge \theta^0 \wedge \theta^2}{r} + \text{Cos}[\theta] r \text{Sin}[\theta] \theta^3 \wedge \theta^2 \wedge \theta^2 \right) + \\
& \frac{1}{2} \text{Csc}[\theta] \left(\frac{m \text{Sin}[\theta]^2 \theta^0 \wedge \theta^3 \wedge \theta^2}{r} + \text{Cos}[\theta] r \text{Sin}[\theta] \theta^2 \wedge \theta^3 \wedge \theta^2 - \right. \\
& \quad \left. \text{Sin}[\theta]^2 \left(\theta^3 \wedge \theta^1 \wedge \theta^2 + \text{Cot}[\theta] r \theta^3 \wedge \theta^2 \wedge \theta^2 \right) \right) +
\end{aligned}$$

$$\begin{aligned}
& \frac{1}{2} \sin[\theta] \left(\theta^1 \wedge \theta^2 \wedge \theta^3 + \frac{m \theta^2 \wedge \theta^0 \wedge \theta^3}{r} - \cos[\theta] r \sin[\theta] \theta^3 \wedge \theta^3 \wedge \theta^3 \right) - \\
& \frac{1}{2} \sin[\theta] \left(\frac{m \theta^0 \wedge \theta^2 \wedge \theta^3}{r} - \theta^2 \wedge \theta^1 \wedge \theta^3 - \cos[\theta] r \sin[\theta] \theta^3 \wedge \theta^3 \wedge \theta^3 \right), \\
S_2 = & -\frac{m \sin[\theta] \theta^1 \wedge \theta^0 \wedge \theta^3}{r} - \left(-1 + \frac{2m}{r} \right) \sin[\theta] \theta^1 \wedge \theta^3 \wedge \theta^0 + \\
& \frac{m \csc[\theta] \left(-\left((2m-r) r \sin[\theta]^2 \theta^0 \wedge \theta^3 \wedge \theta^0 \right) - r^2 \sin[\theta]^2 \theta^1 \wedge \theta^3 \wedge \theta^0 \right)}{2r^3} + \\
& \sin[\theta] \theta^1 \wedge \theta^3 \wedge \theta^1 - \cos[\theta] \left(-1 + \frac{2m}{r} \right) r \theta^2 \wedge \theta^3 \wedge \theta^0 + \\
& \cos[\theta] r \theta^2 \wedge \theta^3 \wedge \theta^1 + m \cos[\theta] \theta^3 \wedge \theta^0 \wedge \theta^2 - \\
& \frac{1}{2} \csc[\theta] \left(-\frac{m(2m-r) \sin[\theta]^2 \theta^3 \wedge \theta^0 \wedge \theta^0}{r^2} + \frac{m \sin[\theta]^2 \theta^3 \wedge \theta^1 \wedge \theta^0}{r} \right) + \\
& \cos[\theta] \left(-1 + \frac{2m}{r} \right) r \theta^3 \wedge \theta^2 \wedge \theta^0 - \frac{1}{2} \csc[\theta] \left(-1 + \frac{2m}{r} \right) \left(\sin[\theta]^2 \theta^1 \wedge \theta^3 \wedge \theta^0 + \right. \\
& \left. \cos[\theta] r \sin[\theta] \theta^2 \wedge \theta^3 \wedge \theta^0 + \frac{m \sin[\theta]^2 \theta^3 \wedge \theta^0 \wedge \theta^0}{r} + \cos[\theta] r \sin[\theta] \theta^3 \wedge \theta^2 \wedge \theta^0 \right) + \\
& \frac{1}{2} \csc[\theta] \left(-1 + \frac{2m}{r} \right) \left(\frac{m \sin[\theta]^2 \theta^0 \wedge \theta^3 \wedge \theta^0}{r} + \cos[\theta] r \sin[\theta] \theta^2 \wedge \theta^3 \wedge \theta^0 - \right. \\
& \left. \sin[\theta]^2 \left(\theta^3 \wedge \theta^1 \wedge \theta^0 + \cot[\theta] r \theta^3 \wedge \theta^2 \wedge \theta^0 \right) \right) - \cos[\theta] r \theta^3 \wedge \theta^2 \wedge \theta^1 + \\
& \frac{1}{2} \csc[\theta] \left(\sin[\theta]^2 \theta^1 \wedge \theta^3 \wedge \theta^1 + \cos[\theta] r \sin[\theta] \theta^2 \wedge \theta^3 \wedge \theta^1 + \frac{m \sin[\theta]^2 \theta^3 \wedge \theta^0 \wedge \theta^1}{r} + \right. \\
& \left. \cos[\theta] r \sin[\theta] \theta^3 \wedge \theta^2 \wedge \theta^1 \right) - \frac{1}{2} \csc[\theta] \left(\frac{m \sin[\theta]^2 \theta^0 \wedge \theta^3 \wedge \theta^1}{r} + \right. \\
& \left. \cos[\theta] r \sin[\theta] \theta^2 \wedge \theta^3 \wedge \theta^1 - \sin[\theta]^2 \left(\theta^3 \wedge \theta^1 \wedge \theta^1 + \cot[\theta] r \theta^3 \wedge \theta^2 \wedge \theta^1 \right) \right), \\
S_3 = & -\frac{m \sin[\theta] \left(-\left((2m-r) r \theta^0 \wedge \theta^2 \wedge \theta^0 \right) - r^2 \theta^1 \wedge \theta^2 \wedge \theta^0 \right)}{2r^3} - \\
& \frac{1}{2} r \sin[\theta] \left(-\frac{m \theta^1 \wedge \theta^0 \wedge \theta^2}{r^2} - \frac{m \cot[\theta] \theta^2 \wedge \theta^0 \wedge \theta^2}{r} \right) + \\
& \frac{1}{2} r \sin[\theta] \left(\frac{m \theta^1 \wedge \theta^0 \wedge \theta^2}{r^2} + \frac{m \cot[\theta] \theta^2 \wedge \theta^0 \wedge \theta^2}{r} \right) + \\
& \frac{1}{2} \sin[\theta] \left(\frac{m(-2m+r) \theta^2 \wedge \theta^0 \wedge \theta^0}{r^2} + \frac{m \theta^2 \wedge \theta^1 \wedge \theta^0}{r} \right) - \\
& \frac{\left(-1 + \frac{2m}{r} \right) \sin[\theta] \left(-r \theta^1 \wedge \theta^2 \wedge \theta^0 - \cot[\theta] r^2 \theta^2 \wedge \theta^2 \wedge \theta^0 \right)}{2r} +
\end{aligned}$$

$$\begin{aligned}
& \frac{\left(-1 + \frac{2m}{r}\right) \sin[\theta] \left(r \theta^1 \wedge \theta^2 \wedge \theta^0 + \cot[\theta] r^2 \theta^2 \wedge \theta^2 \wedge \theta^0\right)}{2r} + \\
& \frac{\sin[\theta] \left(-r \theta^1 \wedge \theta^2 \wedge \theta^1 - \cot[\theta] r^2 \theta^2 \wedge \theta^2 \wedge \theta^1\right)}{2r} - \\
& \frac{\sin[\theta] \left(r \theta^1 \wedge \theta^2 \wedge \theta^1 + \cot[\theta] r^2 \theta^2 \wedge \theta^2 \wedge \theta^1\right)}{2r} + m \cos[\theta] \sin[\theta]^2 \theta^3 \wedge \theta^0 \wedge \theta^3 + \\
& \frac{1}{2} \left(-1 + \frac{2m}{r}\right) \sin[\theta] \left(\theta^1 \wedge \theta^2 \wedge \theta^0 + \frac{m \theta^2 \wedge \theta^0 \wedge \theta^0}{r} - \cos[\theta] r \sin[\theta] \theta^3 \wedge \theta^3 \wedge \theta^0\right) - \\
& \frac{1}{2} \left(-1 + \frac{2m}{r}\right) \sin[\theta] \left(\frac{m \theta^0 \wedge \theta^2 \wedge \theta^0}{r} - \theta^2 \wedge \theta^1 \wedge \theta^0 - \cos[\theta] r \sin[\theta] \theta^3 \wedge \theta^3 \wedge \theta^0\right) - \\
& \frac{1}{2} \sin[\theta] \left(\theta^1 \wedge \theta^2 \wedge \theta^1 + \frac{m \theta^2 \wedge \theta^0 \wedge \theta^1}{r} - \cos[\theta] r \sin[\theta] \theta^3 \wedge \theta^3 \wedge \theta^1\right) + \\
& \frac{1}{2} \sin[\theta] \left(\frac{m \theta^0 \wedge \theta^2 \wedge \theta^1}{r} - \theta^2 \wedge \theta^1 \wedge \theta^1 - \cos[\theta] r \sin[\theta] \theta^3 \wedge \theta^3 \wedge \theta^1\right) \Big\}
\end{aligned}$$

In[*]:= % /. Coframe → dx;

In[*]:= ToValues /@%

Out[*]=

$$\begin{aligned}
\left\{ S_\theta = \frac{m \sin[\theta] d[r] \wedge d[\theta] \wedge d[\phi]}{r} - \frac{m \cos[\theta] d[r] \wedge d[\phi] \wedge d[v]}{r^2} - \frac{m \sin[\theta] d[r] \wedge d[\phi] \wedge d[\theta]}{r} + \right. \\
& \frac{m \sin[\theta] \left(-r^2 d[r] \wedge d[\theta] \wedge d[\phi] - (2m - r) r d[v] \wedge d[\theta] \wedge d[\phi]\right)}{2r^3} - \\
& \frac{m \cos[\theta] d[v] \wedge d[\phi] \wedge d[r]}{r^2} + \frac{m \cos[\theta] \left(-1 + \frac{2m}{r}\right) d[v] \wedge d[\phi] \wedge d[v]}{r^2} - \\
& \frac{m \csc[\theta] \left(-r^2 \sin[\theta]^2 d[r] \wedge d[\phi] \wedge d[\theta] - (2m - r) r \sin[\theta]^2 d[v] \wedge d[\phi] \wedge d[\theta]\right)}{2r^3} - \\
& \frac{1}{2} \sin[\theta] \left(\frac{m d[\theta] \wedge d[r] \wedge d[\phi]}{r} + \frac{m (-2m + r) d[\theta] \wedge d[v] \wedge d[\phi]}{r^2}\right) + \\
& \frac{1}{2} \csc[\theta] \left(\frac{m \sin[\theta]^2 d[\phi] \wedge d[r] \wedge d[\theta]}{r} - \frac{m (2m - r) \sin[\theta]^2 d[\phi] \wedge d[v] \wedge d[\theta]}{r^2}\right) + \\
& \frac{1}{2r^2} \csc[\theta] \left(\cos[\theta] r \sin[\theta] d[r] \wedge d[\phi] \wedge d[r] + \cos[\theta]^2 r^2 d[\theta] \wedge d[\phi] \wedge d[r] + \right. \\
& \left. \cos[\theta] r \sin[\theta] d[\phi] \wedge d[r] \wedge d[r] + (2m - r) r \sin[\theta]^2 d[\phi] \wedge d[\theta] \wedge d[r]\right) - \frac{1}{2r^2} \\
& \csc[\theta] \left(-\cos[\theta] r \sin[\theta] d[r] \wedge d[\phi] \wedge d[r] - r (-2m + r) \sin[\theta]^2 d[\theta] \wedge d[\phi] \wedge d[r] - \right. \\
& \left. \cos[\theta] r \sin[\theta] (d[\phi] \wedge d[r] \wedge d[r] + \cot[\theta] r d[\phi] \wedge d[\theta] \wedge d[r])\right) + \frac{1}{2r^3} \\
& \csc[\theta] (2m - r) \left(\cos[\theta] r \sin[\theta] d[r] \wedge d[\phi] \wedge d[v] + \cos[\theta]^2 r^2 d[\theta] \wedge d[\phi] \wedge d[v] + \right. \\
& \left. \cos[\theta] r \sin[\theta] d[\phi] \wedge d[r] \wedge d[v] + (2m - r) r \sin[\theta]^2 d[\phi] \wedge d[\theta] \wedge d[v]\right) - \frac{1}{2r^2} \\
& \csc[\theta] \left(-1 + \frac{2m}{r}\right) \left(\cos[\theta] r \sin[\theta] d[r] \wedge d[\phi] \wedge d[v] + \cos[\theta]^2 r^2 d[\theta] \wedge d[\phi] \wedge d[v] + \right.
\end{aligned}$$

$$\begin{aligned}
& \cos[\theta] r \sin[\theta] d[\phi] \wedge d[r] \wedge d[v] + (2m - r) r \sin[\theta]^2 d[\phi] \wedge d[\theta] \wedge d[v] \Big) + \frac{1}{2r^2} \csc[\theta] \\
& \left(-1 + \frac{2m}{r} \right) \left(-\cos[\theta] r \sin[\theta] d[r] \wedge d[\phi] \wedge d[v] - r(-2m + r) \sin[\theta]^2 d[\theta] \wedge d[\phi] \wedge d[v] - \right. \\
& \left. \cos[\theta] r \sin[\theta] (d[\phi] \wedge d[r] \wedge d[v] + \cot[\theta] r d[\phi] \wedge d[\theta] \wedge d[v]) \right) + \frac{1}{2r^3} \csc[\theta] \\
& (-2m + r) \left(-\cos[\theta] r \sin[\theta] d[r] \wedge d[\phi] \wedge d[v] - r(-2m + r) \sin[\theta]^2 d[\theta] \wedge d[\phi] \wedge d[v] - \right. \\
& \left. \cos[\theta] r \sin[\theta] (d[\phi] \wedge d[r] \wedge d[v] + \cot[\theta] r d[\phi] \wedge d[\theta] \wedge d[v]) \right) - \frac{1}{2r} \\
& \csc[\theta] (-2m + r) \left(\sin[\theta]^2 d[r] \wedge d[\phi] \wedge d[\theta] + \cos[\theta] r \sin[\theta] d[\theta] \wedge d[\phi] \wedge d[\theta] + \right. \\
& \left. \frac{m \sin[\theta]^2 d[\phi] \wedge d[v] \wedge d[\theta]}{r} + \cos[\theta] r \sin[\theta] d[\phi] \wedge d[\theta] \wedge d[\theta] \right) - \frac{1}{2r} \\
& \csc[\theta] (2m - r) \left(\frac{m \sin[\theta]^2 d[v] \wedge d[\phi] \wedge d[\theta]}{r} + \cos[\theta] r \sin[\theta] d[\theta] \wedge d[\phi] \wedge d[\theta] - \right. \\
& \left. \sin[\theta]^2 (d[\phi] \wedge d[r] \wedge d[\theta] + \cot[\theta] r d[\phi] \wedge d[\theta] \wedge d[\theta]) \right) - \\
& \frac{(-2m + r) \sin[\theta] \left(\frac{m d[v] \wedge d[\theta] \wedge d[\phi]}{r} - d[\theta] \wedge d[r] \wedge d[\phi] - \cos[\theta] r \sin[\theta] d[\phi] \wedge d[\phi] \wedge d[\phi] \right)}{2r} - \\
& \frac{(2m - r) \sin[\theta] \left(d[r] \wedge d[\theta] \wedge d[\phi] + \frac{m d[\theta] \wedge d[v] \wedge d[\phi]}{r} - \cos[\theta] r \sin[\theta] d[\phi] \wedge d[\phi] \wedge d[\phi] \right)}{2r}, \\
S_1 = & \frac{m \sin[\theta] d[v] \wedge d[\theta] \wedge d[\phi]}{r} - \frac{m \cos[\theta] d[v] \wedge d[\phi] \wedge d[v]}{r^2} - \\
& \frac{m \sin[\theta] d[v] \wedge d[\phi] \wedge d[\theta]}{r} - \frac{m \sin[\theta] d[\theta] \wedge d[v] \wedge d[\phi]}{r} + \sin[\theta] d[\theta] \wedge d[\phi] \wedge d[r] - \\
& \left(-1 + \frac{2m}{r} \right) \sin[\theta] d[\theta] \wedge d[\phi] \wedge d[v] + \frac{m \sin[\theta] d[\phi] \wedge d[v] \wedge d[\theta]}{r} - \\
& \sin[\theta] d[\phi] \wedge d[\theta] \wedge d[r] + \left(-1 + \frac{2m}{r} \right) \sin[\theta] d[\phi] \wedge d[\theta] \wedge d[v] - \frac{1}{2r^2} \\
& \csc[\theta] \left(\cos[\theta] r \sin[\theta] d[r] \wedge d[\phi] \wedge d[v] + \cos[\theta]^2 r^2 d[\theta] \wedge d[\phi] \wedge d[v] + \right. \\
& \left. \cos[\theta] r \sin[\theta] d[\phi] \wedge d[r] \wedge d[v] + (2m - r) r \sin[\theta]^2 d[\phi] \wedge d[\theta] \wedge d[v] \right) + \frac{1}{2r^2} \\
& \csc[\theta] \left(-\cos[\theta] r \sin[\theta] d[r] \wedge d[\phi] \wedge d[v] - r(-2m + r) \sin[\theta]^2 d[\theta] \wedge d[\phi] \wedge d[v] - \right. \\
& \left. \cos[\theta] r \sin[\theta] (d[\phi] \wedge d[r] \wedge d[v] + \cot[\theta] r d[\phi] \wedge d[\theta] \wedge d[v]) \right) - \\
& \frac{1}{2} \csc[\theta] \left(\sin[\theta]^2 d[r] \wedge d[\phi] \wedge d[\theta] + \cos[\theta] r \sin[\theta] d[\theta] \wedge d[\phi] \wedge d[\theta] + \right. \\
& \left. \frac{m \sin[\theta]^2 d[\phi] \wedge d[v] \wedge d[\theta]}{r} + \cos[\theta] r \sin[\theta] d[\phi] \wedge d[\theta] \wedge d[\theta] \right) + \\
& \frac{1}{2} \csc[\theta] \left(\frac{m \sin[\theta]^2 d[v] \wedge d[\phi] \wedge d[\theta]}{r} + \cos[\theta] r \sin[\theta] d[\theta] \wedge d[\phi] \wedge d[\theta] - \right. \\
& \left. \sin[\theta]^2 (d[\phi] \wedge d[r] \wedge d[\theta] + \cot[\theta] r d[\phi] \wedge d[\theta] \wedge d[\theta]) \right) - \\
& \frac{1}{2} \sin[\theta] \left(\frac{m d[v] \wedge d[\theta] \wedge d[\phi]}{r} - d[\theta] \wedge d[r] \wedge d[\phi] - \cos[\theta] r \sin[\theta] d[\phi] \wedge d[\phi] \wedge d[\phi] \right) +
\end{aligned}$$

$$\begin{aligned}
& \frac{1}{2} \sin[\theta] \left(d[r] \wedge d[\theta] \wedge d[\phi] + \frac{m d[\theta] \wedge d[v] \wedge d[\phi]}{r} - \cos[\theta] r \sin[\theta] d[\phi] \wedge d[\phi] \wedge d[\phi] \right), \\
S_2 = & - \frac{m \sin[\theta] d[r] \wedge d[v] \wedge d[\phi]}{r} + \sin[\theta] d[r] \wedge d[\phi] \wedge d[r] - \\
& \left(-1 + \frac{2m}{r} \right) \sin[\theta] d[r] \wedge d[\phi] \wedge d[v] + \\
& \frac{m \csc[\theta] \left(-r^2 \sin[\theta]^2 d[r] \wedge d[\phi] \wedge d[v] - (2m - r) r \sin[\theta]^2 d[v] \wedge d[\phi] \wedge d[v] \right)}{2 r^3} + \\
& \cos[\theta] r d[\theta] \wedge d[\phi] \wedge d[r] - \cos[\theta] \left(-1 + \frac{2m}{r} \right) r d[\theta] \wedge d[\phi] \wedge d[v] - \\
& \frac{1}{2} \csc[\theta] \left(\frac{m \sin[\theta]^2 d[\phi] \wedge d[r] \wedge d[v]}{r} - \frac{m (2m - r) \sin[\theta]^2 d[\phi] \wedge d[v] \wedge d[v]}{r^2} \right) + \\
& m \cos[\theta] d[\phi] \wedge d[v] \wedge d[\theta] - \cos[\theta] r d[\phi] \wedge d[\theta] \wedge d[r] + \\
& \frac{1}{2} \csc[\theta] \left(\sin[\theta]^2 d[r] \wedge d[\phi] \wedge d[r] + \cos[\theta] r \sin[\theta] d[\theta] \wedge d[\phi] \wedge d[r] + \right. \\
& \quad \left. \frac{m \sin[\theta]^2 d[\phi] \wedge d[v] \wedge d[r]}{r} + \cos[\theta] r \sin[\theta] d[\phi] \wedge d[\theta] \wedge d[r] \right) - \\
& \frac{1}{2} \csc[\theta] \left(\frac{m \sin[\theta]^2 d[v] \wedge d[\phi] \wedge d[r]}{r} + \cos[\theta] r \sin[\theta] d[\theta] \wedge d[\phi] \wedge d[r] - \right. \\
& \quad \left. \sin[\theta]^2 (d[\phi] \wedge d[r] \wedge d[r] + \cot[\theta] r d[\phi] \wedge d[\theta] \wedge d[r]) \right) + \\
& \cos[\theta] \left(-1 + \frac{2m}{r} \right) r d[\phi] \wedge d[\theta] \wedge d[v] - \frac{1}{2} \csc[\theta] \left(-1 + \frac{2m}{r} \right) \\
& \left(\sin[\theta]^2 d[r] \wedge d[\phi] \wedge d[v] + \cos[\theta] r \sin[\theta] d[\theta] \wedge d[\phi] \wedge d[v] + \right. \\
& \quad \left. \frac{m \sin[\theta]^2 d[\phi] \wedge d[v] \wedge d[v]}{r} + \cos[\theta] r \sin[\theta] d[\phi] \wedge d[\theta] \wedge d[v] \right) + \\
& \frac{1}{2} \csc[\theta] \left(-1 + \frac{2m}{r} \right) \left(\frac{m \sin[\theta]^2 d[v] \wedge d[\phi] \wedge d[v]}{r} + \cos[\theta] r \sin[\theta] d[\theta] \wedge d[\phi] \wedge d[v] - \right. \\
& \quad \left. \sin[\theta]^2 (d[\phi] \wedge d[r] \wedge d[v] + \cot[\theta] r d[\phi] \wedge d[\theta] \wedge d[v]) \right), \\
S_3 = & - \frac{m \sin[\theta] \left(-r^2 d[r] \wedge d[\theta] \wedge d[v] - (2m - r) r d[v] \wedge d[\theta] \wedge d[v] \right)}{2 r^3} + \\
& \frac{1}{2} \sin[\theta] \left(\frac{m d[\theta] \wedge d[r] \wedge d[v]}{r} + \frac{m (-2m + r) d[\theta] \wedge d[v] \wedge d[v]}{r^2} \right) - \\
& \frac{1}{2} r \sin[\theta] \left(-\frac{m d[r] \wedge d[v] \wedge d[\theta]}{r^2} - \frac{m \cot[\theta] d[\theta] \wedge d[v] \wedge d[\theta]}{r} \right) + \\
& \frac{1}{2} r \sin[\theta] \left(\frac{m d[r] \wedge d[v] \wedge d[\theta]}{r^2} + \frac{m \cot[\theta] d[\theta] \wedge d[v] \wedge d[\theta]}{r} \right) + \\
& \frac{\sin[\theta] \left(-r d[r] \wedge d[\theta] \wedge d[r] - \cot[\theta] r^2 d[\theta] \wedge d[\theta] \wedge d[r] \right)}{2 r} - \\
& \frac{\sin[\theta] \left(r d[r] \wedge d[\theta] \wedge d[r] + \cot[\theta] r^2 d[\theta] \wedge d[\theta] \wedge d[r] \right)}{2 r} -
\end{aligned}$$

$$\begin{aligned}
& \frac{\left(-1 + \frac{2m}{r}\right) \sin[\theta] \left(-r d[r] \wedge d[\theta] \wedge d[v] - \cot[\theta] r^2 d[\theta] \wedge d[\theta] \wedge d[v]\right)}{2r} + \\
& \frac{\left(-1 + \frac{2m}{r}\right) \sin[\theta] \left(r d[r] \wedge d[\theta] \wedge d[v] + \cot[\theta] r^2 d[\theta] \wedge d[\theta] \wedge d[v]\right)}{2r} + \\
& m \cos[\theta] \sin[\theta]^2 d[\phi] \wedge d[v] \wedge d[\phi] + \\
& \frac{1}{2} \sin[\theta] \left(\frac{m d[v] \wedge d[\theta] \wedge d[r]}{r} - d[\theta] \wedge d[r] \wedge d[r] - \cos[\theta] r \sin[\theta] d[\phi] \wedge d[\phi] \wedge d[r] \right) - \\
& \frac{1}{2} \sin[\theta] \left(d[r] \wedge d[\theta] \wedge d[r] + \frac{m d[\theta] \wedge d[v] \wedge d[r]}{r} - \cos[\theta] r \sin[\theta] d[\phi] \wedge d[\phi] \wedge d[r] \right) - \\
& \frac{1}{2} \left(-1 + \frac{2m}{r}\right) \sin[\theta] \left(\frac{m d[v] \wedge d[\theta] \wedge d[v]}{r} - d[\theta] \wedge d[r] \wedge d[v] - \right. \\
& \quad \left. \cos[\theta] r \sin[\theta] d[\phi] \wedge d[\phi] \wedge d[v] \right) + \frac{1}{2} \left(-1 + \frac{2m}{r}\right) \sin[\theta] \\
& \quad \left(d[r] \wedge d[\theta] \wedge d[v] + \frac{m d[\theta] \wedge d[v] \wedge d[v]}{r} - \cos[\theta] r \sin[\theta] d[\phi] \wedge d[\phi] \wedge d[v] \right) \Big\}
\end{aligned}$$

In[*]:= Map[ToCanonical, %, {2}]

[\[aplica a todos\]](#)

Out[*]=

$$\begin{aligned}
& \{s_0 = \sin[\theta] d[r] \wedge d[\theta] \wedge d[\phi], s_1 = 2 \sin[\theta] d[r] \wedge d[\theta] \wedge d[\phi] + \sin[\theta] d[v] \wedge d[\theta] \wedge d[\phi], \\
& s_2 = -\sin[\theta] d[r] \wedge d[v] \wedge d[\phi] + \cos[\theta] r d[r] \wedge d[\theta] \wedge d[\phi] - \\
& m \cos[\theta] d[v] \wedge d[\theta] \wedge d[\phi] + \cos[\theta] r d[v] \wedge d[\theta] \wedge d[\phi], s_3 = \sin[\theta] d[r] \wedge d[v] \wedge d[\theta]\}
\end{aligned}$$

In[*]:= Map[Simplify, %, {2}]

[\[apl... simplifica\]](#)

Out[*]=

$$\begin{aligned}
& \{s_0 = \sin[\theta] d[r] \wedge d[\theta] \wedge d[\phi], s_1 = \sin[\theta] (2 d[r] \wedge d[\theta] \wedge d[\phi] + d[v] \wedge d[\theta] \wedge d[\phi]), \\
& s_2 = -\sin[\theta] d[r] \wedge d[v] \wedge d[\phi] + \cos[\theta] (r d[r] \wedge d[\theta] \wedge d[\phi] + (-m + r) d[v] \wedge d[\theta] \wedge d[\phi]), \\
& s_3 = \sin[\theta] d[r] \wedge d[v] \wedge d[\theta]\}
\end{aligned}$$

In[*]:= SparlingEF = %;

In[*]:= % /. {r[] -> 2 mass}

Out[*]=

$$\{s_0 = 0, s_1 = \sin[\theta] d[v] \wedge d[\theta] \wedge d[\phi], s_2 = m \cos[\theta] d[v] \wedge d[\theta] \wedge d[\phi], s_3 = 0\}$$

In[*]:= SparlingHorizon = %;

In[*]:= % // InputForm

[\[forma de entrada\]](#)

Out[*]//InputForm=

$$\begin{aligned}
& \{S[\{0, -EF\}] == 0, S[\{1, -EF\}] == \sin[\theta[]] * \text{Diff}[v[], PD] \wedge \text{Diff}[\theta[], PD] \wedge \text{Diff}[\phi[], PD], \\
& S[\{2, -EF\}] == \text{mass} * \cos[\theta[]] * \text{Diff}[v[], PD] \wedge \text{Diff}[\theta[], PD] \wedge \text{Diff}[\phi[], PD], S[\{3, -E
\end{aligned}$$

In[*]:= Hodge[metricTensor] /@ SparlingEF

Out[*]=

$$\begin{aligned} \{ * [S_0] = \text{Sin}[\theta] (* [d[r] \wedge d[\theta] \wedge d[\phi]]) , * [S_1] = \\ \text{Sin}[\theta] (2 (* [d[r] \wedge d[\theta] \wedge d[\phi]]) + * [d[v] \wedge d[\theta] \wedge d[\phi]]) , * [S_2] = \\ -\text{Sin}[\theta] (* [d[r] \wedge d[v] \wedge d[\phi]]) + \\ \text{Cos}[\theta] (r (* [d[r] \wedge d[\theta] \wedge d[\phi]]) + (-m + r) (* [d[v] \wedge d[\theta] \wedge d[\phi]])) , * [S_3] = \\ \text{Sin}[\theta] (* [d[r] \wedge d[v] \wedge d[\theta]]) \} \end{aligned}$$

In[*]:= Map[ExpandHodgeDual[#, dx[M4], metricTensor] &, %, {2}]

[aplica a todos](#)

Out[*]=

$$\begin{aligned} \{ * [S_0] = \text{Sin}[\theta] \left(\frac{(2m - r) dx^0}{r \sqrt{r^4 \text{Sin}[\theta]^2}} - \frac{dx^1}{\sqrt{r^4 \text{Sin}[\theta]^2}} \right) , * [S_1] = \\ \text{Sin}[\theta] \left(\frac{dx^0}{\sqrt{r^4 \text{Sin}[\theta]^2}} + 2 \left(\frac{(2m - r) dx^0}{r \sqrt{r^4 \text{Sin}[\theta]^2}} - \frac{dx^1}{\sqrt{r^4 \text{Sin}[\theta]^2}} \right) \right) , * [S_2] = \\ \text{Cos}[\theta] \left(\frac{(-m + r) dx^0}{\sqrt{r^4 \text{Sin}[\theta]^2}} + r \left(\frac{(2m - r) dx^0}{r \sqrt{r^4 \text{Sin}[\theta]^2}} - \frac{dx^1}{\sqrt{r^4 \text{Sin}[\theta]^2}} \right) \right) + \frac{r^2 \text{Sin}[\theta] dx^2}{\sqrt{r^4 \text{Sin}[\theta]^2}} , * [S_3] = \\ \frac{\text{Sin}[\theta] \sqrt{r^4 \text{Sin}[\theta]^2} dx^3}{r^2} \} \end{aligned}$$

In[*]:= Map[PowerExpand, %, {2}]

[apl...](#) [expande potencias](#)

Out[*]=

$$\begin{aligned} \{ * [S_0] = \text{Sin}[\theta] \left(\frac{\text{Csc}[\theta] (2m - r) dx^0}{r^3} - \frac{\text{Csc}[\theta] dx^1}{r^2} \right) , * [S_1] = \\ \text{Sin}[\theta] \left(\frac{\text{Csc}[\theta] dx^0}{r^2} + 2 \left(\frac{\text{Csc}[\theta] (2m - r) dx^0}{r^3} - \frac{\text{Csc}[\theta] dx^1}{r^2} \right) \right) , * [S_2] = \\ \text{Cos}[\theta] \left(\frac{\text{Csc}[\theta] (-m + r) dx^0}{r^2} + r \left(\frac{\text{Csc}[\theta] (2m - r) dx^0}{r^3} - \frac{\text{Csc}[\theta] dx^1}{r^2} \right) \right) + dx^2 , * [S_3] = \\ \text{Sin}[\theta]^2 dx^3 \} \end{aligned}$$

In[*]:= ToValues //@%

Out[*]=

$$\begin{aligned} \{ * [S_0] = \left(\frac{\text{Csc}[\theta] d[v] (2m - r)}{r^3} - \frac{\text{Csc}[\theta] d[r]}{r^2} \right) \text{Sin}[\theta] , * [S_1] = \\ \left(2 \left(\frac{\text{Csc}[\theta] d[v] (2m - r)}{r^3} - \frac{\text{Csc}[\theta] d[r]}{r^2} \right) + \frac{\text{Csc}[\theta] d[v]}{r^2} \right) \text{Sin}[\theta] , * [S_2] = \\ d[\theta] + \text{Cos}[\theta] \left(\left(\frac{\text{Csc}[\theta] d[v] (2m - r)}{r^3} - \frac{\text{Csc}[\theta] d[r]}{r^2} \right) r + \frac{\text{Csc}[\theta] d[v] (-m + r)}{r^2} \right) , * [S_3] = \\ d[\phi] \text{Sin}[\theta]^2 \} \end{aligned}$$

```
In[*]:= Map[FullSimplify, %, {2}]
```

[\[apl\]](#) [\[simplifica completamente\]](#)

```
Out[*]=
```

$$\left\{ * [S_0] = -\frac{d[r] r + d[v] (-2 m + r)}{r^3}, * [S_1] = -\frac{2 d[r] r + d[v] (-4 m + r)}{r^3}, * [S_2] = d[\theta] + \frac{\text{Cot}[\theta] (m d[v] - d[r] r)}{r^2}, * [S_3] = d[\phi] \text{Sin}[\theta]^2 \right\}$$

```
In[*]:= % /. {r[] -> 2 mass}
```

```
Out[*]=
```

$$\left\{ * [S_0] = 0, * [S_1] = \frac{d[v]}{4 m^2}, * [S_2] = \frac{\text{Cot}[\theta] d[v]}{4 m} + d[\theta], * [S_3] = d[\phi] \text{Sin}[\theta]^2 \right\}$$

```
In[*]:= SparlingHorizonDual = %;
```

Define observer:

```
In[*]:= DefTensor[u[a], M4]
```

```
In[*]:= Hodge[metricTensor][S[-a]] \times u[a]
```

```
Out[*]=
```

$$u^a (* [S_a])$$

```
In[*]:= ToBasis[EF][%]
```

```
Out[*]=
```

$$u^a (* [S_a])$$

```
In[*]:= TraceBasisDummy[%]
```

```
Out[*]=
```

$$u^0 (* [S_0]) + u^1 (* [S_1]) + u^2 (* [S_2]) + u^3 (* [S_3])$$

```
In[*]:= % /. {r[] -> 2 mass}
```

```
Out[*]=
```

$$u^0 (* [S_0]) + u^1 (* [S_1]) + u^2 (* [S_2]) + u^3 (* [S_3])$$

```
In[*]:= % /. (SparlingHorizonDual /. Equal -> Rule)
```

[\[igual\]](#) [\[regla\]](#)

```
Out[*]=
```

$$\frac{d[v] u^1}{4 m^2} + \left(\frac{\text{Cot}[\theta] d[v]}{4 m} + d[\theta] \right) u^2 + d[\phi] \text{Sin}[\theta]^2 u^3$$

```
In[*]:= Collect[%, Diff, Simplify]
```

[\[agrupa coefi...\]](#) [\[diff\]](#) [\[simplifica\]](#)

```
Out[*]=
```

$$d[\theta] u^2 + \frac{d[v] (u^1 + m \text{Cot}[\theta] u^2)}{4 m^2} + d[\phi] \text{Sin}[\theta]^2 u^3$$

```
In[*]:= CoefficientArrays[%, {Diff[v[]], Diff[r[]], Diff[\theta[]], Diff[\phi[]]}]
```

[\[arreglos de coeficientes\]](#) [\[diff\]](#) [\[diff\]](#) [\[diff\]](#) [\[diff\]](#)

```
Out[*]=
```

$$\{0, \text{SparseArray}\left[\begin{array}{c} \text{+} \\ \text{[red bar]} \end{array} \text{Specified elements: 3} \right. \\ \left. \text{Dimensions: \{4\}} \right\}$$

```

In[*]:= Last@% // Normal
      último      normal
Out[*]=

$$\left\{ \frac{u^1 + m \cot[\theta] u^2}{4 m^2}, \theta, u^2, \sin[\theta]^2 u^3 \right\}$$


In[*]:= CTensor[%, {-EF}]
Out[*]=

$$\text{CTensor}\left[\left\{ \frac{u^1 + m \cot[\theta] u^2}{4 m^2}, \theta, u^2, \sin[\theta]^2 u^3 \right\}, \{-EF\}, \theta\right]$$


In[*]:= flux = %;

In[*]:= flux[-b] × metricTensor[a, b] × flux[-a]
Out[*]=

$$\frac{u^{2^2}}{r^2} + \frac{\sin[\theta]^2 u^{3^2}}{r^2}$$


In[*]:= FullSimplify[%]
      simplifica completamente
Out[*]=

$$\frac{u^{2^2} + \sin[\theta]^2 u^{3^2}}{r^2}$$


```

Schwarzschild in Kerr-Schild coordinates

```

In[*]:= SparlingForm
Out[*]=

$$S_a = -\frac{1}{2} \eta^{bcde} \Gamma[\nabla]_{ba} \wedge \Gamma[\nabla]_{cd} \wedge \theta_e - \frac{1}{2} \eta_a{}^{bcd} \Gamma[\nabla]_b{}^e \wedge \Gamma[\nabla]_{ec} \wedge \theta_d$$


In[*]:= % /. MakeRule[{Coframe[M4][a], Coframe[M4][a]}, MetricOn → All, ContractMetrics → False]
      todo      falso
Out[*]=

$$S_a = -\frac{1}{2} \eta^{bcde} g_{fe} \Gamma[\nabla]_{ba} \wedge \Gamma[\nabla]_{cd} \wedge \theta^f - \frac{1}{2} \eta_a{}^{bcd} g_{fd} \Gamma[\nabla]_b{}^e \wedge \Gamma[\nabla]_{ec} \wedge \theta^f$$


In[*]:= SparlingFormUp = %
Out[*]=

$$S_a = -\frac{1}{2} \eta^{bcde} g_{fe} \Gamma[\nabla]_{ba} \wedge \Gamma[\nabla]_{cd} \wedge \theta^f - \frac{1}{2} \eta_a{}^{bcd} g_{fd} \Gamma[\nabla]_b{}^e \wedge \Gamma[\nabla]_{ec} \wedge \theta^f$$


In[*]:= << xAct`xIdeal`

-----

Package xAct`xIdeal` version 0.1.0, {2025, 10, 29}

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-----

```

```

In[*]:= coords = "KerrSchildCoordinates";

In[*]:= sol = "Schwarzschild";

In[*]:= xActSolutionData[{"Schwarzschild", coords}, "Properties"]
                                         |propiedades
Out[*]=
{CoordinateAssumptions, CoordinateNames, Metric, ParameterNames, ParameterAssumptions,
  ParameterValues, ScalarFunctionNames, ScalarFunctionValues, SignatureOfMetric}

In[*]:= xActSolutionData[{"Schwarzschild", coords}, "CoordinateNames"]
Out[*]=
{t, r,  $\theta$ ,  $\phi$ }

In[*]:= xActSolutionData[{"Schwarzschild", coords}, "SignatureOfMetric"]
Out[*]=
{3, 1, 0}

In[*]:= $DefInfoQ = False;
                                         |falso

In[*]:= $CVVerbose = False;
                                         |falso

In[*]:= DefChart[KS, M4, {0, 1, 2, 3}, {t[], r[],  $\theta$ [],  $\phi$ []}]

In[*]:= xActSolutionData[{sol, coords}, "ParameterNames"]
Out[*]=
{m,  $\epsilon$ }

In[*]:= xActSolutionData[{sol, coords}, "ParameterAssumptions"]
Out[*]=
Function[
  {xAct`xIdeal`Private`coords, xAct`xIdeal`Private`params, xAct`xIdeal`Private`scfuncs},
  With[{xAct`xIdeal`Private`m = xAct`xIdeal`Private`params[[1]],
    epsilon = xAct`xIdeal`Private`params[[2]]},
    xAct`xIdeal`Private`m > 0 && (epsilon === 1 || epsilon === -1)] ]

In[*]:= DefConstantSymbol[mass, PrintAs  $\rightarrow$  "m"]; DefConstantSymbol[ $\epsilon$ ]

In[*]:= metricTensor = xActSolutionData[{sol, coords  $\rightarrow$  KS,
  "ScalarFunctionValues"  $\rightarrow$  {}, "ParameterValues"  $\rightarrow$  {mass, 1}}, "Metric"]
Out[*]=
CTensor[{{{-1 +  $\frac{2m}{r}$ ,  $\frac{2m}{r}$ , 0, 0}, { $\frac{2m}{r}$ , 1 +  $\frac{2m}{r}$ , 0, 0},
  {0, 0,  $r^2$ , 0}, {0, 0, 0,  $r^2 \sin[\theta]^2$ }}, {-KS, -KS}, 0]

In[*]:= SetCMetric[metricTensor, KS, SignatureOfMetric  $\rightarrow$  {3, 1, 0}]

```

... SetCMetric: There are already metrics {metricg} in vbundle $\mathbb{T}M_4$.

```
In[•]:= cd = CovDOfMetric[metricTensor]
```

Out[•]=

$$\begin{aligned} & \text{CCovD} \left[\text{PDKS}, \right. \\ & \text{CTensor} \left[\left\{ \left\{ \left\{ \frac{2 m^2}{r^3}, \frac{m (2 m + r)}{r^3}, 0, 0 \right\}, \left\{ \frac{m (2 m + r)}{r^3}, \frac{2 m (m + r)}{r^3}, 0, 0 \right\}, \{0, 0, -2 m, 0\}, \right. \right. \\ & \quad \left. \{0, 0, 0, -2 m \sin[\theta]^2\} \right\}, \left\{ \left\{ \frac{m (-2 m + r)}{r^3}, -\frac{2 m^2}{r^3}, 0, 0 \right\}, \right. \\ & \quad \left. \left\{ -\frac{2 m^2}{r^3}, -\frac{m (2 m + r)}{r^3}, 0, 0 \right\}, \{0, 0, 2 m - r, 0\}, \{0, 0, 0, (2 m - r) \sin[\theta]^2\} \right\}, \\ & \quad \left\{ \{0, 0, 0, 0\}, \left\{ 0, 0, \frac{1}{r}, 0 \right\}, \left\{ 0, \frac{1}{r}, 0, 0 \right\}, \{0, 0, 0, -\cos[\theta] \sin[\theta]\} \right\}, \\ & \quad \left\{ \{0, 0, 0, 0\}, \left\{ 0, 0, 0, \frac{1}{r} \right\}, \{0, 0, 0, \cot[\theta]\}, \left\{ 0, \frac{1}{r}, \cot[\theta], 0 \right\} \right\}, \{KS, -KS, -KS\}, 0 \Big], \\ & \text{CTensor} \left[\left\{ \left\{ \left\{ -1 + \frac{2 m}{r}, \frac{2 m}{r}, 0, 0 \right\}, \left\{ \frac{2 m}{r}, 1 + \frac{2 m}{r}, 0, 0 \right\}, \{0, 0, r^2, 0\}, \{0, 0, 0, r^2 \sin[\theta]^2\} \right\}, \right. \right. \\ & \quad \left. \left. \{-KS, -KS\}, 0 \right] \right] \end{aligned}$$

This should work automatically:

```
In[ ]:= $Metrics = Reverse[$Metrics];
```

invierte orden

```
In[*]:= TangentM4 /: MetricsOfVBundle[TangentM4] = Reverse[MetricsOfVBundle[TangentM4]];
```

```
In[•]:= MetricsOfVBundle[TangentM4]
```

Out[•]=

$$\left\{ \text{CTensor}\left[\left\{\left\{-1 + \frac{2m}{r}, \frac{2m}{r}, 0, 0\right\}, \left\{\frac{2m}{r}, 1 + \frac{2m}{r}, 0, 0\right\}, \{0, 0, r^2, 0\}, \{0, 0, 0, r^2 \sin^2[\theta]\right\}, \right. \right. \\ \left. \left. \{-KS, -KS\}, 0\right], \text{metricg}\right\}$$

```
In[•]:= Ricci[cd]
```

Out[]=

Zero

```
In[•]:= KerrSolutionQ[metricTensor]
```

```
** ReportCompute: Metric[1, 1]
```

Already cached

```
** ReportCompute: DMetric[-1, -1, -1]
```

Already cached

```
** ReportCompute: DDMetric [-1, -1, -1, -1]
```

Constructed in 0.000882 seconds

Applied Simplify in 3.91627 seconds

```
** ReportCompute: Christoffel[-1, -1, -1]
```

Already cached

```
** ReportCompute: Christoffel[1, -1, -1]
```

Already cached

```

** ReportCompute: CovDOfMetric
  Already cached
** ReportCompute: Riemann[-1, -1, -1, -1]
  Constructed in 0.000522 seconds
  Applied Simplify in 0.019547 seconds
  Stored in 0.000533 seconds and 20160 bytes
** ReportCompute: Riemann[-1, -1, -1, 1]
  Constructed in 0.001381 seconds
  Applied Simplify in 0.034507 seconds
  Stored in 0.00043 seconds and 16544 bytes
** ReportCompute: Ricci[-1, -1]
  Constructed in 0.00011 seconds
  Applied Simplify in 0.006133 seconds
  Stored in 0.000101 seconds and 424 bytes
** ReportCompute: RicciScalar[]
  Constructed in 0.000061 seconds
  Applied Simplify in 0.014171 seconds
  Stored in 0.000116 seconds and 64 bytes
** ReportCompute: Weyl[-1, -1, -1, -1]
  Constructed in 0.000354 seconds
  Applied Simplify in 0.028644 seconds
  Stored in 0.000621 seconds and 20160 bytes
Vacuum
** ReportCompute: computing Weyl concomitant "WeylDual" in 0.018954 seconds:
** ReportCompute: applying Simplify to Weyl concomitant "WeylDual" in 0.009458 seconds:
** ReportCompute: computing Weyl concomitant "WeylSelfDual" in 0.000590 seconds:
** ReportCompute: applying Simplify to Weyl concomitant "WeylSelfDual" in 0.018727 seconds:
** ReportCompute: computing metric concomitant "G2Form" in 0.002027 seconds:
** ReportCompute: applying Simplify to metric concomitant "G2Form" in 0.014203 seconds:
** ReportCompute: computing Weyl concomitant "WeylSelfDual2" in 0.231054 seconds:
** ReportCompute: applying Simplify to Weyl concomitant "WeylSelfDual2" in 0.031721 seconds:
** ReportCompute: computing Weyl concomitant "TraceWeylSelfDual2" in 0.235367 seconds:
** ReportCompute: applying Simplify
  to Weyl concomitant "TraceWeylSelfDual2" in 0.000205 seconds:
** ReportCompute: computing Weyl concomitant "WeylSelfDual3" in 0.014256 seconds:
** ReportCompute: applying Simplify to Weyl concomitant "WeylSelfDual3" in 0.031656 seconds:
** ReportCompute: computing Weyl concomitant "TraceWeylSelfDual3" in 0.236269 seconds:
** ReportCompute: applying Simplify
  to Weyl concomitant "TraceWeylSelfDual3" in 0.000201 seconds:
Type D

```

Kerr NUT

Complex Kerr

Kerr

4.92637

Out[*]=

True

In[*]:= SparlingFormUp

Out[*]=

$$S_a = -\frac{1}{2} \eta^{bcde} g_{fe} \Gamma[\nabla]_{ba} \wedge \Gamma[\nabla]_{cd} \wedge \theta^f - \frac{1}{2} \eta_a^{bcd} g_{fd} \Gamma[\nabla]_b^e \wedge \Gamma[\nabla]_{ec} \wedge \theta^f$$

In[*]:= ToBasis[KS] /@ %

Out[*]=

$$S_a = -\frac{1}{2} \eta^{bcde} g_{fe} \Gamma[\nabla]_{ba} \wedge \Gamma[\nabla]_{cd} \wedge \theta^f - \frac{1}{2} \eta_a^{bcd} g_{fd} \Gamma[\nabla]_b^e \wedge \Gamma[\nabla]_{ec} \wedge \theta^f$$

In[*]:= TraceBasisDummy /@ %;

In[*]:= % /. epsilonmetricg → epsilon[metricTensor];

2.37295

In[*]:= % /. CD → cd;

1.94766

In[*]:= ToValues /@ %;

In[*]:= PowerExpand //@ %;

[\[expande potencias\]](#)

1.87368

In[*]:= ComponentArray@%;

In[*]:= % /. metricg → metricTensor

Out[*]=

$$\left\{ \begin{aligned} S_0 &= \frac{m \cos[\theta] \left(-1 + \frac{2m}{r}\right) \theta^0 \wedge \theta^3 \wedge \theta^0}{r^2} + \frac{2m^2 \cos[\theta] \theta^0 \wedge \theta^3 \wedge \theta^1}{r^3} + \frac{m \sin[\theta] \theta^1 \wedge \theta^2 \wedge \theta^3}{r} - \frac{m^2 \dots}{r^4} + \dots, \\ S_1 &= -\frac{m \sin[\theta] \left(-r \dots\right)}{2r^2} + \dots + \frac{\dots}{r} - \frac{m \dots}{r}, \quad S_{\dots} = \dots, \\ S_3 &= \dots + \frac{m \dots}{r} \left(\theta^{\dots} \wedge \theta^{\dots} \wedge \theta^{\dots} \wedge \theta^{\dots} \right) \end{aligned} \right\}$$

La expresión completa no está disponible (tamaño de memoria original: 267.7 kB)

In[*]:= % /. Coframe → dx;

In[*]:= ToValues /@ %

Out[*]=

$$\left\{ \begin{aligned} S_0 &= \frac{m \sin[\theta] d[r] \wedge d[\theta] \wedge d[\phi]}{r} + \frac{m \cos[\theta] \left(1 + \frac{2m}{r}\right) d[r] \wedge d[\phi] \wedge d[r]}{r^2} + \\ &\quad \frac{2m^2 \cos[\theta] d[r] \wedge d[\phi] \wedge d[t]}{r^3} - \frac{m \sin[\theta] d[r] \wedge d[\phi] \wedge d[\theta]}{r} - \\ &\quad \frac{m^2 \sin[\theta] (-2(m-r) r d[r] \wedge d[\theta] \wedge d[\phi] - (2m-r) r d[t] \wedge d[\theta] \wedge d[\phi])}{r^4} + \end{aligned} \right.$$

$$\begin{aligned}
& \frac{2 m^2 \cos [\theta] d[t] \wedge d[\phi] \wedge d[r]}{r^3} + \frac{m \cos [\theta] \left(-1 + \frac{2 m}{r}\right) d[t] \wedge d[\phi] \wedge d[t]}{r^2} + \\
& \frac{m^2 \csc [\theta] \left(-2 (m-r) r \sin [\theta]^2 d[r] \wedge d[\phi] \wedge d[\theta] - (2 m-r) r \sin [\theta]^2 d[t] \wedge d[\phi] \wedge d[\theta]\right)}{r^4} + \\
& \frac{m \sin [\theta] \left(-\frac{2 m^2 d[\theta] \wedge d[r] \wedge d[\phi]}{r^2} + \frac{m (-2 m+r) d[\theta] \wedge d[t] \wedge d[\phi]}{r^2}\right)}{r} - \\
& \frac{m \csc [\theta] \left(-\frac{2 m^2 \sin [\theta]^2 d[\phi] \wedge d[r] \wedge d[\theta]}{r^2} - \frac{m (2 m-r) \sin [\theta]^2 d[\phi] \wedge d[t] \wedge d[\theta]}{r^2}\right)}{r} + \\
& \frac{1}{r^4} 2 m^2 \csc [\theta] \left(\cos [\theta] r \sin [\theta] d[r] \wedge d[\phi] \wedge d[r] + \cos [\theta]^2 r^2 d[\theta] \wedge d[\phi] \wedge d[r] + \cos [\theta] r \right. \\
& \quad \left. \sin [\theta] d[\phi] \wedge d[r] \wedge d[r] + (2 m-r) r \sin [\theta]^2 d[\phi] \wedge d[\theta] \wedge d[r]\right) - \frac{1}{2 r^3} \csc [\theta] \\
& \quad \left(1 + \frac{2 m}{r}\right) (2 m-r) \left(\cos [\theta] r \sin [\theta] d[r] \wedge d[\phi] \wedge d[r] + \cos [\theta]^2 r^2 d[\theta] \wedge d[\phi] \wedge d[r] + \right. \\
& \quad \left. \cos [\theta] r \sin [\theta] d[\phi] \wedge d[r] \wedge d[r] + (2 m-r) r \sin [\theta]^2 d[\phi] \wedge d[\theta] \wedge d[r]\right) - \frac{1}{r^4} \\
& \quad 2 m^2 \csc [\theta] \left(-\cos [\theta] r \sin [\theta] d[r] \wedge d[\phi] \wedge d[r] - r (-2 m+r) \sin [\theta]^2 d[\theta] \wedge d[\phi] \wedge d[r] - \right. \\
& \quad \left. \cos [\theta] r \sin [\theta] (d[\phi] \wedge d[r] \wedge d[r] + \cot [\theta] r d[\phi] \wedge d[\theta] \wedge d[r])\right) - \\
& \quad \frac{1}{2 r^3} \csc [\theta] \left(1 + \frac{2 m}{r}\right) (-2 m+r) \left(-\cos [\theta] r \sin [\theta] d[r] \wedge d[\phi] \wedge d[r] - \right. \\
& \quad \left. r (-2 m+r) \sin [\theta]^2 d[\theta] \wedge d[\phi] \wedge d[r] - \right. \\
& \quad \left. \cos [\theta] r \sin [\theta] (d[\phi] \wedge d[r] \wedge d[r] + \cot [\theta] r d[\phi] \wedge d[\theta] \wedge d[r])\right) - \\
& \quad \frac{1}{r^4} m \csc [\theta] (2 m-r) \left(\cos [\theta] r \sin [\theta] d[r] \wedge d[\phi] \wedge d[t] + \cos [\theta]^2 r^2 d[\theta] \wedge d[\phi] \wedge d[t] + \right. \\
& \quad \left. \cos [\theta] r \sin [\theta] d[\phi] \wedge d[r] \wedge d[t] + (2 m-r) r \sin [\theta]^2 d[\phi] \wedge d[\theta] \wedge d[t]\right) + \\
& \quad \frac{1}{r^3} m \csc [\theta] \left(-1 + \frac{2 m}{r}\right) \left(\cos [\theta] r \sin [\theta] d[r] \wedge d[\phi] \wedge d[t] + \cos [\theta]^2 r^2 d[\theta] \wedge d[\phi] \wedge d[t] + \right. \\
& \quad \left. \cos [\theta] r \sin [\theta] d[\phi] \wedge d[r] \wedge d[t] + (2 m-r) r \sin [\theta]^2 d[\phi] \wedge d[\theta] \wedge d[t]\right) - \\
& \quad \frac{1}{r^3} m \csc [\theta] \left(-1 + \frac{2 m}{r}\right) \left(-\cos [\theta] r \sin [\theta] d[r] \wedge d[\phi] \wedge d[t] - r (-2 m+r) \sin [\theta]^2 d[\theta] \wedge \right. \\
& \quad \left. d[\phi] \wedge d[t] - \cos [\theta] r \sin [\theta] (d[\phi] \wedge d[r] \wedge d[t] + \cot [\theta] r d[\phi] \wedge d[\theta] \wedge d[t])\right) - \\
& \quad \frac{1}{r^4} m \csc [\theta] (-2 m+r) \left(-\cos [\theta] r \sin [\theta] d[r] \wedge d[\phi] \wedge d[t] - r (-2 m+r) \sin [\theta]^2 d[\theta] \wedge \right. \\
& \quad \left. d[\phi] \wedge d[t] - \cos [\theta] r \sin [\theta] (d[\phi] \wedge d[r] \wedge d[t] + \cot [\theta] r d[\phi] \wedge d[\theta] \wedge d[t])\right) - \\
& \quad \frac{1}{2 r} \csc [\theta] (-2 m+r) \left(\sin [\theta]^2 d[r] \wedge d[\phi] \wedge d[\theta] + \cos [\theta] r \sin [\theta] d[\theta] \wedge d[\phi] \wedge d[\theta] + \right. \\
& \quad \left. \frac{m (2 m-r) \sin [\theta]^2 d[\phi] \wedge d[r] \wedge d[\theta]}{r^2} - \right. \\
& \quad \left. \frac{2 m^2 \sin [\theta]^2 (2 d[\phi] \wedge d[r] \wedge d[\theta] + d[\phi] \wedge d[t] \wedge d[\theta])}{r^2} + \right. \\
& \quad \left. \cos [\theta] r \sin [\theta] d[\phi] \wedge d[\theta] \wedge d[\theta]\right) - \frac{1}{2 r} \csc [\theta] (2 m-r) \\
& \quad \left(\frac{m \left(-((2 m-r) r \sin [\theta]^2 d[r] \wedge d[\phi] \wedge d[\theta]) - 2 m r \sin [\theta]^2 d[t] \wedge d[\phi] \wedge d[\theta])\right)}{r^3} + \right.
\end{aligned}$$

$$\begin{aligned}
& \cos[\theta] r \sin[\theta] d[\theta] \wedge d[\phi] \wedge d[\theta] - \\
& \sin[\theta]^2 (d[\phi] \wedge d[r] \wedge d[\theta] + \cot[\theta] r d[\phi] \wedge d[\theta] \wedge d[\theta]) \Big) - \\
& \frac{1}{2r} (-2m+r) \sin[\theta] \left(\frac{m(-(2m-r) r d[r] \wedge d[\theta] \wedge d[\phi]) - 2m r d[t] \wedge d[\theta] \wedge d[\phi])}{r^3} - \right. \\
& \left. d[\theta] \wedge d[r] \wedge d[\phi] - \cos[\theta] r \sin[\theta] d[\phi] \wedge d[\phi] \wedge d[\phi] \right) - \\
& \frac{1}{2r} (2m-r) \sin[\theta] \left(d[r] \wedge d[\theta] \wedge d[\phi] - \frac{m(-2m+r) d[\theta] \wedge d[r] \wedge d[\phi]}{r^2} - \right. \\
& \left. \frac{2m^2 (2 d[\theta] \wedge d[r] \wedge d[\phi] + d[\theta] \wedge d[t] \wedge d[\phi])}{r^2} - \cos[\theta] r \sin[\theta] d[\phi] \wedge d[\phi] \wedge d[\phi] \right), \\
S_1 = & - \frac{2m^2 \cos[\theta] d[r] \wedge d[\phi] \wedge d[r]}{r^3} - \frac{m \cos[\theta] \left(-1 + \frac{2m}{r}\right) d[r] \wedge d[\phi] \wedge d[t]}{r^2} - \\
& \frac{m \sin[\theta] (-2 r d[r] \wedge d[\theta] \wedge d[\phi] - r d[t] \wedge d[\theta] \wedge d[\phi])}{2 r^2} + \\
& \frac{m \sin[\theta] (2 r d[r] \wedge d[\theta] \wedge d[\phi] + r d[t] \wedge d[\theta] \wedge d[\phi])}{2 r^2} - \\
& \frac{m (2m+r) \sin[\theta] (-2 (m-r) r d[r] \wedge d[\theta] \wedge d[\phi] - (2m-r) r d[t] \wedge d[\theta] \wedge d[\phi])}{2 r^4} - \\
& \frac{1}{2 r^4} m \csc[\theta] \left(1 + \frac{2m}{r}\right) \\
& (-2 \cos[\theta] r^2 \sin[\theta] d[r] \wedge d[\phi] \wedge d[r] - \cos[\theta] r^2 \sin[\theta] d[t] \wedge d[\phi] \wedge d[r]) + \\
& \frac{1}{2 r^4} m \csc[\theta] \left(1 + \frac{2m}{r}\right) \\
& (2 \cos[\theta] r^2 \sin[\theta] d[r] \wedge d[\phi] \wedge d[r] + \cos[\theta] r^2 \sin[\theta] d[t] \wedge d[\phi] \wedge d[r]) - \\
& \frac{m^2 \csc[\theta] (-2 \cos[\theta] r^2 \sin[\theta] d[r] \wedge d[\phi] \wedge d[t] - \cos[\theta] r^2 \sin[\theta] d[t] \wedge d[\phi] \wedge d[t])}{r^5} + \\
& \frac{m^2 \csc[\theta] (2 \cos[\theta] r^2 \sin[\theta] d[r] \wedge d[\phi] \wedge d[t] + \cos[\theta] r^2 \sin[\theta] d[t] \wedge d[\phi] \wedge d[t])}{r^5} + \\
& \frac{m \csc[\theta] (-2 r \sin[\theta]^2 d[r] \wedge d[\phi] \wedge d[\theta] - r \sin[\theta]^2 d[t] \wedge d[\phi] \wedge d[\theta])}{2 r^2} - \\
& \frac{m \csc[\theta] (2 r \sin[\theta]^2 d[r] \wedge d[\phi] \wedge d[\theta] + r \sin[\theta]^2 d[t] \wedge d[\phi] \wedge d[\theta])}{2 r^2} + \\
& \frac{1}{2 r^4} m \csc[\theta] (2m+r) \\
& (-2 (m-r) r \sin[\theta]^2 d[r] \wedge d[\phi] \wedge d[\theta] - (2m-r) r \sin[\theta]^2 d[t] \wedge d[\phi] \wedge d[\theta]) - \\
& \frac{m \sin[\theta] d[\theta] \wedge d[t] \wedge d[\phi]}{2 r} - \frac{m \sin[\theta] (2 d[\theta] \wedge d[r] \wedge d[\phi] + d[\theta] \wedge d[t] \wedge d[\phi])}{2 r} + \\
& \frac{(2m+r) \sin[\theta] \left(-\frac{2m^2 d[\theta] \wedge d[r] \wedge d[\phi]}{r^2} + \frac{m(-2m+r) d[\theta] \wedge d[t] \wedge d[\phi]}{r^2}\right)}{2 r} - \\
& \frac{2m \sin[\theta] d[\theta] \wedge d[\phi] \wedge d[r]}{r} -
\end{aligned}$$

$$\begin{aligned}
& \left(-1 + \frac{2m}{r} \right) \sin[\theta] d[\theta] \wedge d[\phi] \wedge d[t] + \frac{m \sin[\theta] d[\phi] \wedge d[t] \wedge d[\theta]}{2r} + \\
& \frac{m \sin[\theta] (2d[\phi] \wedge d[r] \wedge d[\theta] + d[\phi] \wedge d[t] \wedge d[\theta])}{2r} - \\
& \frac{\csc[\theta] (2m+r) \left(-\frac{2m^2 \sin[\theta]^2 d[\phi] \wedge d[r] \wedge d[\theta]}{r^2} - \frac{m(2m-r) \sin[\theta]^2 d[\phi] \wedge d[t] \wedge d[\theta]}{r^2} \right)}{2r} + \\
& \frac{2m \sin[\theta] d[\phi] \wedge d[\theta] \wedge d[r]}{r} - \\
& \frac{1}{r^3} m \csc[\theta] \left(-1 - \frac{2m}{r} \right) \left(\cos[\theta] r \sin[\theta] d[r] \wedge d[\phi] \wedge d[r] + \cos[\theta]^2 r^2 d[\theta] \wedge d[\phi] \wedge d[r] + \right. \\
& \quad \left. \cos[\theta] r \sin[\theta] d[\phi] \wedge d[r] \wedge d[r] + (2m-r) r \sin[\theta]^2 d[\phi] \wedge d[\theta] \wedge d[r] \right) - \\
& \frac{1}{r^3} m \csc[\theta] \left(1 + \frac{2m}{r} \right) \left(\cos[\theta] r \sin[\theta] d[r] \wedge d[\phi] \wedge d[r] + \cos[\theta]^2 r^2 d[\theta] \wedge d[\phi] \wedge d[r] + \right. \\
& \quad \left. \cos[\theta] r \sin[\theta] d[\phi] \wedge d[r] \wedge d[r] + (2m-r) r \sin[\theta]^2 d[\phi] \wedge d[\theta] \wedge d[r] \right) + \\
& \frac{1}{r^3} m \csc[\theta] \left(1 + \frac{2m}{r} \right) \left(-\cos[\theta] r \sin[\theta] d[r] \wedge d[\phi] \wedge d[r] - r(-2m+r) \sin[\theta]^2 d[\theta] \wedge \right. \\
& \quad \left. d[\phi] \wedge d[r] - \cos[\theta] r \sin[\theta] (d[\phi] \wedge d[r] \wedge d[r] + \cot[\theta] r d[\phi] \wedge d[\theta] \wedge d[r]) \right) - \\
& \frac{1}{r^4} m \csc[\theta] (2m+r) \left(-\cos[\theta] r \sin[\theta] d[r] \wedge d[\phi] \wedge d[r] - r(-2m+r) \sin[\theta]^2 d[\theta] \wedge \right. \\
& \quad \left. d[\phi] \wedge d[r] - \cos[\theta] r \sin[\theta] (d[\phi] \wedge d[r] \wedge d[r] + \cot[\theta] r d[\phi] \wedge d[\theta] \wedge d[r]) \right) + \\
& \left(-1 + \frac{2m}{r} \right) \sin[\theta] d[\phi] \wedge d[\theta] \wedge d[t] - \frac{1}{r^4} 2m^2 \csc[\theta] \left(\cos[\theta] r \sin[\theta] d[r] \wedge d[\phi] \wedge d[t] + \right. \\
& \quad \left. \cos[\theta]^2 r^2 d[\theta] \wedge d[\phi] \wedge d[t] + \cos[\theta] r \sin[\theta] d[\phi] \wedge d[r] \wedge d[t] + \right. \\
& \quad \left. (2m-r) r \sin[\theta]^2 d[\phi] \wedge d[\theta] \wedge d[t] \right) - \frac{1}{2r^2} \csc[\theta] \left(-1 - \frac{2m}{r} \right) \\
& \left(-1 + \frac{2m}{r} \right) \left(\cos[\theta] r \sin[\theta] d[r] \wedge d[\phi] \wedge d[t] + \cos[\theta]^2 r^2 d[\theta] \wedge d[\phi] \wedge d[t] + \right. \\
& \quad \left. \cos[\theta] r \sin[\theta] d[\phi] \wedge d[r] \wedge d[t] + (2m-r) r \sin[\theta]^2 d[\phi] \wedge d[\theta] \wedge d[t] \right) + \frac{1}{r^4} \\
& 2m^2 \csc[\theta] \left(-\cos[\theta] r \sin[\theta] d[r] \wedge d[\phi] \wedge d[t] - r(-2m+r) \sin[\theta]^2 d[\theta] \wedge d[\phi] \wedge d[t] - \right. \\
& \quad \left. \cos[\theta] r \sin[\theta] (d[\phi] \wedge d[r] \wedge d[t] + \cot[\theta] r d[\phi] \wedge d[\theta] \wedge d[t]) \right) - \\
& \frac{1}{2r^3} \csc[\theta] \left(-1 + \frac{2m}{r} \right) (2m+r) \left(-\cos[\theta] r \sin[\theta] d[r] \wedge d[\phi] \wedge d[t] - \right. \\
& \quad \left. r(-2m+r) \sin[\theta]^2 d[\theta] \wedge d[\phi] \wedge d[t] - \right. \\
& \quad \left. \cos[\theta] r \sin[\theta] (d[\phi] \wedge d[r] \wedge d[t] + \cot[\theta] r d[\phi] \wedge d[\theta] \wedge d[t]) \right) + \\
& \frac{1}{r} m \csc[\theta] \left(\sin[\theta]^2 d[r] \wedge d[\phi] \wedge d[\theta] + \cos[\theta] r \sin[\theta] d[\theta] \wedge d[\phi] \wedge d[\theta] + \right. \\
& \quad \left. \frac{m(2m-r) \sin[\theta]^2 d[\phi] \wedge d[r] \wedge d[\theta]}{r^2} - \right. \\
& \quad \left. \frac{2m^2 \sin[\theta]^2 (2d[\phi] \wedge d[r] \wedge d[\theta] + d[\phi] \wedge d[t] \wedge d[\theta])}{r^2} + \right. \\
& \quad \left. \cos[\theta] r \sin[\theta] d[\phi] \wedge d[\theta] \wedge d[\theta] \right) - \\
& \frac{1}{r} m \csc[\theta] \left(\frac{m \left(-((2m-r) r \sin[\theta]^2 d[r] \wedge d[\phi] \wedge d[\theta]) - 2m r \sin[\theta]^2 d[t] \wedge d[\phi] \wedge d[\theta]) \right)}{r^3} + \right.
\end{aligned}$$

$$\begin{aligned}
& \cos[\theta] r \sin[\theta] d[\theta] \wedge d[\phi] \wedge d[\theta] - \\
& \sin[\theta]^2 (d[\phi] \wedge d[r] \wedge d[\theta] + \cot[\theta] r d[\phi] \wedge d[\theta] \wedge d[\theta]) \Big) + \\
& \frac{1}{r} m \sin[\theta] \left(\frac{m (-((2m-r) r d[r] \wedge d[\theta] \wedge d[\phi]) - 2 m r d[t] \wedge d[\theta] \wedge d[\phi])}{r^3} - \right. \\
& \left. d[\theta] \wedge d[r] \wedge d[\phi] - \cos[\theta] r \sin[\theta] d[\phi] \wedge d[\phi] \wedge d[\phi] \right) - \\
& \frac{1}{r} m \sin[\theta] \left(d[r] \wedge d[\theta] \wedge d[\phi] - \frac{m (-2m+r) d[\theta] \wedge d[r] \wedge d[\phi]}{r^2} - \right. \\
& \left. \frac{2 m^2 (2 d[\theta] \wedge d[r] \wedge d[\phi] + d[\theta] \wedge d[t] \wedge d[\phi])}{r^2} - \cos[\theta] r \sin[\theta] d[\phi] \wedge d[\phi] \wedge d[\phi] \right), \\
S_2 = & - \frac{m \sin[\theta] d[r] \wedge d[t] \wedge d[\phi]}{2 r} - \frac{m \sin[\theta] (2 d[r] \wedge d[r] \wedge d[\phi] + d[r] \wedge d[t] \wedge d[\phi])}{2 r} - \\
& \frac{2 m \sin[\theta] d[r] \wedge d[\phi] \wedge d[r]}{r} - \\
& \left(-1 + \frac{2 m}{r} \right) \sin[\theta] d[r] \wedge d[\phi] \wedge d[t] - \frac{1}{2} r^2 \sin[\theta] \\
& \left(- \frac{m \left(- \frac{2 m^2 d[r] \wedge d[r] \wedge d[\phi]}{r^2} - \frac{m (2 m+r) d[t] \wedge d[r] \wedge d[\phi]}{r^2} \right)}{r^3} + \frac{m \left(\frac{m (2 m-r) d[r] \wedge d[t] \wedge d[\phi]}{r^2} + \frac{2 m^2 d[t] \wedge d[t] \wedge d[\phi]}{r^2} \right)}{r^3} \right) + \\
& \frac{1}{2} r^2 \sin[\theta] \left(\frac{m \left(- \frac{2 m (m-r) d[r] \wedge d[r] \wedge d[\phi]}{r^2} - \frac{m (2 m-r) d[t] \wedge d[r] \wedge d[\phi]}{r^2} \right)}{r^3} + \right. \\
& \left. \frac{m \left(- \frac{m (-2 m+r) (2 d[r] \wedge d[r] \wedge d[\phi] + d[r] \wedge d[t] \wedge d[\phi])}{r^2} + \frac{2 m^2 (2 d[t] \wedge d[r] \wedge d[\phi] + d[t] \wedge d[t] \wedge d[\phi])}{r^2} \right)}{r^3} \right) - \\
& \frac{1}{2 r^3} m \csc[\theta] \left(1 + \frac{2 m}{r} \right) (-2 (m-r) r \sin[\theta]^2 d[r] \wedge d[\phi] \wedge d[r] - \\
& (2 m-r) r \sin[\theta]^2 d[t] \wedge d[\phi] \wedge d[r]) - \\
& \frac{m^2 \csc[\theta] (-2 (m-r) r \sin[\theta]^2 d[r] \wedge d[\phi] \wedge d[t] - (2 m-r) r \sin[\theta]^2 d[t] \wedge d[\phi] \wedge d[t])}{r^4} - \\
& 2 m \cos[\theta] d[\theta] \wedge d[\phi] \wedge d[r] - \\
& \cos[\theta] \left(-1 + \frac{2 m}{r} \right) r d[\theta] \wedge d[\phi] \wedge d[t] + \frac{1}{2} \csc[\theta] \left(1 + \frac{2 m}{r} \right) \\
& \left(- \frac{2 m^2 \sin[\theta]^2 d[\phi] \wedge d[r] \wedge d[r]}{r^2} - \frac{m (2 m-r) \sin[\theta]^2 d[\phi] \wedge d[t] \wedge d[r]}{r^2} \right) + \\
& \frac{m \csc[\theta] \left(- \frac{2 m^2 \sin[\theta]^2 d[\phi] \wedge d[r] \wedge d[t]}{r^2} - \frac{m (2 m-r) \sin[\theta]^2 d[\phi] \wedge d[t] \wedge d[t]}{r^2} \right)}{r} + \\
& \frac{1}{2} m \cos[\theta] d[\phi] \wedge d[t] \wedge d[\theta] + \\
& \frac{1}{2} m \cos[\theta] (2 d[\phi] \wedge d[r] \wedge d[\theta] + d[\phi] \wedge d[t] \wedge d[\theta]) + \\
& 2 m \cos[\theta] d[\phi] \wedge d[\theta] \wedge d[r] -
\end{aligned}$$

$$\begin{aligned}
& \frac{1}{r} m \operatorname{Csc}[\theta] \left(\operatorname{Sin}[\theta]^2 d[r] \wedge d[\phi] \wedge d[r] + \right. \\
& \quad \operatorname{Cos}[\theta] r \operatorname{Sin}[\theta] d[\theta] \wedge d[\phi] \wedge d[r] + \frac{m(2m-r) \operatorname{Sin}[\theta]^2 d[\phi] \wedge d[r] \wedge d[r]}{r^2} - \\
& \quad \frac{2m^2 \operatorname{Sin}[\theta]^2 (2d[\phi] \wedge d[r] \wedge d[r] + d[\phi] \wedge d[t] \wedge d[r])}{r^2} + \\
& \quad \left. \operatorname{Cos}[\theta] r \operatorname{Sin}[\theta] d[\phi] \wedge d[\theta] \wedge d[r] \right) + \\
& \frac{1}{r} m \operatorname{Csc}[\theta] \left(\frac{m \left(- \left((2m-r) r \operatorname{Sin}[\theta]^2 d[r] \wedge d[\phi] \wedge d[r] \right) - 2mr \operatorname{Sin}[\theta]^2 d[t] \wedge d[\phi] \wedge d[r] \right)}{r^3} + \right. \\
& \quad \operatorname{Cos}[\theta] r \operatorname{Sin}[\theta] d[\theta] \wedge d[\phi] \wedge d[r] - \\
& \quad \left. \operatorname{Sin}[\theta]^2 (d[\phi] \wedge d[r] \wedge d[r] + \operatorname{Cot}[\theta] r d[\phi] \wedge d[\theta] \wedge d[r]) \right) + \\
& \operatorname{Cos}[\theta] \left(-1 + \frac{2m}{r} \right) r d[\phi] \wedge d[\theta] \wedge d[t] - \frac{1}{2} \operatorname{Csc}[\theta] \left(-1 + \frac{2m}{r} \right) \left(\operatorname{Sin}[\theta]^2 d[r] \wedge d[\phi] \wedge d[t] + \right. \\
& \quad \operatorname{Cos}[\theta] r \operatorname{Sin}[\theta] d[\theta] \wedge d[\phi] \wedge d[t] + \frac{m(2m-r) \operatorname{Sin}[\theta]^2 d[\phi] \wedge d[r] \wedge d[t]}{r^2} - \\
& \quad \frac{2m^2 \operatorname{Sin}[\theta]^2 (2d[\phi] \wedge d[r] \wedge d[t] + d[\phi] \wedge d[t] \wedge d[t])}{r^2} + \\
& \quad \left. \operatorname{Cos}[\theta] r \operatorname{Sin}[\theta] d[\phi] \wedge d[\theta] \wedge d[t] \right) + \frac{1}{2} \operatorname{Csc}[\theta] \left(-1 + \frac{2m}{r} \right) \\
& \quad \left(\frac{m \left(- \left((2m-r) r \operatorname{Sin}[\theta]^2 d[r] \wedge d[\phi] \wedge d[t] \right) - 2mr \operatorname{Sin}[\theta]^2 d[t] \wedge d[\phi] \wedge d[t] \right)}{r^3} + \operatorname{Cos}[\theta] r \right. \\
& \quad \left. \operatorname{Sin}[\theta] d[\theta] \wedge d[\phi] \wedge d[t] - \operatorname{Sin}[\theta]^2 (d[\phi] \wedge d[r] \wedge d[t] + \operatorname{Cot}[\theta] r d[\phi] \wedge d[\theta] \wedge d[t]) \right), \\
S_3 = & \frac{1}{2} r^2 \operatorname{Sin}[\theta] \left(- \frac{m \left(- \frac{2m^2 d[r] \wedge d[r] \wedge d[\theta]}{r^2} - \frac{m(2m+r) d[t] \wedge d[r] \wedge d[\theta]}{r^2} \right)}{r^3} + \right. \\
& \left. \frac{m \left(\frac{m(2m-r) d[r] \wedge d[t] \wedge d[\theta]}{r^2} + \frac{2m^2 d[t] \wedge d[t] \wedge d[\theta]}{r^2} \right)}{r^3} \right) - \\
& \frac{1}{2} r^2 \operatorname{Sin}[\theta] \left(\frac{m \left(- \frac{2m(m-r) d[r] \wedge d[r] \wedge d[\theta]}{r^2} - \frac{m(2m-r) d[t] \wedge d[r] \wedge d[\theta]}{r^2} \right)}{r^3} + \right. \\
& \left. \frac{m \left(- \frac{m(-2m+r) (2d[r] \wedge d[r] \wedge d[\theta] + d[r] \wedge d[t] \wedge d[\theta])}{r^2} + \frac{2m^2 (2d[t] \wedge d[r] \wedge d[\theta] + d[t] \wedge d[t] \wedge d[\theta])}{r^2} \right)}{r^3} \right) + \\
& \frac{m \left(1 + \frac{2m}{r} \right) \operatorname{Sin}[\theta] (-2(m-r) r d[r] \wedge d[\theta] \wedge d[r] - (2m-r) r d[t] \wedge d[\theta] \wedge d[r])}{2r^3} + \\
& \frac{m^2 \operatorname{Sin}[\theta] (-2(m-r) r d[r] \wedge d[\theta] \wedge d[t] - (2m-r) r d[t] \wedge d[\theta] \wedge d[t])}{r^4} -
\end{aligned}$$

$$\begin{aligned}
& \frac{1}{2} \left(1 + \frac{2m}{r} \right) \sin[\theta] \left(-\frac{2m^2 d[\theta] \wedge d[r] \wedge d[r]}{r^2} + \frac{m(-2m+r) d[\theta] \wedge d[t] \wedge d[r]}{r^2} \right) - \\
& \frac{m \sin[\theta] \left(-\frac{2m^2 d[\theta] \wedge d[r] \wedge d[t]}{r^2} + \frac{m(-2m+r) d[\theta] \wedge d[t] \wedge d[t]}{r^2} \right)}{r} + \\
& \frac{1}{2} r \sin[\theta] \left(\frac{m d[r] \wedge d[t] \wedge d[\theta]}{r^2} + \frac{m \cot[\theta] d[\theta] \wedge d[t] \wedge d[\theta]}{r} \right) - \\
& \frac{1}{2} r \sin[\theta] \left(-\frac{m(2d[r] \wedge d[r] \wedge d[\theta] + d[r] \wedge d[t] \wedge d[\theta])}{r^2} - \right. \\
& \quad \left. \frac{m \cot[\theta] (2d[\theta] \wedge d[r] \wedge d[\theta] + d[\theta] \wedge d[t] \wedge d[\theta])}{r} \right) - \\
& \frac{m \sin[\theta] (-r d[r] \wedge d[\theta] \wedge d[r] - \cot[\theta] r^2 d[\theta] \wedge d[\theta] \wedge d[r])}{r^2} + \\
& \frac{m \sin[\theta] (r d[r] \wedge d[\theta] \wedge d[r] + \cot[\theta] r^2 d[\theta] \wedge d[\theta] \wedge d[r])}{r^2} - \\
& \frac{(-1 + \frac{2m}{r}) \sin[\theta] (-r d[r] \wedge d[\theta] \wedge d[t] - \cot[\theta] r^2 d[\theta] \wedge d[\theta] \wedge d[t])}{2r} + \\
& \frac{(-1 + \frac{2m}{r}) \sin[\theta] (r d[r] \wedge d[\theta] \wedge d[t] + \cot[\theta] r^2 d[\theta] \wedge d[\theta] \wedge d[t])}{2r} + \\
& \frac{1}{2} m \cos[\theta] \sin[\theta]^2 d[\phi] \wedge d[t] \wedge d[\phi] + \\
& \frac{1}{2} m \cos[\theta] \sin[\theta]^2 (2d[\phi] \wedge d[r] \wedge d[\phi] + d[\phi] \wedge d[t] \wedge d[\phi]) - \\
& \frac{1}{r} m \sin[\theta] \left(\frac{m(-((2m-r) r d[r] \wedge d[\theta] \wedge d[r]) - 2m r d[t] \wedge d[\theta] \wedge d[r])}{r^3} - \right. \\
& \quad \left. d[\theta] \wedge d[r] \wedge d[r] - \cos[\theta] r \sin[\theta] d[\phi] \wedge d[\phi] \wedge d[r] \right) + \\
& \frac{1}{r} m \sin[\theta] \left(d[r] \wedge d[\theta] \wedge d[r] - \frac{m(-2m+r) d[\theta] \wedge d[r] \wedge d[r]}{r^2} - \right. \\
& \quad \left. \frac{2m^2 (2d[\theta] \wedge d[r] \wedge d[r] + d[\theta] \wedge d[t] \wedge d[r])}{r^2} - \cos[\theta] r \sin[\theta] d[\phi] \wedge d[\phi] \wedge d[r] \right) - \\
& \frac{1}{2} \left(-1 + \frac{2m}{r} \right) \sin[\theta] \left(\frac{m(-((2m-r) r d[r] \wedge d[\theta] \wedge d[t]) - 2m r d[t] \wedge d[\theta] \wedge d[t])}{r^3} - \right. \\
& \quad \left. d[\theta] \wedge d[r] \wedge d[t] - \cos[\theta] r \sin[\theta] d[\phi] \wedge d[\phi] \wedge d[t] \right) + \\
& \frac{1}{2} \left(-1 + \frac{2m}{r} \right) \sin[\theta] \left(d[r] \wedge d[\theta] \wedge d[t] - \frac{m(-2m+r) d[\theta] \wedge d[r] \wedge d[t]}{r^2} - \right. \\
& \quad \left. \frac{2m^2 (2d[\theta] \wedge d[r] \wedge d[t] + d[\theta] \wedge d[t] \wedge d[t])}{r^2} - \cos[\theta] r \sin[\theta] d[\phi] \wedge d[\phi] \wedge d[t] \right) \Big\}
\end{aligned}$$

In[*]:= Map[ToCanonical, %, {2}]

[aplica a todos](#)

```
Out[*]= { S0 == Sin[θ] d[r] ∧ d[θ] ∧ d[φ], S1 == Sin[θ] d[t] ∧ d[θ] ∧ d[φ],
          S2 == -Sin[θ] d[r] ∧ d[t] ∧ d[φ] - m Cos[θ] d[r] ∧ d[θ] ∧ d[φ] -
          m Cos[θ] d[t] ∧ d[θ] ∧ d[φ] + Cos[θ] r d[t] ∧ d[θ] ∧ d[φ], S3 == Sin[θ] d[r] ∧ d[t] ∧ d[θ] }
```

```
In[*]:= Map[Simplify, %, {2}]
[apl]· [simplifica]
```

```
Out[*]= { S0 == Sin[θ] d[r] ∧ d[θ] ∧ d[φ], S1 == Sin[θ] d[t] ∧ d[θ] ∧ d[φ],
          S2 == -Sin[θ] d[r] ∧ d[t] ∧ d[φ] - Cos[θ] (m d[r] ∧ d[θ] ∧ d[φ] + (m - r) d[t] ∧ d[θ] ∧ d[φ]),
          S3 == Sin[θ] d[r] ∧ d[t] ∧ d[θ] }
```

```
In[*]:= SparlingEF = %;
```

```
In[*]:= % /. {r[] → 2 mass}
```

```
Out[*]= { S0 == 0, S1 == Sin[θ] d[t] ∧ d[θ] ∧ d[φ], S2 == m Cos[θ] d[t] ∧ d[θ] ∧ d[φ], S3 == 0 }
```

```
In[*]:= SparlingHorizon = %;
```

```
In[*]:= Hodge[metricTensor] /@ SparlingEF
```

```
Out[*]= { * [S0] == Sin[θ] (* [d[r] ∧ d[θ] ∧ d[φ]]), * [S1] ==
          Sin[θ] (* [d[t] ∧ d[θ] ∧ d[φ]]), * [S2] == -Sin[θ] (* [d[r] ∧ d[t] ∧ d[φ]]) -
          Cos[θ] (m (* [d[r] ∧ d[θ] ∧ d[φ]]) + (m - r) (* [d[t] ∧ d[θ] ∧ d[φ]])), * [S3] ==
          Sin[θ] (* [d[r] ∧ d[t] ∧ d[θ]]) }
```

```
In[*]:= Map[ExpandHodgeDual[#, dx[M4], metricTensor] &, %, {2}]
[aplica a todos]
```

```
Out[*]= { * [S0] == Sin[θ]  $\left( \frac{(2m - r) dx^0}{r \sqrt{r^4 \sin^2[\theta]}} + \frac{2m dx^1}{r \sqrt{r^4 \sin^2[\theta]}} \right)$ , * [S1] ==
          Sin[θ]  $\left( -\frac{2m dx^0}{r \sqrt{r^4 \sin^2[\theta]}} - \frac{(2m + r) dx^1}{r \sqrt{r^4 \sin^2[\theta]}} \right)$ , * [S2] ==
          -Cos[θ]  $\left( m \left( \frac{(2m - r) dx^0}{r \sqrt{r^4 \sin^2[\theta]}} + \frac{2m dx^1}{r \sqrt{r^4 \sin^2[\theta]}} \right) + (m - r) \left( -\frac{2m dx^0}{r \sqrt{r^4 \sin^2[\theta]}} - \frac{(2m + r) dx^1}{r \sqrt{r^4 \sin^2[\theta]}} \right) \right) +$ 
           $\frac{r^2 \sin[\theta] dx^2}{\sqrt{r^4 \sin^2[\theta]}}$ , * [S3] ==  $\frac{\sin[\theta] \sqrt{r^4 \sin^2[\theta]} dx^3}{r^2} \}$ 
```

In[*]:= Map[PowerExpand, %, {2}]

[\[apl\] · \[expande potencias\]](#)

Out[*]=

$$\left\{ * [S_0] = \sin[\theta] \left(\frac{\csc[\theta] (2m-r) dx^0}{r^3} + \frac{2m \csc[\theta] dx^1}{r^3} \right), * [S_1] = \sin[\theta] \left(-\frac{2m \csc[\theta] dx^0}{r^3} - \frac{\csc[\theta] (2m+r) dx^1}{r^3} \right), * [S_2] = -\cos[\theta] \left(m \left(\frac{\csc[\theta] (2m-r) dx^0}{r^3} + \frac{2m \csc[\theta] dx^1}{r^3} \right) + (m-r) \left(-\frac{2m \csc[\theta] dx^0}{r^3} - \frac{\csc[\theta] (2m+r) dx^1}{r^3} \right) \right) + dx^2, * [S_3] = \sin[\theta]^2 dx^3 \right\}$$

In[*]:= ToValues // @ %

Out[*]=

$$\left\{ * [S_0] = \left(\frac{2m \csc[\theta] d[r]}{r^3} + \frac{\csc[\theta] d[t] (2m-r)}{r^3} \right) \sin[\theta], * [S_1] = \left(-\frac{2m \csc[\theta] d[t]}{r^3} - \frac{\csc[\theta] d[r] (2m+r)}{r^3} \right) \sin[\theta], * [S_2] = d[\theta] - \cos[\theta] \left(m \left(\frac{2m \csc[\theta] d[r]}{r^3} + \frac{\csc[\theta] d[t] (2m-r)}{r^3} \right) + (m-r) \left(-\frac{2m \csc[\theta] d[t]}{r^3} - \frac{\csc[\theta] d[r] (2m+r)}{r^3} \right) \right), * [S_3] = d[\phi] \sin[\theta]^2 \right\}$$

In[*]:= Map[FullSimplify, %, {2}]

[\[apl\] · \[simplifica completamente\]](#)

Out[*]=

$$\left\{ * [S_0] = \frac{2m (d[r] + d[t]) - d[t] r}{r^3}, * [S_1] = -\frac{2m d[t] + d[r] (2m+r)}{r^3}, * [S_2] = d[\theta] - \frac{\cot[\theta] (m d[t] + d[r] (m+r))}{r^2}, * [S_3] = d[\phi] \sin[\theta]^2 \right\}$$

In[*]:= % /. {r[] → 2 mass}

Out[*]=

$$\left\{ * [S_0] = 0, * [S_1] = -\frac{d[t]}{4 m^2}, * [S_2] = -\frac{\cot[\theta] d[t]}{4 m} + d[\theta], * [S_3] = d[\phi] \sin[\theta]^2 \right\}$$

In[*]:= SparlingHorizonDual = %;

Define observer:

In[*]:= DefTensor[u[a], M4]

In[*]:= Hodge[metricTensor][S[-a]] × u[a]

Out[*]=

$$u^a (* [S_a])$$

In[*]:= ToBasis[KS][%]

Out[*]=

$$u^a (* [S_a])$$

```

In[ ]:= TraceBasisDummy[%]
Out[ ]:=

$$u^0 \left( * [S_0] \right) + u^1 \left( * [S_1] \right) + u^2 \left( * [S_2] \right) + u^3 \left( * [S_3] \right)$$


In[ ]:= % /. {r[] → 2 mass}
Out[ ]:=

$$u^0 \left( * [S_0] \right) + u^1 \left( * [S_1] \right) + u^2 \left( * [S_2] \right) + u^3 \left( * [S_3] \right)$$


In[ ]:= % /. (SparlingHorizonDual /. Equal → Rule)
Out[ ]:=

$$-\frac{d[t] u^1}{4 m^2} + \left( -\frac{\text{Cot}[\theta] d[t]}{4 m} + d[\theta] \right) u^2 + d[\phi] \text{Sin}[\theta]^2 u^3$$


In[ ]:= Collect[%, _Diff, Simplify]
Out[ ]:=

$$d[\theta] u^2 - \frac{d[t] \left( u^1 + m \text{Cot}[\theta] u^2 \right)}{4 m^2} + d[\phi] \text{Sin}[\theta]^2 u^3$$


In[ ]:= CoefficientArrays[%, {Diff[t[]], Diff[r[]], Diff[\theta[]], Diff[\phi[]]}]
Out[ ]:=

$$\{0, \text{SparseArray}\left[\begin{array}{c} \text{Specified elements: 3} \\ \text{Dimensions: } \{4\} \end{array}\right]\}$$


In[ ]:= Last@% // Normal
Out[ ]:=

$$\left\{ -\frac{u^1 + m \text{Cot}[\theta] u^2}{4 m^2}, \theta, u^2, \text{Sin}[\theta]^2 u^3 \right\}$$


In[ ]:= CTensor[%, {-KS}]
Out[ ]:=

$$\text{CTensor}\left[\left\{ -\frac{u^1 + m \text{Cot}[\theta] u^2}{4 m^2}, \theta, u^2, \text{Sin}[\theta]^2 u^3 \right\}, \{-KS\}, \theta\right]$$


In[ ]:= flux = %;
In[ ]:= flux[-b] × metricTensor[a, b] × flux[-a]
Out[ ]:=

$$\frac{u^{2^2}}{r^2} + \frac{\left(-1 - \frac{2m}{r}\right) \left(u^1 + m \text{Cot}[\theta] u^2\right)^2}{16 m^4} + \frac{\text{Sin}[\theta]^2 u^{3^2}}{r^2}$$


In[ ]:= % /. {r[] → 2 mass}
Out[ ]:=

$$\frac{u^{2^2}}{4 m^2} - \frac{\left(u^1 + m \text{Cot}[\theta] u^2\right)^2}{8 m^4} + \frac{\text{Sin}[\theta]^2 u^{3^2}}{4 m^2}$$


In[ ]:= FullSimplify[%]
Out[ ]:=

$$\frac{2 m^2 u^{2^2} - \left(u^1 + m \text{Cot}[\theta] u^2\right)^2 + 2 m^2 \text{Sin}[\theta]^2 u^{3^2}}{8 m^4}$$


```


In[]:= **ToBasis**[KS][u[a] × **metricTensor**[-a, -b] × u[b]]

Out[]:=

$$-e_a^0 e_b^0 u^a u^b + e_a^1 e_b^1 u^a u^b + \frac{2m e_a^0 e_b^0 u^a u^b}{r} + \frac{2m e_a^1 e_b^0 u^a u^b}{r} + \frac{2m e_a^0 e_b^1 u^a u^b}{r} + \frac{2m e_a^1 e_b^1 u^a u^b}{r} + e_a^2 e_b^2 r^2 u^a u^b + e_a^3 e_b^3 r^2 \sin[\theta]^2 u^a u^b$$

In[]:= **ToValues@TraceBasisDummy@%**

Out[]:=

$$-u^0{}^2 + \frac{2m u^0{}^2}{r} + \frac{4m u^0 u^1}{r} + u^1{}^2 + \frac{2m u^1{}^2}{r} + r^2 u^2{}^2 + r^2 \sin[\theta]^2 u^3{}^2$$

In[]:= **FullSimplify@%**

[simplifica completamente](#)

Out[]:=

$$\frac{(2m - r) u^0{}^2 + 4m u^0 u^1 + (2m + r) u^1{}^2 + r^3 (u^2{}^2 + \sin[\theta]^2 u^3{}^2)}{r}$$

In[]:= **% /. {r[] → 2 mass}**

Out[]:=

$$\frac{4m u^0 u^1 + 4m u^1{}^2 + 8m^3 (u^2{}^2 + \sin[\theta]^2 u^3{}^2)}{2m}$$

In[]:= **x == (r - 1) E^r**

[núme](#)

Out[]:=

$$x == e^r (-1 + r)$$

In[]:= **Reduce**[% , r, Reals]

[reduce](#)

Out[]:=

$$(x == -1 \&\& r == 0) \mid \mid (x == 0 \&\& r == 1) \mid \mid \left(-1 \leq x < 0 \&\& \left(r == 1 + \text{ProductLog}\left[-1, \frac{x}{e}\right] \mid \mid r == 1 + \text{ProductLog}\left[\frac{x}{e}\right] \right) \right) \mid \mid \left(x > 0 \&\& r == 1 + \text{ProductLog}\left[\frac{x}{e}\right] \right)$$

In[]:= **? ProductLog**

[producto logaritmo](#)

Out[]:=

Symbol i

ProductLog[z] gives the principal solution for w in z = we^w.

ProductLog[k, z] gives the kth solution.

Documentation [Local »](#) | [Web »](#)

Attributes {Listable, Protected, ReadProtected}

Full Name System`ProductLog

^

R

In[]:= ? Reduce

reduce

Out[]:=

Symbol i

Reduce[*expr*, *vars*] reduces the statement *expr* by solving equations or inequalities for *vars* and eliminating quantifiers.

Reduce[*expr*, *vars*, *dom*] does the reduction over the domain *dom*. Common choices of *dom* are Reals, Integers, and Complexes.

Documentation [Local »](#) | [Web »](#)

Options ▶ Backsubstitution → False ... (7 total)

Attributes {Protected}

Full Name System`Reduce

^

In[]:= ? ProductLog

producto logaritmo

Out[]:=

Symbol i

ProductLog[z] gives the principal solution for *w* in $z = we^w$.

ProductLog[k, z] gives the k^{th} solution.

▼

Schwarzschild in Kruskal coordinates

In[]:= SparlingForm

Out[]:=

$$S_a = -\frac{1}{2} \eta^{bcde} \Gamma[\nabla]_{ba} \wedge \Gamma[\nabla]_{cd} \wedge \Theta_e - \frac{1}{2} \eta_a^{bcd} \Gamma[\nabla]_b^e \wedge \Gamma[\nabla]_{ec} \wedge \Theta_d$$

In[]:= % /. MakeRule[{Coframe[M4][a], Coframe[M4][a]}, MetricOn → All, ContractMetrics → False]

todo

false

Out[]:=

$$S_a = -\frac{1}{2} \eta^{bcde} g_{fe} \Gamma[\nabla]_{ba} \wedge \Gamma[\nabla]_{cd} \wedge \Theta^f - \frac{1}{2} \eta_a^{bcd} g_{fd} \Gamma[\nabla]_b^e \wedge \Gamma[\nabla]_{ec} \wedge \Theta^f$$

In[]:= SparlingFormUp = %

Out[]:=

$$S_a = -\frac{1}{2} \eta^{bcde} g_{fe} \Gamma[\nabla]_{ba} \wedge \Gamma[\nabla]_{cd} \wedge \Theta^f - \frac{1}{2} \eta_a^{bcd} g_{fd} \Gamma[\nabla]_b^e \wedge \Gamma[\nabla]_{ec} \wedge \Theta^f$$

```

In[*]:= << xAct`xIdeal`

-----

Package xAct`xIdeal` version 0.1.0, {2025, 10, 29}

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it under certain conditions. See the General Public License for details.

-----

In[*]:= coords = "KruskalSzekeresCoordinates";

In[*]:= sol = "Schwarzschild";

In[*]:= xActSolutionData[{sol, coords}, "Properties"]
      |propiedades

Out[*]=
{CoordinateAssumptions, CoordinateNames, Metric, ParameterNames, ParameterAssumptions,
ParameterValues, ScalarFunctionNames, ScalarFunctionValues, SignatureOfMetric}

In[*]:= xActSolutionData[{sol, coords}, "CoordinateNames"]
Out[*]=
{T, X,  $\theta$ ,  $\phi$ }

In[*]:= xActSolutionData[{sol, coords}, "SignatureOfMetric"]
Out[*]=
{3, 1, 0}

In[*]:= $DefInfoQ = False;
      |false

In[*]:= $CVVerbose = False;
      |false

In[*]:= DefChart[KS, M4, {0, 1, 2, 3}, {T[], X[],  $\theta$ [],  $\phi$ []}]

In[*]:= xActSolutionData[{sol, coords}, "ParameterNames"]
Out[*]=
{m}

In[*]:= DefConstantSymbol[mass, PrintAs  $\rightarrow$  "m"];

In[*]:= xActSolutionData[{sol, coords}, "ScalarFunctionNames"]
Out[*]=
{R}

In[*]:= DefScalarFunction[R]

In[*]:= xActSolutionData[{sol, coords}, "ScalarFunctionValues"] [
      {T[], X[],  $\theta$ [],  $\phi$ []}, {mass}, {R}]

Out[*]=
{Function[{xAct`xIdeal`Private`T$, xAct`xIdeal`Private`X$},
R[xAct`xIdeal`Private`T$, xAct`xIdeal`Private`X$]]}

```

```

In[ ]:= xActSolutionData[{sol, coords}, "ParameterAssumptions"]
Out[ ]:=
Function[{xAct`xIdeal`Private`coords,
  xAct`xIdeal`Private`params, xAct`xIdeal`Private`scfuncs}, With[
  {xAct`xIdeal`Private`m = xAct`xIdeal`Private`params[[1]], xAct`xIdeal`Private`m > 0}]

In[ ]:= metricTensor = FullSimplify@xActSolutionData[{sol, coords → KS,
  |simplifica completamente
  "ScalarFunctionValues" → {R}, "ParameterValues" → {mass, 1}}, "Metric"]
Out[ ]:=
CTensor[{{{- $\frac{32 e^{-\frac{R[T, X]}{2 m}} m^3}{R[T, X]}$ , 0, 0, 0}, {0,  $\frac{32 e^{-\frac{R[T, X]}{2 m}} m^3}{R[T, X]}$ , 0, 0},
  {0, 0,  $R[T, X]^2$ , 0}, {0, 0, 0,  $R[T, X]^2 \sin[\theta]^2$ }}, {-KS, -KS}, 0]

In[ ]:= xActSolutionData[{sol, coords}, "CoordinateAssumptions"][{T, X,  $\theta$ ,  $\phi$ }, {mass}, {R}]
Out[ ]:=
 $-T^2 + X^2 == e^{\frac{R[T, X]}{2 m}} \left( -1 + \frac{R[T, X]}{2 m} \right) \&\& \pi > \theta > 0 \&\& 0 < \phi < 2 \pi$ 

In[ ]:= First[%]
|primero
Out[ ]:=
 $-T^2 + X^2 == e^{\frac{R[T, X]}{2 m}} \left( -1 + \frac{R[T, X]}{2 m} \right)$ 

In[ ]:= Reduce[%, R[T, X], Reals]
|reduce
1.08967
Out[ ]:=

$$\left( m \neq 0 \&\& \left( T < 0 \&\& \left( X == -\sqrt{T^2} \mid \mid X == \sqrt{T^2} \right) \right) \mid \mid \right.$$


$$\left. (T == 0 \&\& X == 0) \mid \mid (T > 0 \&\& \left( X == -\sqrt{T^2} \mid \mid X == \sqrt{T^2} \right)) \right) \&\& R[T, X] == 2 m \mid \mid$$


$$\left( m \neq 0 \&\& T < -1 \&\& X < -\sqrt{T^2} \&\& R[T, X] == 2 m + 2 m \text{ProductLog}\left[\frac{-T^2 + X^2}{e}\right] \right) \mid \mid$$


$$\left( m \neq 0 \&\& T < -1 \&\& -\sqrt{T^2} < X \leq -\sqrt{-1 + T^2} \&\& R[T, X] == 2 m + 2 m \text{ProductLog}\left[-1, \frac{-T^2 + X^2}{e}\right] \right) \mid \mid$$


$$\left( m \neq 0 \&\& T < -1 \&\& -\sqrt{T^2} < X \leq -\sqrt{-1 + T^2} \&\& R[T, X] == 2 m + 2 m \text{ProductLog}\left[\frac{-T^2 + X^2}{e}\right] \right) \mid \mid$$


$$\left( m \neq 0 \&\& T < -1 \&\& \sqrt{-1 + T^2} \leq X < \sqrt{T^2} \&\& R[T, X] == 2 m + 2 m \text{ProductLog}\left[-1, \frac{-T^2 + X^2}{e}\right] \right) \mid \mid$$


$$\left( m \neq 0 \&\& T < -1 \&\& \sqrt{-1 + T^2} \leq X < \sqrt{T^2} \&\& R[T, X] == 2 m + 2 m \text{ProductLog}\left[\frac{-T^2 + X^2}{e}\right] \right) \mid \mid$$


$$\left( m \neq 0 \&\& T < -1 \&\& X > \sqrt{T^2} \&\& R[T, X] == 2 m + 2 m \text{ProductLog}\left[\frac{-T^2 + X^2}{e}\right] \right) \mid \mid$$


$$\left( m \neq 0 \&\& -1 \leq T \leq 1 \&\& X < -\sqrt{T^2} \&\& R[T, X] == 2 m + 2 m \text{ProductLog}\left[\frac{-T^2 + X^2}{e}\right] \right) \mid \mid$$


$$\left( m < 0 \&\& -1 \leq T < 0 \&\& -\sqrt{T^2} < X < \sqrt{T^2} \&\& R[T, X] == 2 m + 2 m \text{ProductLog}\left[-1, \frac{-T^2 + X^2}{e}\right] \right) \mid \mid$$


```

$$\begin{aligned}
& \left(m < 0 \ \&\& \ 0 < T \leq 1 \ \&\& \ -\sqrt{T^2} < X < \sqrt{T^2} \ \&\& \ R[T, X] == 2 m + 2 m \text{ProductLog}\left[-1, \frac{-T^2 + X^2}{e}\right] \right) || \\
& \left(m > 0 \ \&\& \ -1 \leq T < 0 \ \&\& \ -\sqrt{T^2} < X < \sqrt{T^2} \ \&\& \ R[T, X] == 2 m + 2 m \text{ProductLog}\left[-1, \frac{-T^2 + X^2}{e}\right] \right) || \\
& \left(m > 0 \ \&\& \ 0 < T \leq 1 \ \&\& \ -\sqrt{T^2} < X < \sqrt{T^2} \ \&\& \ R[T, X] == 2 m + 2 m \text{ProductLog}\left[-1, \frac{-T^2 + X^2}{e}\right] \right) || \\
& \left(m < 0 \ \&\& \ -1 \leq T < 0 \ \&\& \ -\sqrt{T^2} < X < \sqrt{T^2} \ \&\& \ R[T, X] == 2 m + 2 m \text{ProductLog}\left[\frac{-T^2 + X^2}{e}\right] \right) || \\
& \left(m < 0 \ \&\& \ 0 < T \leq 1 \ \&\& \ -\sqrt{T^2} < X < \sqrt{T^2} \ \&\& \ R[T, X] == 2 m + 2 m \text{ProductLog}\left[\frac{-T^2 + X^2}{e}\right] \right) || \\
& \left(m > 0 \ \&\& \ -1 \leq T < 0 \ \&\& \ -\sqrt{T^2} < X < \sqrt{T^2} \ \&\& \ R[T, X] == 2 m + 2 m \text{ProductLog}\left[\frac{-T^2 + X^2}{e}\right] \right) || \\
& \left(m > 0 \ \&\& \ 0 < T \leq 1 \ \&\& \ -\sqrt{T^2} < X < \sqrt{T^2} \ \&\& \ R[T, X] == 2 m + 2 m \text{ProductLog}\left[\frac{-T^2 + X^2}{e}\right] \right) || \\
& \left(m \neq 0 \ \&\& \ -1 \leq T \leq 1 \ \&\& \ X > \sqrt{T^2} \ \&\& \ R[T, X] == 2 m + 2 m \text{ProductLog}\left[\frac{-T^2 + X^2}{e}\right] \right) || \\
& \left(m \neq 0 \ \&\& \ T > 1 \ \&\& \ X < -\sqrt{T^2} \ \&\& \ R[T, X] == 2 m + 2 m \text{ProductLog}\left[\frac{-T^2 + X^2}{e}\right] \right) || \\
& \left(m \neq 0 \ \&\& \ T > 1 \ \&\& \ -\sqrt{T^2} < X \leq -\sqrt{-1 + T^2} \ \&\& \ R[T, X] == 2 m + 2 m \text{ProductLog}\left[-1, \frac{-T^2 + X^2}{e}\right] \right) || \\
& \left(m \neq 0 \ \&\& \ T > 1 \ \&\& \ -\sqrt{T^2} < X \leq -\sqrt{-1 + T^2} \ \&\& \ R[T, X] == 2 m + 2 m \text{ProductLog}\left[\frac{-T^2 + X^2}{e}\right] \right) || \\
& \left(m \neq 0 \ \&\& \ T > 1 \ \&\& \ \sqrt{-1 + T^2} \leq X < \sqrt{T^2} \ \&\& \ R[T, X] == 2 m + 2 m \text{ProductLog}\left[-1, \frac{-T^2 + X^2}{e}\right] \right) || \\
& \left(m \neq 0 \ \&\& \ T > 1 \ \&\& \ \sqrt{-1 + T^2} \leq X < \sqrt{T^2} \ \&\& \ R[T, X] == 2 m + 2 m \text{ProductLog}\left[\frac{-T^2 + X^2}{e}\right] \right) || \\
& \left(m \neq 0 \ \&\& \ T > 1 \ \&\& \ X > \sqrt{T^2} \ \&\& \ R[T, X] == 2 m + 2 m \text{ProductLog}\left[\frac{-T^2 + X^2}{e}\right] \right) || \\
& \left(m \neq 0 \ \&\& \ \left(\left(T \leq -1 \ \&\& \ \left(X == -\sqrt{-1 + T^2} \ || \ X == \sqrt{-1 + T^2} \right) \right) || \right. \right. \\
& \quad \left. \left. \left(T \geq 1 \ \&\& \ \left(X == -\sqrt{-1 + T^2} \ || \ X == \sqrt{-1 + T^2} \right) \right) \right) \ \&\& \ R[T, X] == 0 \right)
\end{aligned}$$

In[*]:= red = %;

We are interested in the BH interior, so we need this branch:

In[*]:= red[[1, 2, 3]]

Out[*]=

$$T \neq 0 \ \&\& \ -\sqrt{T^2} < X < \sqrt{T^2} \ \&\& \ R[T, X] == 2 m - 2 m \text{ProductLog}\left[e T^2 - e X^2\right]$$

```
In[*]:= metricTensor = FullSimplify@xActSolutionData[{sol, coords → KS, "ScalarFunctionValues" →
|simplifica completamente
{Function[{T, X}, 2 mass + 2 mass ProductLog[(-T^2 + X^2) / E]}],
|producto logaritmo |número e
"ParameterValues" → {mass, 1}}, "Metric"]
```

```
Out[*]:=
CTensor[{{{- 16 m^2
|e
1+ProductLog[-T^2+X^2] - T^2 + X^2
, 0, 0, 0},
{0, 16 m^2
|e
1+ProductLog[-T^2+X^2] - T^2 + X^2
, 0, 0}, {0, 0, 4 (m + m ProductLog[-T^2 + X^2 / e])^2, 0},
{0, 0, 0, 4 (m + m ProductLog[-T^2 + X^2 / e])^2 Sin[θ]^2}}, {-KS, -KS}, 0]
```

```
In[*]:= Unprotect[ProductLog];
|desprotege |producto logaritmo
ProductLog /: ScalarFunctionQ[ProductLog] = True;
|producto logaritmo |producto logari... |verdade
Protect[ProductLog];
|protege |producto logaritmo
```

```
In[*]:= UnsetCMetric[metricTensor]
```

```
In[*]:= SetCMetric[metricTensor, KS, SignatureOfMetric → {3, 1, 0}]
```

SetCMetric: There are already metrics {metricg} in vbundle TM4.

```
In[*]:= cd = CovDOfMetric[metricTensor];
```

This should work automatically:

```
In[*]:= $Metrics = Reverse[$Metrics];
|invierte orden
```

```
In[*]:= TangentM4 /: MetricsOfVBundle[TangentM4] = Reverse[MetricsOfVBundle[TangentM4]];
|invierte orden
```

```
In[*]:= MetricsOfVBundle[TangentM4]
```

```
Out[*]:=
CTensor[{{{- 16 m^2
|e
1+ProductLog[-T^2+X^2] - T^2 + X^2
, 0, 0, 0},
{0, 16 m^2
|e
1+ProductLog[-T^2+X^2] - T^2 + X^2
, 0, 0}, {0, 0, 4 (m + m ProductLog[-T^2 + X^2 / e])^2, 0},
{0, 0, 0, 4 (m + m ProductLog[-T^2 + X^2 / e])^2 Sin[θ]^2}}, {-KS, -KS}, 0, metricg}
```

```
In[*]:= Ricci[cd]
```

```
2.3986
```

```
Out[*]=
```

$$\text{CTensor}\left[\left\{\left\{\frac{1}{\left(1 + \text{ProductLog}\left[\frac{-T^2 + X^2}{e}\right]\right)^4}\right.\right.\right.$$

$$2\left(-\frac{2 \text{ProductLog}\left[\frac{-T^2 + X^2}{e}\right]}{T^2 - X^2} - \frac{1}{\left(e^{1 + \text{ProductLog}\left[\frac{-T^2 + X^2}{e}\right]} - T^2 + X^2\right)^2} \left(1 + \text{ProductLog}\left[\frac{-T^2 + X^2}{e}\right]\right)\right.$$

$$\left.\left.\left.\left(e^{1 + \text{ProductLog}\left[\frac{-T^2 + X^2}{e}\right]} \text{ProductLog}\left[\frac{-T^2 + X^2}{e}\right]^3 + 4 \text{ProductLog}\left[\frac{-T^2 + X^2}{e}\right] (T^2 - X^2) + \right.\right.\right.$$

$$\left.\left.\text{ProductLog}\left[\frac{-T^2 + X^2}{e}\right]^2 \left(4 e^{1 + \text{ProductLog}\left[\frac{-T^2 + X^2}{e}\right]} + T^2 - X^2\right) + \right.\right.$$

$$\left.\left.2 \left(e^{1 + \text{ProductLog}\left[\frac{-T^2 + X^2}{e}\right]} - T^2 + X^2\right)\right)\right], 0, 0, 0\right\},$$

$$\left\{0, \frac{1}{\left(1 + \text{ProductLog}\left[\frac{-T^2 + X^2}{e}\right]\right)^4} 2\left(\frac{2 \text{ProductLog}\left[\frac{-T^2 + X^2}{e}\right]}{T^2 - X^2} + \frac{1}{\left(e^{1 + \text{ProductLog}\left[\frac{-T^2 + X^2}{e}\right]} - T^2 + X^2\right)^2}\right.\right.$$

$$\left.\left.\left(1 + \text{ProductLog}\left[\frac{-T^2 + X^2}{e}\right]\right)\left(e^{1 + \text{ProductLog}\left[\frac{-T^2 + X^2}{e}\right]} \text{ProductLog}\left[\frac{-T^2 + X^2}{e}\right]^3 + \right.\right.$$

$$\left.\left.4 \text{ProductLog}\left[\frac{-T^2 + X^2}{e}\right] (T^2 - X^2) + \text{ProductLog}\left[\frac{-T^2 + X^2}{e}\right]^2 \right.\right.$$

$$\left.\left.\left(4 e^{1 + \text{ProductLog}\left[\frac{-T^2 + X^2}{e}\right]} + T^2 - X^2\right) + 2 \left(e^{1 + \text{ProductLog}\left[\frac{-T^2 + X^2}{e}\right]} - T^2 + X^2\right)\right)\right],$$

$$0, 0\}, \{0, 0, 0, 0\}, \{0, 0, 0, 0\}, \{-KS, -KS\}, 0]$$

```
In[*]:= FullSimplify //@ %
```

```
|simplifica completamente
```

```
Out[*]=
```

```
Zero
```

```
In[*]:= KerrSolutionQ[metricTensor]
```

```
** ReportCompute: Metric[1, 1]
```

```
Already cached
```

```
** ReportCompute: DMetric[-1, -1, -1]
```

```
Already cached
```

```
** ReportCompute: DDMetric[-1, -1, -1, -1]
```

```
Constructed in 0.000882 seconds
```

```
Applied Simplify in 3.91627 seconds
```

```

** ReportCompute: Christoffel[-1, -1, -1]
    Already cached
** ReportCompute: Christoffel[1, -1, -1]
    Already cached
** ReportCompute: CovDOfMetric
    Already cached
** ReportCompute: Riemann[-1, -1, -1, -1]
    Constructed in 0.000522 seconds
    Applied Simplify in 0.019547 seconds
    Stored in 0.000533 seconds and 20160 bytes
** ReportCompute: Riemann[-1, -1, -1, 1]
    Constructed in 0.001381 seconds
    Applied Simplify in 0.034507 seconds
    Stored in 0.00043 seconds and 16544 bytes
** ReportCompute: Ricci[-1, -1]
    Constructed in 0.00011 seconds
    Applied Simplify in 0.006133 seconds
    Stored in 0.000101 seconds and 424 bytes
** ReportCompute: RicciScalar[]
    Constructed in 0.000061 seconds
    Applied Simplify in 0.014171 seconds
    Stored in 0.000116 seconds and 64 bytes
** ReportCompute: Weyl[-1, -1, -1, -1]
    Constructed in 0.000354 seconds
    Applied Simplify in 0.028644 seconds
    Stored in 0.000621 seconds and 20160 bytes
Vacuum
** ReportCompute: computing Weyl concomitant "WeylDual" in 0.018954 seconds:
** ReportCompute: applying Simplify to Weyl concomitant "WeylDual" in 0.009458 seconds:
** ReportCompute: computing Weyl concomitant "WeylSelfDual" in 0.000590 seconds:
** ReportCompute: applying Simplify to Weyl concomitant "WeylSelfDual" in 0.018727 seconds:
** ReportCompute: computing metric concomitant "G2Form" in 0.002027 seconds:
** ReportCompute: applying Simplify to metric concomitant "G2Form" in 0.014203 seconds:
** ReportCompute: computing Weyl concomitant "WeylSelfDual2" in 0.231054 seconds:
** ReportCompute: applying Simplify to Weyl concomitant "WeylSelfDual2" in 0.031721 seconds:
** ReportCompute: computing Weyl concomitant "TraceWeylSelfDual2" in 0.235367 seconds:
** ReportCompute: applying Simplify
    to Weyl concomitant "TraceWeylSelfDual2" in 0.000205 seconds:
** ReportCompute: computing Weyl concomitant "WeylSelfDual3" in 0.014256 seconds:

```


** ReportCompute: applying Simplify to Weyl concomitant "WeylSelfDual3" in 0.031656 seconds:

** ReportCompute: computing Weyl concomitant "TraceWeylSelfDual3" in 0.236269 seconds:

** ReportCompute: applying Simplify
to Weyl concomitant "TraceWeylSelfDual3" in 0.000201 seconds:

Type D

Kerr NUT

Complex Kerr

Kerr

4.92637

Out[*]=

True

In[*]:= SparlingFormUp

Out[*]=

$$S_a = -\frac{1}{2} \eta^{bcde} g_{fe} \Gamma[\nabla]_{ba} \wedge \Gamma[\nabla]_{cd} \wedge \theta^f - \frac{1}{2} \eta_a^{bcd} g_{fd} \Gamma[\nabla]_b^e \wedge \Gamma[\nabla]_{ec} \wedge \theta^f$$

In[*]:= ToBasis[KS] /@%

Out[*]=

$$S_a = -\frac{1}{2} \eta^{bcde} g_{fe} \Gamma[\nabla]_{ba} \wedge \Gamma[\nabla]_{cd} \wedge \theta^f - \frac{1}{2} \eta_a^{bcd} g_{fd} \Gamma[\nabla]_b^e \wedge \Gamma[\nabla]_{ec} \wedge \theta^f$$

In[*]:= TraceBasisDummy /@%;

In[*]:= % /. epsilonmetricg → epsilon[metricTensor];

14.8069

In[*]:= % /. CD → cd;

15.9366

In[*]:= ToValues /@%;

In[*]:= PowerExpand //@%;

[\[expande potencias\]](#)

18.5859

In[*]:= ComponentArray@%;

1.95783

In[*]:= % /. metricg → metricTensor;

In[*]:= % /. Coframe → dx;

In[]:= ToValues /@ %

Out[]:=

$$\left\{ \begin{aligned} S_0 &= \frac{16 m^2 \text{ProductLog}\left[\frac{-T^2+X^2}{e}\right]^2 \sin[\theta] T^2 d[\phi] \wedge d[\phi] \wedge d[X]}{\left(1+\text{ProductLog}\left[\frac{-T^2+X^2}{e}\right]\right)^2 (T^2-X^2)^2} - \\ &\frac{16 m^2 \text{ProductLog}\left[\frac{1}{e}\right]^2 \sin[\theta] T^2 d[\phi] \wedge d[\phi] \wedge d[X]}{\left(1+\text{ProductLog}\left[\frac{1}{e}\right]\right)^2 (T^2-X^2)^2} + \dots 33 \dots + \frac{\text{Csc}[\theta] \left(\frac{8 \dots 5 \dots X}{\left(1+\text{ProductLog}\left[\frac{1}{e}\right]\right)^2} + \dots 4 \dots \right)}{2 m^2 \left(1+\text{ProductLog}\left[\frac{1}{e}\right]\right)^2} - \\ &\frac{\text{Csc}[\theta] \left(\dots 1 \dots \right)^2 \left(\frac{8 m^2 \cos[\theta] \text{ProductLog}\left[\frac{-T^2+X^2}{e}\right] \sin[\theta] d[\phi] \wedge d[\phi] \wedge d[X]}{-T^2+X^2} - \frac{8 \dots 1 \dots 2 \dots 3 \dots T \left(-T \frac{1}{1+\text{ProductLog}\left[\frac{1}{e}\right]} + \dots 1 \dots \right)}{\left(1+\text{ProductLog}\left[\frac{1}{e}\right]\right) \left(1+\text{ProductLog}\left[\frac{1}{e}\right]\right)} + \dots 1 \dots + \frac{\text{Csc}[\theta] \left(\frac{8 \dots 1 \dots X}{-T^2+X^2} + \frac{8 \dots 1 \dots X}{T^2-X^2} + \frac{2 \dots 2 \dots \left(1+\text{ProductLog}\left[\frac{1}{e}\right]\right)}{T^2-X^2} \right)}{\left(1+\text{ProductLog}\left[\frac{1}{e}\right]\right)^2} \right)}{2 m^2 \left(1+\text{ProductLog}\left[\frac{-T^2+X^2}{e}\right]\right)^2} \right\}, \\ S_{\dots 49 \dots} &= \dots 1 \dots, S_{\dots 49 \dots} = \dots 1 \dots, S_3 = \dots 1 \dots \end{aligned} \right\}$$

La expresión completa no está disponible (tamaño de memoria original: 0.9 MB)

In[]:= Map[ToCanonical, %, {2}]

[aplica a todos](#)

11.9396

Out[]:=

$$\left\{ \begin{aligned} S_0 &= \frac{32 m^2 \text{ProductLog}\left[\frac{-T^2+X^2}{e}\right]^2 \sin[\theta] T^2 d[X] \wedge d[\theta] \wedge d[\phi]}{\left(1+\text{ProductLog}\left[\frac{-T^2+X^2}{e}\right]\right)^2 (T^2-X^2)^2} + \dots 70 \dots, \\ S_1 &= \frac{32 m^2 \text{ProductLog}\left[\frac{1}{e}\right]^2 \sin[\theta] d[T] \wedge d[\theta] \wedge d[\phi] X^2}{\left(1+\text{ProductLog}\left[\frac{1}{e}\right]\right)^2 (-T^2+X^2)^2} + \frac{32 \dots 5 \dots X}{\dots 1 \dots} + \dots 66 \dots, \\ S_2 &= \dots 1 \dots, S_3 = \frac{16 m^2 \text{ProductLog}\left[\frac{-T^2+X^2}{e}\right]^2 \sin[\theta] T^2 d[T] \wedge d[X] \wedge d[\theta]}{\left(1+\text{ProductLog}\left[\frac{-T^2+X^2}{e}\right]\right)^2 (T^2-X^2)^2} + \\ &\frac{16 m^2 \text{ProductLog}\left[\frac{-T^2+X^2}{e}\right]^2 \sin[\theta] d[T] \wedge d[X] \wedge d[\theta] X^2}{\left(1+\text{ProductLog}\left[\frac{-T^2+X^2}{e}\right]\right)^2 (T^2-X^2) (-T^2+X^2)} + \dots 37 \dots + \frac{8 m^2 \text{ProductLog}\left[\frac{-T^2+X^2}{e}\right]^2 \sin[\theta] d[T] \wedge d[X] \wedge d[\theta] X^2}{\left(1+\text{ProductLog}\left[\frac{-T^2+X^2}{e}\right]\right) (-T^2+X^2) \left(1+\text{ProductLog}\left[\frac{-T^2+X^2}{e}\right] - T^2+X^2\right)} \end{aligned} \right\}$$

La expresión completa no está disponible (tamaño de memoria original: 0.6 MB)

In[]:= Map[Simplify, %, {2}]

[apl...](#) [simplifica](#)

14.0441

Out[]:=

$$\left\{ \begin{aligned} S_0 &= \\ &- \left(\left(16 m^2 \left(2 \cos[\theta] \left(2 + 3 \text{ProductLog}\left[\frac{-T^2+X^2}{e}\right] + \text{ProductLog}\left[\frac{-T^2+X^2}{e}\right]^2 \right) T d[T] \wedge d[X] \wedge d[\phi] \right. \right. \right. \\ &\quad \left. \left. \left(T^2 - X^2 \right)^2 + \sin[\theta] \left(-2 T d[T] \wedge d[\theta] \wedge d[\phi] X + d[X] \wedge d[\theta] \wedge d[\phi] \left(T^2 + X^2 \right) \right) \right. \right. \\ &\quad \left. \left. \left(2 \left(T^2 - X^2 \right)^2 + 2 \text{ProductLog}\left[\frac{-T^2+X^2}{e}\right] \left(T^2 - X^2 \right)^2 - \text{ProductLog}\left[\frac{-T^2+X^2}{e}\right]^3 \right. \right. \right. \\ &\quad \left. \left. \left(T^2 - X^2 \right) \left(e^{1+\text{ProductLog}\left[\frac{-T^2+X^2}{e}\right]} - T^2 + X^2 \right) - \text{ProductLog}\left[\frac{-T^2+X^2}{e}\right]^2 \right. \right. \\ &\quad \left. \left. \left(e^{2 \left(1+\text{ProductLog}\left[\frac{-T^2+X^2}{e}\right]} \right) + e^{1+\text{ProductLog}\left[\frac{-T^2+X^2}{e}\right]} \left(T^2 - X^2 \right) - 2 \left(T^2 - X^2 \right)^2 \right) \right) \right) \right) / \end{aligned} \right\}$$

$$\begin{aligned}
& \left(\left(1 + \text{ProductLog} \left[\frac{-T^2 + X^2}{e} \right] \right)^2 (T^2 - X^2)^2 \left(e^{1 + \text{ProductLog} \left[\frac{-T^2 + X^2}{e} \right]} - T^2 + X^2 \right)^2 \right), \\
S_1 = & \left(16 m^2 \left(\text{ProductLog} \left[\frac{-T^2 + X^2}{e} \right]^3 \text{Sin}[\theta] (T^2 - X^2) \left(e^{1 + \text{ProductLog} \left[\frac{-T^2 + X^2}{e} \right]} - T^2 + X^2 \right) \right. \right. \\
& \left(-2 T d[X] \wedge d[\theta] \wedge d[\phi] X + d[T] \wedge d[\theta] \wedge d[\phi] (T^2 + X^2) \right) - \\
& 2 (T^2 - X^2)^2 \left(-2 (\text{Cos}[\theta] d[T] \wedge d[X] \wedge d[\phi] + \text{Sin}[\theta] T d[X] \wedge d[\theta] \wedge d[\phi]) X + \right. \\
& \left. \text{Sin}[\theta] d[T] \wedge d[\theta] \wedge d[\phi] (T^2 + X^2) \right) - 2 \text{ProductLog} \left[\frac{-T^2 + X^2}{e} \right] (T^2 - X^2)^2 \\
& \left(- (3 \text{Cos}[\theta] d[T] \wedge d[X] \wedge d[\phi] + 2 \text{Sin}[\theta] T d[X] \wedge d[\theta] \wedge d[\phi]) X + \right. \\
& \left. \text{Sin}[\theta] d[T] \wedge d[\theta] \wedge d[\phi] (T^2 + X^2) \right) + \\
& \left. \text{ProductLog} \left[\frac{-T^2 + X^2}{e} \right]^2 \left(\text{Sin}[\theta] d[T] \wedge d[\theta] \wedge d[\phi] (T^2 + X^2) \right. \right. \\
& \left(e^{2 (1 + \text{ProductLog} \left[\frac{-T^2 + X^2}{e} \right])} + e^{1 + \text{ProductLog} \left[\frac{-T^2 + X^2}{e} \right]} (T^2 - X^2) - 2 (T^2 - X^2)^2 \right) + \\
& \left. 2 X \left(\text{Cos}[\theta] d[T] \wedge d[X] \wedge d[\phi] (T^2 - X^2)^2 + \text{Sin}[\theta] T d[X] \wedge d[\theta] \wedge d[\phi] \right. \right. \\
& \left. \left. \left(-e^{2 (1 + \text{ProductLog} \left[\frac{-T^2 + X^2}{e} \right])} + 2 (T^2 - X^2)^2 + e^{1 + \text{ProductLog} \left[\frac{-T^2 + X^2}{e} \right]} (-T^2 + X^2) \right) \right) \right) \Bigg) / \\
& \left(\left(1 + \text{ProductLog} \left[\frac{-T^2 + X^2}{e} \right] \right)^2 (T^2 - X^2)^2 \left(e^{1 + \text{ProductLog} \left[\frac{-T^2 + X^2}{e} \right]} - T^2 + X^2 \right)^2 \right), \\
S_2 = & \left(4 m^2 \left(\text{Cos}[\theta] \text{ProductLog} \left[\frac{-T^2 + X^2}{e} \right]^5 (T d[X] \wedge d[\theta] \wedge d[\phi] - d[T] \wedge d[\theta] \wedge d[\phi] X) (T^2 - X^2) \right. \right. \\
& \left(e^{1 + \text{ProductLog} \left[\frac{-T^2 + X^2}{e} \right]} - T^2 + X^2 \right) + 2 \text{ProductLog} \left[\frac{-T^2 + X^2}{e} \right]^2 \left(e^{1 + \text{ProductLog} \left[\frac{-T^2 + X^2}{e} \right]} - T^2 + X^2 \right) \\
& \left(\text{Cos}[\theta] (T d[X] \wedge d[\theta] \wedge d[\phi] - d[T] \wedge d[\theta] \wedge d[\phi] X) \left(3 e^{1 + \text{ProductLog} \left[\frac{-T^2 + X^2}{e} \right]} + 5 T^2 - 5 X^2 \right) - \right. \\
& \left. 2 \text{Sin}[\theta] d[T] \wedge d[X] \wedge d[\phi] \left(e^{1 + \text{ProductLog} \left[\frac{-T^2 + X^2}{e} \right]} + 4 T^2 - 4 X^2 \right) \right) + \\
& 2 \text{ProductLog} \left[\frac{-T^2 + X^2}{e} \right]^3 \left(e^{1 + \text{ProductLog} \left[\frac{-T^2 + X^2}{e} \right]} - T^2 + X^2 \right) \\
& \left(\text{Cos}[\theta] (T d[X] \wedge d[\theta] \wedge d[\phi] - d[T] \wedge d[\theta] \wedge d[\phi] X) \left(3 e^{1 + \text{ProductLog} \left[\frac{-T^2 + X^2}{e} \right]} + 4 T^2 - 4 X^2 \right) - \right. \\
& \left. 2 \text{Sin}[\theta] d[T] \wedge d[X] \wedge d[\phi] \left(e^{1 + \text{ProductLog} \left[\frac{-T^2 + X^2}{e} \right]} + 3 T^2 - 3 X^2 \right) \right) + \\
& 2 \text{ProductLog} \left[\frac{-T^2 + X^2}{e} \right]^4 \left(e^{1 + \text{ProductLog} \left[\frac{-T^2 + X^2}{e} \right]} - T^2 + X^2 \right) \\
& \left(\text{Cos}[\theta] (T d[X] \wedge d[\theta] \wedge d[\phi] - d[T] \wedge d[\theta] \wedge d[\phi] X) \left(e^{1 + \text{ProductLog} \left[\frac{-T^2 + X^2}{e} \right]} + 2 T^2 - 2 X^2 \right) - \right. \\
& \left. 2 \text{Sin}[\theta] d[T] \wedge d[X] \wedge d[\phi] (T^2 - X^2) \right) + \\
& (T^2 - X^2) \left(8 \text{Sin}[\theta] d[T] \wedge d[X] \wedge d[\phi] (T^2 - X^2) + \text{Cos}[\theta] \right. \\
& \left. (T d[X] \wedge d[\theta] \wedge d[\phi] - d[T] \wedge d[\theta] \wedge d[\phi] X) \left(2 e^{1 + \text{ProductLog} \left[\frac{-T^2 + X^2}{e} \right]} - 7 T^2 + 7 X^2 \right) \right) + \\
& \left. \text{ProductLog} \left[\frac{-T^2 + X^2}{e} \right] \left(8 \text{Sin}[\theta] d[T] \wedge d[X] \wedge d[\phi] (T^2 - X^2)^2 + \right. \right.
\end{aligned}$$

$$\begin{aligned}
& \cos[\theta] \left(T d[X] \wedge d[\theta] \wedge d[\phi] - d[T] \wedge d[\theta] \wedge d[\phi] X \right) \\
& \left(2 e^{2 \left(1 + \text{ProductLog}\left[\frac{-T^2 + X^2}{e}\right]} \right) - 7 (T^2 - X^2)^2 \right) \Bigg) \Bigg) \Bigg) / \\
& \left(\left(1 + \text{ProductLog}\left[\frac{-T^2 + X^2}{e}\right] \right)^3 (T^2 - X^2) \left(e^{1 + \text{ProductLog}\left[\frac{-T^2 + X^2}{e}\right]} - T^2 + X^2 \right)^2 \right), \\
S_3 = & \left(16 m^2 \sin[\theta] d[T] \wedge d[X] \wedge d[\theta] \right. \\
& \left(-2 (T^2 - X^2)^2 - 2 \text{ProductLog}\left[\frac{-T^2 + X^2}{e}\right] (T^2 - X^2)^2 + \right. \\
& \text{ProductLog}\left[\frac{-T^2 + X^2}{e}\right]^4 (T^2 - X^2) \left(e^{1 + \text{ProductLog}\left[\frac{-T^2 + X^2}{e}\right]} - T^2 + X^2 \right) + \\
& \text{ProductLog}\left[\frac{-T^2 + X^2}{e}\right]^2 \left(e^{2 \left(1 + \text{ProductLog}\left[\frac{-T^2 + X^2}{e}\right]} \right) + \right. \\
& \left. 3 e^{1 + \text{ProductLog}\left[\frac{-T^2 + X^2}{e}\right]} (T^2 - X^2) - 4 (T^2 - X^2)^2 \right) + \text{ProductLog}\left[\frac{-T^2 + X^2}{e}\right]^3 \\
& \left. \left(e^{2 \left(1 + \text{ProductLog}\left[\frac{-T^2 + X^2}{e}\right]} \right) + 2 e^{1 + \text{ProductLog}\left[\frac{-T^2 + X^2}{e}\right]} (T^2 - X^2) - 3 (T^2 - X^2)^2 \right) \right) \Bigg) \Bigg) / \\
& \left(\left(1 + \text{ProductLog}\left[\frac{-T^2 + X^2}{e}\right] \right)^3 (T^2 - X^2) \left(e^{1 + \text{ProductLog}\left[\frac{-T^2 + X^2}{e}\right]} - T^2 + X^2 \right)^2 \right) \Bigg\}
\end{aligned}$$

In[*]:= %128;

In[]:= Map[FullSimplify, %, {2}]

|apl| simplifica completamente

39.1537

Out[]:=

$$\left\{ \begin{aligned} S_0 &= -\frac{1}{\left(1 + \text{ProductLog}\left[\frac{-T^2 + X^2}{e}\right]\right)^4} 16 e^{-2 \left(1 + \text{ProductLog}\left[\frac{-T^2 + X^2}{e}\right]\right)} m^2 \left(2 \cos[\theta] \left(1 + \text{ProductLog}\left[\frac{-T^2 + X^2}{e}\right]\right) \right. \\ &\quad \left. \left(2 + \text{ProductLog}\left[\frac{-T^2 + X^2}{e}\right]\right) T d[T] \wedge d[X] \wedge d[\phi] + \left(1 + \text{ProductLog}\left[\frac{-T^2 + X^2}{e}\right]\right)^3 \right. \\ &\quad \left. \sin[\theta] \left(-2 T d[T] \wedge d[\theta] \wedge d[\phi] X + d[X] \wedge d[\theta] \wedge d[\phi] (T^2 + X^2)\right) \right), \\ S_1 &= \frac{1}{\left(1 + \text{ProductLog}\left[\frac{-T^2 + X^2}{e}\right]\right)^3} 16 e^{-2 \left(1 + \text{ProductLog}\left[\frac{-T^2 + X^2}{e}\right]\right)} m^2 \\ &\quad \left(2 \cos[\theta] \left(2 + \text{ProductLog}\left[\frac{-T^2 + X^2}{e}\right]\right) d[T] \wedge d[X] \wedge d[\phi] X - \left(1 + \text{ProductLog}\left[\frac{-T^2 + X^2}{e}\right]\right)^2 \right. \\ &\quad \left. \sin[\theta] \left(-2 T d[X] \wedge d[\theta] \wedge d[\phi] X + d[T] \wedge d[\theta] \wedge d[\phi] (T^2 + X^2)\right) \right), \\ S_2 &= \left(4 m^2 \left(1 + \text{ProductLog}\left[\frac{-T^2 + X^2}{e}\right]\right) (T - X) (T + X) \left(4 \sin[\theta] d[T] \wedge d[X] \wedge d[\phi] - \right. \right. \\ &\quad \left. \left. \cos[\theta] \left(1 + \text{ProductLog}\left[\frac{-T^2 + X^2}{e}\right]\right) (T d[X] \wedge d[\theta] \wedge d[\phi] - d[T] \wedge d[\theta] \wedge d[\phi] X) \right) \right) / \\ &\quad \left(e^{1 + \text{ProductLog}\left[\frac{-T^2 + X^2}{e}\right]} - T^2 + X^2 \right)^2, \quad S_3 = -\frac{16 m^2 \text{ProductLog}\left[\frac{-T^2 + X^2}{e}\right]^2 \sin[\theta] d[T] \wedge d[X] \wedge d[\theta]}{\left(1 + \text{ProductLog}\left[\frac{-T^2 + X^2}{e}\right]\right) (T^2 - X^2)} \end{aligned} \right\}$$

In[]:= SparlingEF = %;

In[]:= % /. {T[] -> X[]}

Power: Infinite expression $\frac{1}{0}$ encountered. ⓘ

Infinity: Indeterminate expression $0 m^2 \text{ComplexInfinity} \sin[\theta] d[X] \wedge d[X] \wedge d[\theta]$ encountered. ⓘ

Out[]:=

$$\left\{ \begin{aligned} S_0 &= -\frac{64 m^2 \cos[\theta] d[X] \wedge d[X] \wedge d[\phi] X}{e^2}, \\ S_1 &= \frac{64 m^2 \cos[\theta] d[X] \wedge d[X] \wedge d[\phi] X}{e^2}, \quad S_2 = 0, \quad S_3 = \text{Indeterminate} \end{aligned} \right\}$$

In[]:= ProductLog[0]

|producto logaritmo

Out[]:=

0

In[]:= SparlingHorizon = %;

In[*]:= Hodge[metricTensor] /@ SparlingEF

Out[*]=

$$\left\{ * [S_0] = \sin[\theta] (* [d[r] \wedge d[\theta] \wedge d[\phi]]), * [S_1] = \sin[\theta] (* [d[t] \wedge d[\theta] \wedge d[\phi]]), * [S_2] = -\sin[\theta] (* [d[r] \wedge d[t] \wedge d[\phi]]) - \cos[\theta] (m (* [d[r] \wedge d[\theta] \wedge d[\phi]]) + (m-r) (* [d[t] \wedge d[\theta] \wedge d[\phi]])), * [S_3] = \sin[\theta] (* [d[r] \wedge d[t] \wedge d[\theta]]) \right\}$$

In[*]:= Map[ExpandHodgeDual[#, dx[M4], metricTensor] &, %, {2}]

[\[aplica a todos\]](#)

Out[*]=

$$\left\{ * [S_0] = \sin[\theta] \left(\frac{(2m-r) dx^0}{r \sqrt{r^4 \sin[\theta]^2}} + \frac{2m dx^1}{r \sqrt{r^4 \sin[\theta]^2}} \right), * [S_1] = \sin[\theta] \left(-\frac{2m dx^0}{r \sqrt{r^4 \sin[\theta]^2}} - \frac{(2m+r) dx^1}{r \sqrt{r^4 \sin[\theta]^2}} \right), * [S_2] = -\cos[\theta] \left(m \left(\frac{(2m-r) dx^0}{r \sqrt{r^4 \sin[\theta]^2}} + \frac{2m dx^1}{r \sqrt{r^4 \sin[\theta]^2}} \right) + (m-r) \left(-\frac{2m dx^0}{r \sqrt{r^4 \sin[\theta]^2}} - \frac{(2m+r) dx^1}{r \sqrt{r^4 \sin[\theta]^2}} \right) \right) + \frac{r^2 \sin[\theta] dx^2}{\sqrt{r^4 \sin[\theta]^2}}, * [S_3] = \frac{\sin[\theta] \sqrt{r^4 \sin[\theta]^2} dx^3}{r^2} \right\}$$

In[*]:= Map[PowerExpand, %, {2}]

[\[apl... \[expande potencias\]](#)

Out[*]=

$$\left\{ * [S_0] = \sin[\theta] \left(\frac{\csc[\theta] (2m-r) dx^0}{r^3} + \frac{2m \csc[\theta] dx^1}{r^3} \right), * [S_1] = \sin[\theta] \left(-\frac{2m \csc[\theta] dx^0}{r^3} - \frac{\csc[\theta] (2m+r) dx^1}{r^3} \right), * [S_2] = -\cos[\theta] \left(m \left(\frac{\csc[\theta] (2m-r) dx^0}{r^3} + \frac{2m \csc[\theta] dx^1}{r^3} \right) + (m-r) \left(-\frac{2m \csc[\theta] dx^0}{r^3} - \frac{\csc[\theta] (2m+r) dx^1}{r^3} \right) \right) + dx^2, * [S_3] = \sin[\theta]^2 dx^3 \right\}$$

In[*]:= ToValues //@ %

Out[*]=

$$\left\{ * [S_0] = \left(\frac{2m \csc[\theta] d[r]}{r^3} + \frac{\csc[\theta] d[t] (2m-r)}{r^3} \right) \sin[\theta], * [S_1] = \left(-\frac{2m \csc[\theta] d[t]}{r^3} - \frac{\csc[\theta] d[r] (2m+r)}{r^3} \right) \sin[\theta], * [S_2] = d[\theta] - \cos[\theta] \left(m \left(\frac{2m \csc[\theta] d[r]}{r^3} + \frac{\csc[\theta] d[t] (2m-r)}{r^3} \right) + (m-r) \left(-\frac{2m \csc[\theta] d[t]}{r^3} - \frac{\csc[\theta] d[r] (2m+r)}{r^3} \right) \right), * [S_3] = d[\phi] \sin[\theta]^2 \right\}$$

```
In[*]:= Map[FullSimplify, %, {2}]
```

[apl]· [simplifica completamente]

```
Out[*]=
```

$$\left\{ * [S_0] = \frac{2 m (d[r] + d[t]) - d[t] r}{r^3}, * [S_1] = -\frac{2 m d[t] + d[r] (2 m + r)}{r^3}, * [S_2] = \right. \\ \left. d[\theta] - \frac{\text{Cot}[\theta] (m d[t] + d[r] (m + r))}{r^2}, * [S_3] = d[\phi] \text{Sin}[\theta]^2 \right\}$$

```
In[*]:= % /. {r[] -> 2 mass}
```

```
Out[*]=
```

$$\left\{ * [S_0] = 0, * [S_1] = -\frac{d[t]}{4 m^2}, * [S_2] = -\frac{\text{Cot}[\theta] d[t]}{4 m} + d[\theta], * [S_3] = d[\phi] \text{Sin}[\theta]^2 \right\}$$

```
In[*]:= SparlingHorizonDual = %;
```

Define observer:

```
In[*]:= DefTensor[u[a], M4]
```

```
In[*]:= Hodge[metricTensor][S[-a]] × u[a]
```

```
Out[*]=
```

$$u^a (* [S_a])$$

```
In[*]:= ToBasis[KS][%]
```

```
Out[*]=
```

$$u^a (* [S_a])$$

```
In[*]:= TraceBasisDummy[%]
```

```
Out[*]=
```

$$u^0 (* [S_0]) + u^1 (* [S_1]) + u^2 (* [S_2]) + u^3 (* [S_3])$$

```
In[*]:= % /. {r[] -> 2 mass}
```

```
Out[*]=
```

$$u^0 (* [S_0]) + u^1 (* [S_1]) + u^2 (* [S_2]) + u^3 (* [S_3])$$

```
In[*]:= % /. (SparlingHorizonDual /. Equal -> Rule)
```

[igual] [regla]

```
Out[*]=
```

$$-\frac{d[t] u^1}{4 m^2} + \left(-\frac{\text{Cot}[\theta] d[t]}{4 m} + d[\theta] \right) u^2 + d[\phi] \text{Sin}[\theta]^2 u^3$$

```
In[*]:= Collect[%, Diff, Simplify]
```

[agrupa coefi]· [diff] [simplifica]

```
Out[*]=
```

$$d[\theta] u^2 - \frac{d[t] (u^1 + m \text{Cot}[\theta] u^2)}{4 m^2} + d[\phi] \text{Sin}[\theta]^2 u^3$$

```
In[*]:= CoefficientArrays[%, {Diff[t[]], Diff[r[]], Diff[θ[]], Diff[φ[]]}]
```

[arreglos de coeficientes] [diff] [diff] [diff] [diff]

```
Out[*]=
```

$$\left\{ 0, \text{SparseArray}\left[\begin{array}{c} \text{+} \quad \text{■} \quad \text{■} \quad \text{■} \\ \text{Specified elements: 3} \\ \text{Dimensions: {4}} \end{array} \right] \right\}$$

In[*]:= **Last@%** // **Normal**
 último normal

Out[*]=

$$\left\{ -\frac{u^1 + m \cot[\theta] u^2}{4 m^2}, \theta, u^2, \sin[\theta]^2 u^3 \right\}$$

In[*]:= **CTensor**[% , {-KS}]

Out[*]=

$$\text{CTensor}\left[\left\{-\frac{u^1 + m \cot[\theta] u^2}{4 m^2}, \theta, u^2, \sin[\theta]^2 u^3\right\}, \{-KS\}, \theta\right]$$

In[*]:= **flux** = %;

In[*]:= **flux**[-b] × **metricTensor**[a, b] × **flux**[-a]

Out[*]=

$$\frac{u^{2^2}}{r^2} + \frac{\left(-1 - \frac{2m}{r}\right) \left(u^1 + m \cot[\theta] u^2\right)^2}{16 m^4} + \frac{\sin[\theta]^2 u^{3^2}}{r^2}$$

In[*]:= % /. {r[] → 2 mass}

Out[*]=

$$\frac{u^{2^2}}{4 m^2} - \frac{\left(u^1 + m \cot[\theta] u^2\right)^2}{8 m^4} + \frac{\sin[\theta]^2 u^{3^2}}{4 m^2}$$

In[*]:= **FullSimplify**[%]

simplifica completamente

Out[*]=

$$\frac{2 m^2 u^{2^2} - \left(u^1 + m \cot[\theta] u^2\right)^2 + 2 m^2 \sin[\theta]^2 u^{3^2}}{8 m^4}$$

In[*]:= **ToBasis**[KS] [u[a] × **metricTensor**[-a, -b] × u[b]]

Out[*]=

$$\begin{aligned} & -e_a^\theta e_b^\theta u^a u^b + e_a^1 e_b^1 u^a u^b + \frac{2 m e_a^\theta e_b^\theta u^a u^b}{r} + \frac{2 m e_a^1 e_b^\theta u^a u^b}{r} + \\ & \frac{2 m e_a^\theta e_b^1 u^a u^b}{r} + \frac{2 m e_a^1 e_b^1 u^a u^b}{r} + e_a^2 e_b^2 r^2 u^a u^b + e_a^3 e_b^3 r^2 \sin[\theta]^2 u^a u^b \end{aligned}$$

In[*]:= **ToValues@TraceBasisDummy@%**

Out[*]=

$$-u^{\theta^2} + \frac{2 m u^{\theta^2}}{r} + \frac{4 m u^\theta u^1}{r} + u^{1^2} + \frac{2 m u^{1^2}}{r} + r^2 u^{2^2} + r^2 \sin[\theta]^2 u^{3^2}$$

In[*]:= **FullSimplify@%**

simplifica completamente

Out[*]=

$$\frac{(2 m - r) u^{\theta^2} + 4 m u^\theta u^1 + (2 m + r) u^{1^2} + r^3 (u^{2^2} + \sin[\theta]^2 u^{3^2})}{r}$$

In[*]:= % /. {r[] → 2 mass}

Out[*]=

$$\frac{4 m u^\theta u^1 + 4 m u^{1^2} + 8 m^3 (u^{2^2} + \sin[\theta]^2 u^{3^2})}{2 m}$$

In[]:= **x == (r - 1) E^r**
|núme

Out[]:=
 $x == e^r (-1 + r)$

In[]:= **Reduce[%, r, Reals]**
|reduce

Out[]:=

$$\left((x == -1 \&\& r == 0) \mid \mid (x == 0 \&\& r == 1) \mid \mid \left(-1 \leq x < 0 \&\& \left(r == 1 + \text{ProductLog}\left[-1, \frac{x}{e}\right] \mid \mid r == 1 + \text{ProductLog}\left[\frac{x}{e}\right] \right) \right) \mid \mid \left(x > 0 \&\& r == 1 + \text{ProductLog}\left[\frac{x}{e}\right] \right) \right)$$

In[]:= **? ProductLog**
|producto logaritmo

Out[]:=

Symbol i

ProductLog[z] gives the principal solution for w in $z = we^w$.
 ProductLog[k, z] gives the k^{th} solution.

Documentation [Local »](#) | [Web »](#)
 Attributes {Listable, Protected, ReadProtected}
 Full Name System`ProductLog

^

R

In[]:= **? Reduce**
|reduce

Out[]:=

Symbol i

Reduce[*expr*, *vars*] reduces the statement *expr*
 by solving equations or inequalities for *vars* and eliminating quantifiers.
 Reduce[*expr*, *vars*, *dom*] does the reduction over the
 domain *dom*. Common choices of *dom* are Reals, Integers, and Complexes.

Documentation [Local »](#) | [Web »](#)
 Options » Backsubstitution → False ... (7 total)
 Attributes {Protected}
 Full Name System`Reduce

^

In[]:= **? ProductLog**
|producto logaritmo

Out[]:=

Symbol i

ProductLog[z] gives the principal solution for w in $z = we^w$.
 ProductLog[k, z] gives the k^{th} solution.

^

we finally check if the S_3 component in the kruskal-Szekeres coordinates is zero (we prove the limit converges to 0, via 3 ways to make sure)

```
expr = - (16 m^2 ProductLog[ (-T^2 + X^2) / E] ^2 Sin[θ]) /
          |producto logaritmo          |nú... |seno
      ((1 + ProductLog[ (-T^2 + X^2) / E]) (T^2 - X^2));
          |producto logaritmo          |número e
```

(*2. Método A:Límite directo*)

(*Evaluamos el límite cuando la coordenada X se acerca a T*)

```
resultadoLimite = Limit[expr, X → T]
                  |límite
```

(*3. Método B:Usando una variable auxiliar'u' (muy útil para expansiones)*)

(*Hacemos la sustitución $u = -T^2 + X^2$, lo que implica que $T^2 - X^2 = -u$ *)

```
exprU = expr /. { (-T^2 + X^2) → u, (T^2 - X^2) → -u};
```

(*Comprobamos el límite cuando $u \rightarrow 0$ *)

```
Limit[exprU, u → 0]
|límite
```

Out[]=

0

Out[]=

0